

**SE-BRL: A MODALITY-AWARE SOCIAL ENGINEERING BEHAVIOR REPRESENTATION
ARCHITECTURE USING MACHINE LEARNING FOR EXPLAINABLE PHISHING
DETECTION AND RISK-ADAPTIVE DEFENSE**

A Thesis Project

Presented to the Faculty of the
Department of Computer, Information Sciences and Mathematics
University of San Carlos

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SCIENTIA • VIRTUS • DEVOTIO



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APPROVAL SHEET

This thesis proposal entitled: **SE-BRL: A MODALITY-AWARE SOCIAL ENGINEERING BEHAVIOR REPRESENTATION ARCHITECTURE USING MACHINE LEARNING FOR EXPLAINABLE PHISHING DETECTION AND RISK-ADAPTIVE DEFENSE**

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(Proposal Hearing)

SUGGESTIONS	NOT APPLICABLE	COMPLIED WITH	
		YES Page No.	NO Reason/s
1. Standardize the manuscript formatting and presentation to ensure consistency in labeling, headings, spacing, indentation, table and figure formatting, caption placement, terminology, and overall alignment and coherence throughout the revised paper.		Throughout the revised manuscript Chapters 1–4	
2. Properly place the Figure 3.2 label in relation to the diagram and ensure that its figure number, title, and placement are consistent with the formatting of the other figures.		p. 36 Figure 3.2	
3. Revise the SRS Document, Section 2.3, by changing the label “User Classes” to “User Persona” to conform with industry-standard terminology.		p. 9 SRS Document Section 2.3	
4. Revise the SRS scope by separating and clearly identifying the system’s In-Scope and Out-of-Scope functions, boundaries, and exclusions.		p. 7 SRS Document Scope Section	
5. Present the study-specific formulas as properly numbered figures and include them in the manuscript’s figure sequence.		pp. 61–63 Figures 4.4–4.9	
6. Include inference speed and inference latency as explicit metrics in candidate-model evaluation and final model-selection criteria.		pp. 56–60; 62–63 Table 4.8; Section 4.3.5.2; Figures 4.8–4.9	

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7. Align the manuscript and supporting documents with the most updated methodological scope: expert validation is not required for the behavioral dimensions. Remove conflicting references to expert behavioral validation or human behavioral annotation and maintain a coherent deterministic, literature-derived SE-BRL rule specification throughout the study.		Throughout the revised manuscript Especially pp. 4–9, 17–18, 32–40, and 41–53	

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ABSTRACT

Phishing combines technical deception with social-engineering tactics that induce consequential actions such as credential disclosure, financial transactions, or interaction with malicious content. This study proposes the **Social Engineering Behavior Representation Layer (SE-BRL)**, a modality-aware, literature-derived deterministic rule layer for explainable phishing detection and risk-adaptive defense. SE-BRL operationalizes twelve observable indicators under four behavioral dimensions and represents each dimension as **Supported, Absent, or Unavailable** according to artifact assessability. Content-bearing emails and webpages undergo deterministic SE-BRL analysis, while stand-alone URLs and engineered technical records remain within separate U1 technical benchmarks. Phishing detection compares conventional (E1), deterministic SE-BRL (E2), and fused (E3) representations using Logistic Regression, Linear SVM, Random Forest, and XGBoost under leakage-controlled, group-aware partitions and a one-time sealed-test procedure. The explainability framework separates observable artifact evidence from model attribution, while calibrated phishing likelihood, evidence coverage, requested-action severity, corroboration, and uncertainty support deterministic **Informational, Elevated, High, and Review Required** states with bounded defensive responses. A proof-of-concept FastAPI backend, browser extension, and web dashboard demonstrate end-to-end integration. The study excludes expert behavioral annotation and validation, supervised behavioral learning, psychological or demographic profiling, attacker-intent inference, and unsupported behavioral inference from unavailable modalities. Findings are bound to the audited datasets, supported modalities, and defined experimental conditions.

Keywords: phishing detection, social engineering, SE-BRL, behavioral representation, explainable artificial intelligence, risk-adaptive defense

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CHAPTER 1

INTRODUCTION

This chapter establishes the study's foundation by defining the cybersecurity problem, research gap, and proposed Social Engineering Behavior Representation Layer (SE-BRL). It presents phishing as a socio-technical, modality-dependent threat and outlines the research questions, objectives, significance, scope, and delimitations guiding the development and evaluation of the proposed architecture.

1.1 Background of the Study

Phishing remains a persistent socio-technical deception because it couples technical concealment with communication intended to induce consequential action. Modern detection analyzes evidence across multiple channels—message content, routing, URL structure, DOM elements, visual identity, and interaction behavior (Bountakas & Xenakis, 2023; Bustio-Martínez et al., 2026; Prasad & Chandra, 2024). Because technical indicators establish suspicious origin or structure while persuasive content induces pressure, borrows trust, or requests sensitive actions, neither technical irregularity nor persuasive text alone provides a complete artifact representation. Evidence availability varies inherently by modality: emails expose text, sender identities, headers, links, and attachments; webpages present rendered content, logos, DOM forms, and interaction states; and URL-only records provide lexical and host structures lacking conversational context. Recent advances show webpage models identifying brand spoofing and credential harvesting, while URL classifiers achieve strong detection without message text, demonstrating that multimodal frameworks must strictly respect evidence availability boundaries (Lin et al., 2021; Liu et al., 2022; Prasad & Chandra, 2024). Concurrently, machine-learning detectors combine heterogeneous textual, structural, and identity features—such as HELPHED's ensemble learning or recent multimodal persuasion architectures (Bountakas & Xenakis, 2023; Bustio-Martínez et al., 2026). However, high predictive performance on engineered vectors does not imply explicit representation of social-engineering mechanics;

standard feature vectors capture suspicious tokens or DOM elements without formally encoding urgency, impersonation, brand exploitation, confidentiality, or requested-outcome severity. Social-engineering research offers a foundation for formalization: ontologies describe social engineering via explicit entities and relations, while persuasion-focused studies operationalize selected principles as machine-readable labels for classification (Bustio-Martínez et al., 2024; Wang et al., 2021). Yet these frameworks vary in purpose, taxonomy, unit of analysis, and evidence rules. A domain ontology is not an artifact-level feature representation, nor does a persuasion taxonomy cover identity manipulation, brand exploitation, financial requests, or modality-specific unavailability. A defensible representation requires bounded operational definitions, explicit inclusion/exclusion criteria, multi-label support, and traceable evidence. Explainability presents a distinct constraint. Current systems detect imitated brands, infer credential harvesting, generate warnings, or expose features via post-hoc attribution (Bustio-Martínez et al., 2026; Desolda et al., 2025; Lin et al., 2021; Liu et al., 2022). However, artifact evidence (observable content supporting a label) and model attribution (features driving a prediction) address different questions. An explainable architecture must maintain both channels without conflating feature weights with behavioral evidence.

Finally, evaluation integrity requires rigorous experimental design. Security-machine-learning studies demonstrate that random splits, duplicates, source artifacts, and legacy corpora yield inflated performance and weak cross-corpus transfer (Arp et al., 2022; Bhuiyan & Bhuiyan, 2026; Pajola et al., 2025). Behavioral and conventional representations must therefore be evaluated using identical leakage-controlled partitions and consistent preprocessing boundaries. Consequently, this study models observable artifact behaviors, modality constraints, incremental representation value, evidence-grounded explanations, and artifact-level risk processing without inferring recipient susceptibility, psychological states, or user compliance.

1.2 Research Gap

Recent research has established several foundations relevant to the proposed study. Persuasion-oriented approaches can encode selected influence principles for phishing classification; multimodal systems can combine message, URL, and persuasion features; webpage detectors can provide brand- and intention-oriented explanations; and language-model-based tools can generate contextual phishing warnings (Bustio-Martínez et al., 2024; Bustio-Martínez et al., 2026; Desolda et al., 2025; Lin et al., 2021; Liu et al., 2022). The research gap therefore cannot be stated as the complete absence of behavioral, multimodal, or explainable phishing detection.

The more specific representational limitation concerns the absence of a consistently support-aware, artifact-level layer that simultaneously formalizes a bounded set of observable influence cues, trust- and identity-manipulation behaviors, and requested outcomes; permits multiple concurrent dimensions; records traceable artifact evidence; and distinguishes Supported, Absent, and Unavailable states according to modality. Existing systems may explain a target brand, credential-taking intention, influential feature, or persuasion principle, but the reviewed literature does not consistently preserve these outputs within one deterministic evidence representation that can be applied to content-bearing email and webpage artifacts while refusing unsupported behavioral inference from URL-only or incomplete records.

A second limitation concerns controlled empirical attribution of representational value. A fused classifier may perform well without demonstrating whether the added representation contributes information beyond conventional features or whether apparent gains are produced by duplicates, source identity, missingness, or unequal preprocessing. Security-machine-learning research shows that random splits, duplicated records, source artifacts, and legacy corpora can inflate benchmark performance and weaken transfer (Arp et al., 2022; Bhuiyan & Bhuiyan, 2026; Pajola et al., 2025). The incremental contribution of

SE-BRL must therefore be tested through conventional-only, deterministic SE-BRL-only, and fused configurations under identical leakage-controlled conditions.

A third limitation concerns the transition from detection to explanation and response. Model attribution does not by itself establish the observable evidence supporting a behavioral finding, while a phishing probability alone does not encode evidence coverage, requested-action severity, or unresolved uncertainty (Bustio-Martínez et al., 2026; Desolda et al., 2025). A defensible architecture must preserve factual artifact evidence separately from model attribution and connect calibrated phishing likelihood and bounded artifact-level context to auditable defensive-response rules.

The integrated research gap therefore lies in the insufficient establishment of a support-aware and evidence-linked behavioral representation architecture that is deterministically operationalized and software-verified, whose incremental predictive value is evaluated against conventional baselines under leakage-controlled procedures, whose evidence is separated from model attribution, and whose calibrated artifact-level outputs are connected to auditable risk-adaptive responses. The study does not claim expert-validated behavioral ground truth, human behavioral annotation validity, or psychological construct validation.

1.3 Proposed Study

This study proposes the Social Engineering Behavior Representation Layer (SE-BRL): A Modality-Aware Social Engineering Behavior Representation Architecture for Explainable Phishing Detection and Risk-Adaptive Defense. SE-BRL is the behavioral representation component of a broader phishing-defense architecture that integrates conventional features, phishing classification, explainability, artifact-level risk assessment, deterministic decision processing, and proof-of-concept system integration. It complements, rather than replaces, textual, lexical, identity-related, URL-based, and structural phishing features.

SE-BRL operationalizes twelve observable behavioral indicators organized into four primary dimensions. Pressure and Threat Cues include urgency, fear or threat, and scarcity;

Lure and Attention Cues include curiosity and reward or lure; Trust and Identity Manipulation includes authority, impersonation, brand exploitation, and confidentiality or isolation; and Requested Action and Consequence includes calls to action, credential or sensitive-data requests, and financial-action requests. The twelve indicators serve as subordinate evidence indicators, while the four dimensions are deterministic representation states produced by frozen literature-derived evidence rules rather than by human annotation or a separately trained behavioral classifier. Because multiple dimensions may be independently supported within one artifact, SE-BRL remains a multi-label representation.

The architecture is modality-aware. Content-bearing emails and webpages undergo SE-BRL processing only when the required evidence is assessable. Stand-alone URLs and engineered technical records remain in the U1 technical branch and do not receive unsupported language-dependent behavioral values. For each dimension, SE-BRL distinguishes Supported, Absent, and Unavailable evidence and stores traceable evidence references so that missing evidence is not interpreted as behavioral absence.

The study compares conventional-only (E1), deterministic SE-BRL-only (E2), and fused conventional-plus-SE-BRL (E3) representations using identical eligible artifacts, leakage-controlled partitions, fold-local learned preprocessing, candidate classifier families, and evaluation conditions. Four phishing-classification families are evaluated: Logistic Regression, Linear SVM, Random Forest, and XGBoost. Logistic Regression is retained as the primary confirmatory classifier for the principal E3-versus-E1 representation contrast, while the final implementation classifier is selected from development-stage evidence using predeclared predictive, calibration, inference-latency, throughput, resource-efficiency, and reproducibility criteria before the sealed test is opened.

Artifact evidence and model attribution remain separate explanation channels. Artifact evidence identifies the observable content, relation, field, webpage element, or technical condition supporting a finding; model attribution identifies the features influencing the

phishing-classification output. Calibrated phishing likelihood, evidence coverage, requested-action severity, corroboration, and uncertainty are then processed through deterministic rules to produce Informational, Elevated, High, or Review Required states and bounded defensive responses.

The proof-of-concept browser extension, supporting dashboard, and FastAPI backend demonstrate secure integration, modality routing, evidence presentation, decision processing, and end-to-end execution. They are not presented as production-grade security controls, do not measure personal susceptibility or user-behavior change, and do not establish human-level behavioral-classification validity.

1.4 Statement of the Problem

Despite advances in phishing detection, existing approaches do not consistently represent observable social-engineering behaviors as explicit, evidence-linked, modality-bounded computational states whose implementation can be verified independently from the phishing classifier. This limits controlled analysis of the representation's contribution, complementarity with conventional features, and explanatory value. The study therefore asks the following artifact-level technical research questions:

1. To what extent can the frozen literature-derived SE-BRL rule specification consistently represent Supported, Absent, and Unavailable behavioral evidence in eligible email and webpage artifacts while preserving modality boundaries, traceable evidence references, schema conformity, and deterministic repeatability?
2. How do conventional-only (E1), deterministic SE-BRL-only (E2), and fused (E3) feature representations compare in phishing-detection performance under identical leakage-controlled data partitions, preprocessing, candidate classifier families, and evaluation procedures, and which evaluated modality-specific classifier satisfies the predeclared predictive, calibration, inference-latency, throughput, resource-efficiency, and reproducibility criteria for final implementation?

3. To what extent do the explainability components accurately and consistently present observable artifact evidence separately from model-based feature attribution in phishing-detection decisions?
4. To what extent does the context-induced decision mechanism consistently map calibrated phishing likelihood, evidence coverage, requested-outcome severity, corroboration, and uncertainty into risk classifications, review-required states, and predefined risk-adaptive defensive responses?
5. Does the proof-of-concept browser extension and supporting backend satisfy all predefined functional, integration, security, performance, failure-handling, and end-to-end processing requirements for phishing detection, deterministic SE-BRL execution, evidence display, model attribution, risk assessment, decision logic, and bounded interventions?

1.5 Research Objectives

1.5.1 General Objectives

The general objective of this study is to develop and evaluate SE-BRL as a modality-aware, deterministic artifact-evidence representation layer within an explainable phishing-detection and risk-adaptive defense architecture, evaluate its contribution to phishing classification under controlled conditions, preserve evidence separately from model attribution, and demonstrate a bounded proof-of-concept implementation across supported email, webpage, and technical URL evidence environments.

1.5.2 Specific Objectives

1. Develop and software-verify a frozen, modality-aware SE-BRL rule specification and evidence schema that operationalizes four behavioral dimensions through twelve subordinate indicators, distinguishes Supported, Absent, and Unavailable states, preserves traceable evidence references, and enforces modality boundaries.
2. Audit, screen, group, and freeze the eligible datasets; construct E1, E2, E3, and U1 representations; comparatively evaluate Logistic Regression, Linear SVM, Random Forest,

and XGBoost under identical leakage-controlled development conditions; and finalize the implementation classifier using predeclared predictive, calibration, inference-latency, throughput, resource-efficiency, and reproducibility criteria before sealed-test access.

3. Integrate and verify explainability components that separately present observable behavioral and technical artifact evidence and model-based feature attribution associated with phishing-classification decisions.
4. Design and verify a context-induced artifact-risk and decision mechanism that maps calibrated phishing likelihood, evidence coverage, requested-outcome severity, corroboration, and uncertainty into Informational, Elevated, High, or Review Required states and predefined risk-adaptive defensive responses.
5. Implement and functionally validate a proof-of-concept browser extension, FastAPI backend, and supporting dashboard that integrate artifact processing, phishing detection, deterministic SE-BRL representation, evidence presentation, model attribution, risk assessment, decision processing, performance measurement, failure handling, and bounded intervention generation within an end-to-end workflow.

1.6 Significance of the Study

Cybersecurity Researchers and Security Practitioners. The study provides a reproducible framework for representing artifact-level social-engineering evidence as explicit deterministic states rather than informal or expert-annotated labels. The frozen rule specification, modality boundaries, dataset-audit artifacts, controlled representation comparisons, and model-selection records can support replication, refinement, and scrutiny of predictive contribution, calibration, efficiency, explanation quality, and shortcut learning.

End Users. The proof-of-concept demonstrates how phishing analysis can present supporting technical and behavioral artifact evidence, model attribution, bounded defensive responses beyond a binary prediction. The study does not measure user understanding, trust, decision-making, compliance, susceptibility, or resistance to phishing.

Future Researchers. The audited datasets, deterministic SE-BRL rule package, candidate-model results, model-selection records, explanation outputs, risk scenarios, and reproducibility artifacts can support future extensions involving additional modalities, alternative classifiers, human-subject evaluation, external validation, or production-oriented systems without implying that those extensions are established by the present study.

1.7 Scope and Delimitations

This study develops and evaluates SE-BRL, a frozen, literature-derived deterministic rule layer with four behavioral dimensions and twelve subordinate indicators, using content-bearing emails, webpages, stand-alone URLs, and engineered technical records as the unit of analysis. Public datasets undergo provenance, label, content-availability, privacy/licensing, duplication, grouping, and leakage audits before final inclusion. Emails and webpages receive Supported, Absent, or Unavailable behavioral states, while PhiUSIIL and UCI Phishing Websites remain separate U1 technical benchmarks without language-dependent behavioral modeling. Phishing detection compares conventional (E1), deterministic SE-BRL (E2), and fused (E3) representations using Logistic Regression, Linear SVM, Random Forest, and XGBoost under identical fold-isolated controls. Explainability separates artifact evidence from model attribution, while calibrated phishing likelihood, evidence coverage, requested-action severity, corroboration, and uncertainty determine Informational, Elevated, High, or Review Required states and bounded responses. The study excludes expert behavioral annotation or validation, supervised behavioral learning, psychological profiling, attacker-intent inference, live monitoring, payload execution, endpoint defense, and unsupported phishing channels; findings remain bounded to the audited datasets, supported modalities, and experimental conditions.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter synthesizes literature across observable social-engineering behaviors, modality-aware evidence, multimodal detection, behavioral representation, feature fusion, explainability, calibrated risk, adaptive warnings, dataset validity, and defensive systems. Decoupling artifact-level evidence from recipient susceptibility focuses the review on observable content, modality inference boundaries, and uncertainty-constrained decisions. Consequently, this synthesis highlights the central research gap addressed by SE-BRL: the absence of a unified, evidence-linked representation that maintains cross-modality behavioral semantics while unifying conventional features, comparative evaluation, explainable decisions, and risk-adaptive defense.

2.1 Phishing as a Socio-Technical and Multimodal Threat

2.1.1 Technical and Social-Engineering Characteristics of Phishing

Recent literature conceptualizes phishing as an attack coupling technical subterfuge with action-inducing communication. Social-engineering ontologies formalize these threats via relationships among attacker, target, medium, technique, and goal, demonstrating that phishing transcends malicious code or isolated features (Wang et al., 2021). Modern frameworks evaluate URL structural anomalies, identity inconsistencies, DOM elements, visual imitation, and linguistic persuasion as complementary evidence channels (Bustio-Martínez et al., 2024; Liu et al., 2022; Prasad & Chandra, 2024). Technical indicators establish origin and capability, whereas behavioral cues capture compliance mechanisms; SE-BRL preserves both channels without treating either as a self-sufficient explanation.

2.1.2 Phishing Across Email, Webpage, and URL Modalities

Phishing modalities expose different evidence. Email systems can analyze language, identity fields, links, and attachment metadata, whereas webpage systems inspect text, forms,

HTML, screenshots, logos, and interaction dynamics (Al-Subaiey et al., 2024; Bountakas & Xenakis, 2023; Lin et al., 2021; Liu et al., 2022). URL models instead use lexical, structural, and host-derived attributes without rendering content (Buu & Cho, 2025; Prasad & Chandra, 2024; Rashid et al., 2024). These approaches agree on the value of multiple channels but expose a boundary: URL-only records cannot reveal the full persuasive message or requested action. Evidence availability must therefore be established before outputs are compared across artifacts.

2.1.3 Implications for Artifact-Level Phishing Analysis

Artifact-level analysis mandates that claims remain strictly proportional to observed evidence. Webpage models infer brand targeting and credential harvesting from rendered interfaces, whereas URL-only inputs lack content-level evidence (Liu et al., 2022). Furthermore, component-specific persuasion variations prevent generalizing subject-line traits to message bodies (Khadka et al., 2026). Aligning feature semantics with operational availability prevents shortcut learning in security models (Arp et al., 2022). Consequently, SE-BRL remains bounded to observable artifact evidence—neither inferring unobservable psychological states nor equating missing content with behavioral absence.

2.2 Observable Social-Engineering Behaviors in Phishing Artifacts

2.2.1 Influence Cues

Influence cues denote observable content generating pressure, loss anticipation, opportunity, or heightened attention. Persuasion literature shows phishing routinely blends authority, distraction, scarcity, affect, reciprocity, and deception rather than single triggers (Bustio-Martínez et al., 2024; Khadka et al., 2026). While subject lines favor immediate triggers, message bodies construct layered narratives (Khadka et al., 2026). SE-BRL formalizes five influence indicators: urgency, fear/threat, scarcity, curiosity, and reward/lure. Because legitimate notices leverage identical tactics, these indicators register observable influence rather than

inherent maliciousness, requiring joint evaluation with identity, destination, and outcome evidence.

2.2.2 Trust-Manipulation Behaviors

Trust manipulation appropriates credibility from individuals, institutions, brands, or restricted relationships. SE-BRL formalizes four distinct constructs: authority, impersonation, brand exploitation, and confidentiality/isolation.

These constructs are functionally non-equivalent: visual systems detect brand anomalies without sender impersonation, whereas email header anomalies confirm impersonation absent visual branding (Lin et al., 2021; Liu et al., 2022). Social-engineering ontologies similarly differentiate attack techniques, media, targets, and goals (Wang et al., 2021). Because authority and deception frequently co-occur across message components (Khadka et al., 2026), distinct indicators remain defensible only under explicit, evidence-backed inclusion and exclusion criteria.

2.2.3 Requested Outcomes

Requested outcomes capture intended recipient actions, distinguishing general calls-to-action (clicking, replying, downloading), sensitive-data requests (passwords, identity data), and financial requests (transfers, payments). Webpage analysis shows brand identification differs from credential-taking intent, as dynamic interactions reveal requests hidden in screenshots (Liu et al., 2022). Combining text, link, and structural evidence allows action language and technical destinations to corroborate each other (Al-Subaiey et al., 2024; Bountakas & Xenakis, 2023). Thus, while interactive elements are not inherently malicious, preserving requested purposes provides critical semantic context missing from raw structural features.

2.2.4 Behavioral Overlap, Legitimate Uses, and Evidence Boundaries

Behavioral overlap is expected because phishing artifacts combine multiple cues. Khadka et al. (2026) identified recurring persuasion combinations, while Bustio-Martínez et al.

(2024) demonstrated that variable cue detectability precludes collapsing tactics into a singular persuasion score. This supports a multi-label representation where urgency coexists with authority, impersonation, and credential requests. Crucially, it prevents circular inference: cues cannot be inferred solely from a malicious label, as legitimate messages frequently impose deadlines or request actions. SE-BRL decoupled behavior detection from phishing verdicts and excludes recipient psychology, emotional state, or susceptibility (Zhuo et al., 2023). Consequently, a rigorous codebook must ensure each indicator remains observable, bounded, and reproducible across modalities.

2.3 Modality-Aware Evidence in Email, Webpage, and URL Artifacts

2.3.1 Evidence Available in Email Artifacts

Email artifacts can expose subject and body language, sender and reply identities, embedded URLs, header fields, authentication outcomes, formatting, and attachment metadata. Hybrid systems combine these signals because text may miss identity anomalies and structural features may miss persuasive intent (Al-Subaiey et al., 2024; Bountakas & Xenakis, 2023). Message content can support influence cues and requested outcomes, while identity fields can corroborate impersonation or brand exploitation. The scope remains limited to available content and metadata: attachment filename, extension, or MIME information may be used, but attachment contents are not opened or analyzed. Missing bodies and headers are unavailable rather than negative evidence.

2.3.2 Evidence Available in Content-Bearing Webpages

Content-bearing webpages expose visible text, DOM structure, logos, forms, input types, and resource relationships. Phishpedia demonstrates brand identification via visual logos, whereas PhishIntention pairs visual appearance with runtime dynamics to infer brand target and credential-harvesting intent (Lin et al., 2021; Liu et al., 2022). Global-local visual similarity models further leverage multi-scale imagery, though performance remains bounded by reference set coverage and adversarial evasion (Wang et al., 2024). SE-BRL extracts text and

structural elements to evaluate authority, urgency, brand exploitation, and credential requests. However, raw webpage datasets frequently omit required rendered assets; content availability must be verified prior to behavioral pipeline routing.

2.3.3 Evidence Available in URL-Only Records

URL-only records support lexical and structural analysis through length, token patterns, domain and path composition, subdomains, character distributions, similarity indicators, and host-related attributes. PhiUSIIL and recent URL research show that these variables can support strong technical classification and incremental or calibrated detection (Buu & Cho, 2025; Prasad & Chandra, 2024). Nevertheless, cross-domain evaluations show that performance can deteriorate when the target distribution differs from training data, even when URL classifiers perform well within a source (Rashid et al., 2024). A URL can sometimes contain a brand-like token or action-related word, but it cannot reliably establish message-level fear, confidentiality, or a complete credential request. URL-only records should therefore remain valid technical inputs while unsupported language-dependent behaviors are marked unavailable rather than encoded as zero.

2.3.4 Supported, Absent, and Unavailable Behavioral Evidence

A three-state classification prevents absent modalities from masquerading as negative behavioral evidence. Supported indicates required fields contain codebook-compliant evidence; absent denotes an inspected, fully available environment containing no qualifying cues; unavailable specifies missing required fields or modalities. This distinction operationalizes feature-meaning and distribution-alignment principles in security machine learning (Arp et al., 2022) while accounting for component-level persuasion variances (Khadka et al., 2026). Furthermore, it establishes a defensible missingness policy for feature fusion. Instead of permitting collection-dependent missingness to act as a spurious class signal, SE-BRL explicitly encodes evidence availability and links Supported states to verifiable text spans, field relationships, forms, or structural elements.

2.4 Conventional and Multimodal Phishing Detection Approaches

2.4.1 Textual and Linguistic Features

Recent email studies use lexical, semantic, and contextual representations, but generally favor combinations rather than text alone. HELPHED integrates heterogeneous evidence through ensembles, while an interpretable platform combines text processing with explainability (Al-Subaiey et al., 2024; Bountakas & Xenakis, 2023). Persuasion research adds a higher-level semantic representation of influence principles (Bustio-Martínez et al., 2024; Khadka et al., 2026). Conventional vectors preserve detailed vocabulary; behavioral indicators compress language into interpretable constructs. Their contribution must be compared because compression can improve meaning while discarding predictive detail.

2.4.2 URL and Lexical Features

URL classifiers leverage character, token, domain, path, and similarity features for rapid pre-rendering execution. Comparative evaluations confirm tree ensembles and boosting methods excel on engineered phishing datasets, while PhiUSIIL integrates URL and webpage-derived indicators, and fuzzy-calibrated transformers address decision uncertainty (Almujahid et al., 2024; Buu & Cho, 2025; Prasad & Chandra, 2024). However, domain-adaptation failures demonstrate that high within-dataset performance does not ensure cross-corpus transfer (Rashid et al., 2024). Consequently, URL features remain an essential baseline branch, providing technical indicators rather than complete behavioral semantics.

2.4.3 Email Metadata and Identity Features

Metadata and identity features describe relationships among displayed identity, actual address, reply destination, routing, authentication, and links. Contemporary systems combine them with content because legitimate language may originate from inconsistent infrastructure, while technically aligned messages may come from compromised accounts (Al-Subaiey et al., 2024; Bountakas & Xenakis, 2023). Thus, identity evidence is not a self-sufficient verdict. A

mismatch may support impersonation, but the behavior remains distinct from the classifier's use of that feature, supporting separate auditable representations.

2.4.4 Webpage and Structural Features

Webpage detection increasingly integrates visual, structural, and interaction evidence. Phishpedia isolates target brands via visual logos, PhishIntention infers credential-harvesting intent through static and dynamic analysis, and multi-scale similarity models analyze visual correspondence (Lin et al., 2021; Liu et al., 2022; Wang et al., 2024). These frameworks confirm that rendered content provides evidence inaccessible to URL-only models. However, adversarial modifications significantly degrade machine-learning and deep-learning detection rates (Kulkarni et al., 2025). Thus, webpage features expand semantic coverage but remain vulnerable to evasion, reference-set bounds, and distribution shifts.

2.4.5 Machine-Learning and Multimodal Detection Approaches

Recent phishing studies compare linear methods, support vector machines, random forests, boosted trees, deep networks, and transformers. Linear and support-vector models provide baselines for sparse text and URL features, whereas forests and boosting capture nonlinear feature interactions (Almujahid et al., 2024; Bhuiyan & Bhuiyan, 2026; Hannousse & Yahiouche, 2021). Multimodal systems combine message, URL, persuasion, visual, or dynamic evidence (Bustio-Martínez et al., 2025; Liu et al., 2022). Bustio-Martínez et al. (2025) are the closest precedent because they fuse message, URL, and persuasion features with SHAP. This narrows SE-BRL's defensible novelty to support-aware states, exact evidence linkage, three-modality routing, controlled representation comparisons, and risk-adaptive use.

2.4.6 Limitations of Conventional Feature Representations

Conventional representations can be predictive without preserving the social-engineering meaning of an artifact. They are also vulnerable to temporal drift, source artifacts, prevalence changes, and adversarial manipulation. Continual-learning research documents degradation as phishing evolves, cross-domain URL research shows transfer failure,

and recent cross-corpus email analysis demonstrates that pooled benchmark results can conceal artifact-driven learning (Bhuiyan & Bhuiyan, 2026; Ejaz et al., 2023; Rashid et al., 2024). Kulkarni et al. (2025) further show that adversarial page modifications can evade multiple detector families. Behavioral representation does not automatically solve these problems; labels can introduce annotation bias or duplicate the class definition. The literature therefore supports a layered evaluation in which conventional and behavioral channels remain independently inspectable and are tested under the same leakage-controlled conditions.

2.5 Representation of Observable Social-Engineering Behaviors

2.5.1 Social-Engineering Taxonomies and Construct Operationalization

Taxonomies organize social-engineering knowledge, but a machine-readable representation requires observable definitions rather than broad psychological concepts. Wang et al. (2021) formalized social engineering through an ontology linking technique, medium, goal, target, and other entities, while recent phishing studies operationalized persuasion principles in message components (Bustio-Martínez et al., 2024; Khadka et al., 2026). These sources agree that structured categories are useful, yet their labels and analytical purposes differ. SE-BRL therefore does not claim that its twelve indicators reproduce a single established scale. Instead, it organizes recurring observable evidence into four operational dimensions: Pressure and Threat Cues, Lure and Attention Cues, Trust and Identity Manipulation, and Requested Action and Consequence. Each indicator requires a definition, inclusion rule, exclusion rule, minimum evidence, and supported modality, thereby translating theory into artifact-level constructs without diagnosing users or asserting attacker intention.

2.5.2 Behavioral Evidence Codebooks and Operationalization

Prior social-engineering and content-analysis research demonstrates the value of explicit codebooks, operational definitions, evidence units, inclusion and exclusion rules, and procedures for resolving ambiguous cues. Some prior studies establish behavioral labels through human annotation and agreement statistics; those procedures are relevant as examples

of how constructs have been bounded in the literature, but they are not adopted as behavioral ground truth in the present study (Krippendorff, 2018; Bustio-Martínez et al., 2024; Khadka et al., 2026).

For SE-BRL, the reviewed literature is used to derive a frozen deterministic evidence specification. Each subordinate indicator is defined through observable evidence requirements, modality support, minimum qualifying conditions, exclusion rules, and traceable evidence locators. Verification therefore concerns whether the software executes the frozen specification correctly, preserves Supported/Absent/Unavailable distinctions, respects modality boundaries, produces valid evidence references, and repeats deterministically. The study does not claim expert-validated psychological constructs or inter-annotator behavioral ground truth.

2.5.3 Multi-Label Behavioral Representation

Phishing artifacts can contain multiple social-engineering cues simultaneously; urgency can coexist with authority, brand exploitation, credential requests, and calls to action. Multi-label learning literature provides a useful conceptual distinction from mutually exclusive categorization because several labels may be supported in one observation (Zhang & Zhou, 2014). In the present study, however, multi-label semantics describe the deterministic SE-BRL representation rather than a separately trained behavioral classifier.

Each of the four dimensions is evaluated independently against the frozen evidence rules when the required modality fields are assessable. This permits multiple Supported states within one artifact and preserves Unavailable when the evidence environment does not support assessment. Supervised learning is reserved for the downstream phishing-classification task that consumes E1, E2, or E3 features.

2.5.4 Evidence-Linked Behavioral Representation

Evidence linkage connects a behavior assertion to the observable element that justifies it. In email, this may be a phrase, sender-reply mismatch, or destination relation; in webpages, it may be visible language, a credential form, logo-domain inconsistency, or interaction state.

Contemporary explainable phishing systems increasingly provide target-brand, feature, or generated-language explanations (Desolda et al., 2025; Lin et al., 2021; Liu et al., 2022), but these explanation channels do not automatically constitute a validated behavioral codebook. SE-BRL's evidence record is therefore conceptually prior to model attribution: it establishes what was observed and whether the modality supported the state. This design enables audit, rule/evidence consistency analysis, and modality-aware abstention even when the final classifier uses other features more heavily.

2.6 Behavioral and Conventional Feature Fusion in Phishing Detection

2.6.1 Feature-Fusion Approaches in Prior Studies

Recent phishing systems fuse heterogeneous features through vectors, ensembles, or specialized channels. HELPHED integrates email feature groups, PhishIntention combines appearance with dynamics, and Bustio-Martínez et al. (2025) combine message, URL, and persuasion evidence (Bountakas & Xenakis, 2023; Bustio-Martínez et al., 2025; Liu et al., 2022). Fusion broadens coverage, but the studies differ in input availability, semantic granularity, and evaluation design. It may improve discrimination while obscuring modality contribution, missingness, or transferability.

2.6.2 Independent and Incremental Feature Contribution

A fused model alone cannot establish the value of a behavioral layer. Bustio-Martínez et al. (2025) strengthen the literature by comparing multimodal and unimodal representations, while broader security-machine-learning guidance calls for meaningful baselines and controlled comparisons (Arp et al., 2022). For SE-BRL, the literature supports three analytically distinct questions: whether conventional evidence predicts phishing, whether behavior indicators contain independent class information, and whether fusion adds information beyond the conventional baseline. These questions correspond to conventional-only, behavior-only, and fused configurations, but the RRL does not presume which will perform best. Behavioral

features may contribute to recall, explanation, error analysis, or intervention selection even when overall accuracy changes little.

2.6.3 Comparative Evaluation of Feature Representations

Feature representations must be compared under the same partitions, preprocessing boundaries, model families, and tuning opportunities; otherwise, performance differences may reflect unequal experimental conditions. Recent cross-corpus and domain-adaptation studies show that conclusions can reverse when evaluation moves beyond the source distribution (Bhuiyan & Bhuiyan, 2026; Rashid et al., 2024). Comparison should therefore consider discrimination, calibration, modality-specific performance, cross-source transfer, and deterministic rule conformance and evidence-state correctness rather than a single aggregate accuracy. This broader evaluation is essential when outputs will drive risk tiers, because an apparently stronger fused model may be less calibrated or more dependent on corpus artifacts.

2.6.4 Limitations of Existing Fusion Approaches

Fusion introduces risks of duplicate information, source identification, missingness shortcuts, and dominance by one high-dimensional modality. Arp et al. (2022) warn that security models often learn unintended shortcuts, while Bhuiyan and Bhuiyan (2026) show that email classifiers can identify corpus artifacts rather than portable phishing characteristics. The close prior work by Bustio-Martínez et al. (2025) demonstrates explainable persuasion-aware fusion, but it does not fully establish SE-BRL's proposed combination of a twelve-label evidence codebook, support-aware email-webpage-URL routing, exact behavioral evidence distinct from SHAP attribution, and calibrated evidence-to-response logic. This is a narrower and more defensible gap than claiming that multimodal persuasion-aware phishing detection does not exist.

2.7 Explainable Phishing Detection

2.7.1 Inherently Interpretable and Post-Hoc Approaches

Recent phishing research uses both directly interpretable models and post-hoc explanations. Linear and tree-based models can expose coefficients, rules, or split logic, while complex ensembles, deep networks, and language models often require SHAP-like attributions or generated explanations (Al-Subaiey et al., 2024; Bustio-Martínez et al., 2025; Desolda et al., 2025). These approaches agree on the value of intelligible decisions but differ in fidelity. A plausible natural-language warning is not necessarily a faithful account of the classifier, and a feature attribution is not automatically understandable to a user. SE-BRL can improve semantic readability by presenting explicit behaviors, yet the model remains explainable only to the extent that the relationship among evidence, behavior features, and the final decision can be inspected.

2.7.2 Local and Global Model Explanations

Local explanations address one artifact, whereas global explanations describe recurring model reliance. SHAP-based persuasion analysis and interpretable email platforms expose influential features, while brand-identification systems explain a local decision by naming an imitated entity (Al-Subaiey et al., 2024; Bustio-Martínez et al., 2025; Lin et al., 2021; Liu et al., 2022). Local evidence can support a warning; global patterns can reveal shortcuts or implausible dependencies. One global ranking, however, cannot validate every local decision.

2.7.3 Behavioral Evidence and Model Attribution

Behavioral evidence and model attribution answer different questions. Evidence asks whether an artifact satisfies a codebook definition and where that support is located; attribution asks how much a model used a feature when producing its output. A credential request may be clearly supported but receive low attribution because a destination or identity feature dominates. Conversely, a behavior feature may receive high attribution even when the underlying deterministic state is disputed. Generated warnings such as APOLLO illustrate the value of

user-facing explanations, while persuasion-aware SHAP analysis illustrates model-centered attribution (Bustio-Martínez et al., 2025; Desolda et al., 2025). SE-BRL preserves both channels and prevents model importance from being restated as factual behavioral evidence.

2.7.4 Reliability and Limitations of Explanations

Explanation quality includes fidelity, stability, completeness, and alignment with observable evidence.

APOLLO demonstrates that language models can generate understandable phishing warnings, but its participant evaluation also reflects a human-centered scope that differs from purely functional prototype testing (Desolda et al., 2025). The close multimodal study reports SHAP-based interpretability, yet attribution still depends on representation and background data (Bustio-Martínez et al., 2025). Explanations can therefore be useful without being causal or universally reliable. In the present study, explanation validation is bounded to whether displayed behaviors and evidence correspond to the analyzed artifact, whether model contributions are reported as attributions, and whether unavailable evidence is disclosed rather than invented.

2.8 Context-Induced Risk Assessment

2.8.1 Phishing Probability, Confidence, and Calibration

A classifier score, confidence estimate, and operational risk level are not identical. Recent URL research explicitly integrates calibration or uncertainty into model outputs, while cross-corpus studies show that apparently strong scores can become unreliable under changed prevalence or source conditions (Bhuiyan & Bhuiyan, 2026; Buu & Cho, 2025). For SE-BRL, phishing probability is the model's estimated class likelihood under evaluated conditions; calibration concerns whether those estimates correspond to observed frequencies; and confidence concerns the reliability of the decision given available evidence. None is a universal probability that a specific person will be harmed. Calibration is necessary before probability ranges are used in risk tiers, but calibration alone does not encode consequence or requested-action severity.

2.8.2 Artifact-Level Risk Indicators and Severity

Context-induced risk is defined as an artifact-level decision construct integrating four bounded components: estimated phishing likelihood, strength and coverage of supporting evidence, severity of the requested outcome, and uncertainty or conflict among modalities. The requested-outcome literature supports distinguishing a credential form, sensitive-data request, or financial action from a low-consequence link, while webpage-intention systems show that credential-taking intention can be analyzed separately from visual brand identity (Liu et al., 2022). This definition does not estimate organizational loss or personal susceptibility. It provides a transparent basis for deciding whether the artifact warrants information, caution, stronger interruption, or review.

2.8.3 Uncertainty, Thresholds, and Review-Required Decisions

Operational thresholds translate scores into actions, but distribution shift and missing modalities can make escalation unsafe. Cross-domain adaptation, continual learning, and cross-corpus evaluation show that reliability changes outside training conditions (Bhuiyan & Bhuiyan, 2026; Ejaz et al., 2023; Rashid et al., 2024). A review-required state is defensible when probability is intermediate, evidence conflicts, fields are unavailable, or the source is unfamiliar. It is an uncertainty-aware policy rather than a new class. Exact thresholds and calibration procedures remain Chapter 4 decisions.

2.8.4 Boundaries of Artifact-Level Risk Assessment

Human-centered phishing studies examine susceptibility, training, and user response, but those constructs require participant data and cannot be inferred from artifacts alone (Zhuo et al., 2023). SE-BRL therefore excludes demographic inference, personality, cognitive state, emotional vulnerability, and predictions that a particular recipient will comply. It also avoids equating a phishing probability with total harm. The risk assessment is limited to observable artifact conditions, model uncertainty, and requested consequences within the supported

modality. This boundary keeps the construct aligned with the study's data and prevents behavioral representation from becoming an unsupported psychological profile.

2.9 Risk-Adaptive Defensive Responses

2.9.1 Security Warning Design

Recent warning research supports timely, comprehensible, and actionable communication rather than undifferentiated alerts. Greco et al. (2025) found that warning timing and explanation affected responses in an email-client experiment, while Pham et al. (2025) emphasize that design decisions shape how people experience phishing warnings. APOLLO similarly explores generated explanations for phishing emails (Desolda et al., 2025). These studies agree that context matters, but they also show that effectiveness claims require human-subject evidence. SE-BRL can draw on their design principles while limiting its own evaluation to implementation and evidence-response alignment because the present study does not test user behavior.

2.9.2 Contextual and Evidence-Based Interventions

A contextual intervention states the concern and recommends an action matched to evidence. Identity inconsistency may justify independent verification; a credential request on a mismatched destination may justify interruption; and a financial request combined with impersonation may warrant explicit transfer avoidance. Explanation-enhanced warnings support providing reasons rather than bare labels (Desolda et al., 2025; Greco et al., 2025). SE-BRL therefore maps supported behavior, technical evidence, risk tier, and uncertainty to bounded, auditable recommendations.

2.9.3 Warning Fatigue, Overwarning, and Response Escalation

Warning systems must balance interruption against false-positive burden and habituation. Pham et al. (2025) show that users interpret warning design as part of the security experience, and Greco et al. (2025) report that warning timing and explanation should vary with context rather than remain uniform. Bender et al. (2026) further demonstrate the value of

just-in-time feedback in a large organizational experiment, but that result concerns training and user response, not artifact classification. These findings support graded escalation: lower-confidence or lower-severity cases can present caution or verification guidance, while strongly supported credential or financial threats can trigger more interruptive responses. Poor calibration would undermine this design by producing excessive escalation.

2.9.4 Evaluation Boundaries of Defensive Responses

The present proof-of-concept cannot claim reduced click-through, improved comprehension, durable learning, or lower organizational loss because it does not conduct a controlled user study. Those outcomes belong to the human-centered evidence base represented by Greco et al. (2025), Bender et al. (2026), and Desolda et al. (2025). Its defensible evaluation is narrower: whether the correct intervention rule is selected, whether the displayed reason corresponds to supported evidence, whether uncertainty produces review rather than overclaiming, whether warning coverage is complete for defined scenarios, and whether unsafe recommendations are prevented. This distinction separates functional validation from human-effectiveness validation and avoids extending the prototype beyond its tested scope.

2.10 Dataset Quality and Evaluation Validity in Phishing Research

2.10.1 Dataset Provenance, Label Validity, and Class Definitions

Recent research shows that phishing results depend on corpus construction, class definitions, and artifact completeness. Website benchmark design and legitimate-page selection materially affect measured performance and generalization (Hannousse & Yahiouche, 2021; Sánchez-Paniagua et al., 2022). Legacy email sources such as Enron, SpamAssassin, Nazario, and CEAS-08 remain candidates, but their mixed semantics and source artifacts require verification rather than automatic pooling (Bhuiyan & Bhuiyan, 2026). PhiUSIIL is a contemporary URL-oriented benchmark, yet engineered features do not automatically provide raw content for SE-BRL evidence operationalization (Prasad & Chandra, 2024). UCI Phishing

Websites and ISCX-URL2016 likewise belong to the technical branch only after provenance and label audit.

2.10.2 Duplicates, Leakage, and Source-Class Confounding

Exact duplicates, near-duplicate templates, shared campaigns, repeated domains, and source-specific formatting can inflate security-machine-learning performance when related records cross data partitions (Arp et al., 2022; Bhuiyan & Bhuiyan, 2026; Pajola et al., 2025). Dataset handling must therefore occur before model development: duplicate and near-duplicate relationships are grouped conservatively, and a group is prohibited from crossing development, calibration, and sealed-test partitions.

Source-class confounding is especially important when phishing and legitimate email records originate from different corpora. The study consequently retains source-only baselines, normalized-format ablations, source-stratified error reporting, and an explicit limitation where source and class are correlated. Dataset source names, file paths, group identifiers, and target labels are audit or partitioning metadata and are not supplied as predictive features. Phishing class labels are also never used as inputs to deterministic SE-BRL evidence rules.

2.10.3 Dataset Splitting, Generalization, and Cross-Corpus Evaluation

Evaluation validity requires grouping before stratification and preventing a campaign, template, registrable domain, or duplicate family from crossing partitions. The study therefore uses a development, calibration, and sealed-test structure with grouped cross-validation inside the development partition and fold-local fitting of learned preprocessing operations. These controls reduce leakage and preserve a genuinely untouched final evaluation (Arp et al., 2022).

The methodology does not require out-of-fold predictions from a separate behavioral-dimension model because SE-BRL is deterministic. Out-of-fold predictions remain applicable to learned phishing classifiers or probability-calibration procedures when required by the frozen protocol. Claims of generalization remain limited to the evaluated sources and are not extended to unseen populations or all future phishing attacks without external evidence.

2.10.4 Metrics, Calibration, and Reproducibility

Standard accuracy fails under shifting prevalence, asymmetric costs, or threshold adjustments. While evaluations report precision, recall, and F1, calibrated models and prevalence-shift studies show probability quality requires separate evaluation (Almujahid et al., 2024; Bhuiyan & Bhuiyan, 2026; Buu & Cho, 2025). SE-BRL complements binary classification metrics with deterministic rule-conformance and evidence-state verification, calibration assessment, modality-specific error analysis, and confidence intervals where applicable. Reproducibility demands documented acquisition dates, versions, verification decisions, feature extraction, split manifests, and saved configurations—ensuring robust cross-source validity over leak-driven benchmark scores.

2.11 Related Phishing-Defense Systems and Research Prototypes

2.11.1 Browser Safe-Browsing and URL-Reputation Systems

Browser and reputation defenses emphasize destination intelligence, blocking, and warnings, while research systems identify unseen pages from artifact evidence. Phishpedia and PhishIntention add visual brand and credential-taking intention (Lin et al., 2021; Liu et al., 2022), yet adversarial modifications can evade several detector families (Kulkarni et al., 2025). Reputation is therefore complementary to SE-BRL: it provides technical evidence but not a twelve-label, evidence-linked behavioral representation.

2.11.2 Secure Email and Link-Protection Systems

Secure-email controls combine filtering, sender authentication, link analysis, attachment handling, and policy; academic systems similarly fuse text, URL, and metadata (Al-Subaiey et al., 2024; Bountakas & Xenakis, 2023). A verdict or link score, however, is not a structured account of urgency, impersonation, confidentiality, or requested outcome. SE-BRL is therefore positioned as a complementary analytical and decision-support layer, not a replacement for authentication or reputation.

2.11.3 Browser Extensions and Explainable or Behavior-Aware Prototypes

Current prototypes increasingly connect detection with explanation. Phishpedia and PhishIntention provide brand- or intention-based explanations, APOLLO generates warning text, and Bustio-Martínez et al. (2025) combine persuasion features with SHAP (Bustio-Martínez et al., 2025; Desolda et al., 2025; Lin et al., 2021; Liu et al., 2022). These precedents preclude broad novelty claims. The remaining distinction is SE-BRL's combined support-aware taxonomy, evidence status, three-modality routing, evidence-attribution separation, controlled E1-E3 evaluation, calibrated risk, and deterministic response mapping.

2.11.4 Comparative Synthesis of Existing Systems

Existing systems differ in input access, latency, explanation type, and goal. Visual systems explain target identity but require rendered pages and references; email platforms use broader metadata; and persuasion-aware multimodal systems are the closest feature-level precedent (Bustio-Martínez et al., 2025; Lin et al., 2021; Liu et al., 2022). SE-BRL must therefore be defended as an integration and representation contribution, not as the first phishing detector, multimodal system, or explainable warning tool.

2.12 Synthesis of the Reviewed Literature and Research Gap

2.12.1 Convergence, Divergence, and Unresolved Limitations

The literature converges on several key principles: phishing integrates technical deception with social engineering, distinct modalities expose complementary evidence, heterogeneous features expand coverage, and contextual warnings improve intelligibility (Arp et al., 2022; Bhuiyan & Bhuiyan, 2026; Bustio-Martínez et al., 2025; Greco et al., 2025). However, frameworks diverge across taxonomy, fusion, and evaluation. Standard persuasion models lack SE-BRL's twelve subordinate indicators, while unavailable-modality semantics and calibrated response logic remain under-explored. Although human warning studies confirm behavioral efficacy, this prototype strictly validates functional and safety execution.

2.12.2 Integrated Research Gap and Positioning of SE-BRL

The reviewed literature positions SE-BRL between behavior-oriented analysis and conventional phishing detection. Its intended contribution is not a claim that behavioral phishing analysis is absent, but that observable evidence can be organized into a support-aware, modality-bounded, deterministic representation whose evidence is independently traceable and whose incremental predictive contribution can be measured under controlled conditions.

Accordingly, the study positions deterministic SE-BRL as an input representation rather than a separately trained behavioral classifier. Audited datasets and leakage-controlled partitions support E1/E2/E3/U1 experiments; four candidate phishing-classification families are evaluated under predeclared criteria; artifact evidence remains distinct from model attribution; and calibrated classifier output feeds deterministic risk-response logic. This positioning preserves the artifact-level scope and avoids claims of expert-validated behavioral ground truth, recipient susceptibility, or attacker intent.

CHAPTER 3

TECHNICAL BACKGROUND

This chapter presents the theoretical and technical foundations required to understand the Social Engineering Behavior Representation Layer (SE-BRL) and its integration within a modality-aware phishing detection architecture. It moves from phishing artifact evidence and conventional representations to hierarchical behavioral representation, supervised learning, controlled feature comparison, explainability, probability calibration, artifact-level risk, and secure proof-of-concept integration. The discussion is limited to concepts necessary for understanding the methodology in Chapter 4 and does not infer victim susceptibility, psychology, demographic characteristics, or attacker intent.

3.1 Phishing Artifacts, Modalities, and Evidence

Phishing detection is an artifact-centered cybersecurity problem in which deceptive language, identity presentation, network destinations, and interface structures may jointly support a malicious objective. Contemporary systems therefore inspect multiple evidence types rather than relying on one signal. Text, sender relations, URL composition, webpage structure, forms, and brand-domain relations can provide complementary information, while multimodal studies show that technical and persuasion-oriented features may contribute different predictive value (Bustio-Martínez et al., 2026; Opara et al., 2024). The present architecture distinguishes three evidence environments. Content-bearing emails may expose a subject, body, sender and reply-to relations, routing or authentication fields when available, formatting, embedded links, and attachment metadata. Content-bearing webpages may expose visible text, HTML and Document Object Model (DOM) structure, forms, input types, resource relationships, and brand-domain relations. Stand-alone URLs and engineered records primarily expose lexical, structural, host-related, and precomputed technical indicators. A bare URL can be inspected before navigation, but it cannot directly establish complete message-level urgency, fear,

authority, confidentiality, credential solicitation, or financial solicitation (Prasad & Chandra, 2024; Rashid et al., 2024).

Table 3.1 summarizes the ordinary evidence availability of the three artifact environments and establishes the modality boundary that governs later SE-BRL execution.

Table 3.1.

Evidence availability across phishing artifact modalities

Evidence category	Email	Content-bearing webpage	Stand-alone URL / engineered record
Natural-language content	Available when content exists	Available when content exists	Generally unavailable
Identity / sender relation	Available when fields exist	Conditional	Limited to domain strings
HTML/DOM and forms	Limited	Available	Unavailable
URL lexical / structural	Available through embedded links	Available through embedded links	Available
Brand-destination relation	Conditional	Available when fields exist	Limited
Language-dependent SE-BRL evidence	Assessable when required evidence exists	Assessable when required evidence exists	Unavailable unless content evidence is separately present

Note. “Conditional” requires the relevant identity, branding, destination, or structural relation.

As shown in Table 3.1, availability is an evidence-environment property rather than a behavioral decision. Missing or unsupported fields are represented as Unavailable instead of being interpreted as evidence of absence. This distinction prevents URL-only or incomplete artifacts from receiving unsupported language-dependent SE-BRL states.

3.2 Conventional Phishing Feature Representations

Conventional phishing classifiers transform artifact properties into machine-learning-compatible features. Email representations commonly include word or character n-grams, TF-IDF, subject and body patterns, sender and reply-to consistency, available authentication fields, structural indicators, and lexical features from embedded URLs. Webpage representations include visible text, raw HTML, DOM relationships, forms and input types, links and external resources, and brand-domain consistency. URL representations

describe hostname, subdomain, path, query, length, character composition, and host-related values supplied by the source dataset (Al-Subaiey et al., 2024; Opara et al., 2024; Prasad & Chandra, 2024). These features are useful because they capture regularities associated with phishing and legitimate artifacts, but predictive utility does not guarantee semantic completeness. A model may rely on a token, missing header, source format, or engineered indicator without preserving an explicit account of the manipulation expressed in the artifact. Engineered datasets can also omit the raw evidence needed to verify why a feature value exists. In the present scope, archived webpage screenshots may support audit and evidence verification but are not machine-learning inputs because no visual feature extractor is included. Security-machine-learning research further shows that duplicate contamination, unrealistic row-level splits, source artifacts, and repeated test access can inflate performance (Arp et al., 2022). Conventional features therefore remain essential for the E1 baseline and E3 fused model, but they are complemented by a representation that preserves behavior, modality status, and inspectable evidence.

3.3 Social Engineering Behavior Representation Layer

3.3.1 Hierarchical and Multi-Label Behavioral Representation

SE-BRL organizes the twelve observable indicators into four higher-level dimensions so that evidence can be represented consistently without treating the indicators as mutually exclusive classes. Pressure and Threat Cues comprise urgency, fear or threat, and scarcity; Lure and Attention Cues comprise curiosity and reward or lure; Trust and Identity Manipulation comprises authority, impersonation, brand exploitation, and confidentiality or isolation; and Requested Action and Consequence comprises calls to action, credential or sensitive-data requests, and financial-action requests.

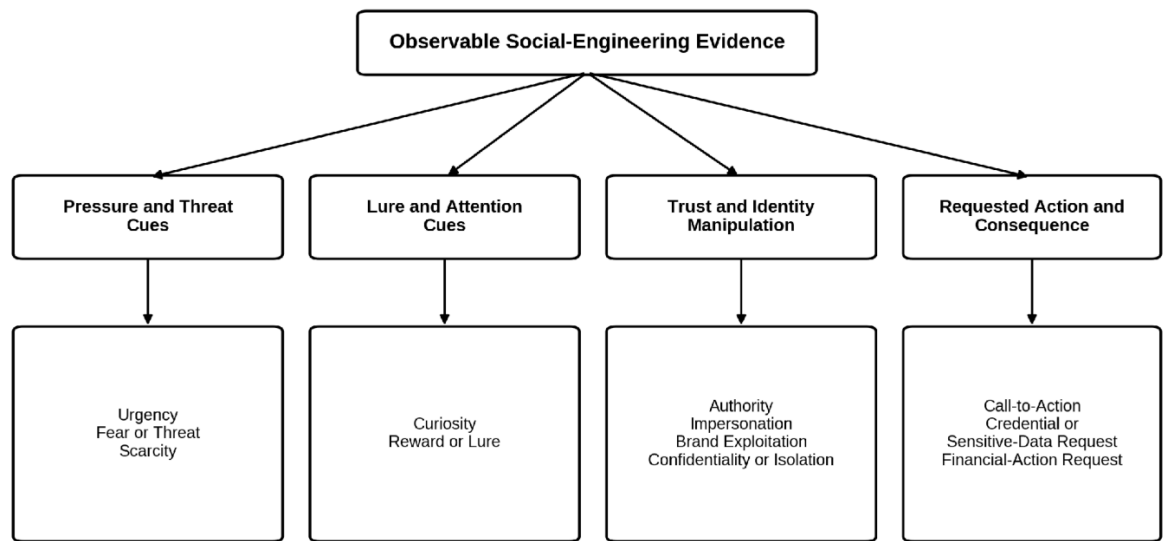
The four dimensions are not annotation targets or outputs of a separately trained behavioral classifier. Instead, each dimension is a deterministic representation state produced by frozen evidence rules whose inputs are observable artifact fields. The subordinate indicators

supply fine-grained evidence locators and can coexist within and across dimensions. This preserves multi-label semantics while keeping the representation grounded in assessable artifact evidence.

Figure 3.1 illustrates the deterministic hierarchy.

Figure 3.1.

Hierarchical Structure of the SE-BRL Behavioral Dimensions



Note. The diagram represents deterministic evidence operationalization and does not imply expert annotation or a learned behavioral-dimension classifier.

As shown in Figure 3.1, the subordinate indicators provide observable evidence to frozen rules, which in turn produce Supported, Absent, or Unavailable dimension states. The resulting representation is therefore traceable to the artifact and bounded by modality rather than learned from behavioral ground-truth labels.

3.3.2 Evidence States and Machine-Compatible Encoding

SE-BRL uses two deterministic values per dimension: a support value and an availability value. Availability indicates whether the required evidence environment permits assessment; support indicates whether qualifying evidence is present when the

dimension is assessable. The combination distinguishes Supported, Absent, and Unavailable states without requiring a learned behavioral confidence.

The fixed-width representation is consumed directly by E2 and appended to conventional features in E3. Evidence text, locators, rule identifiers, and other audit records remain available for explanation and verification but are not inserted as raw predictive values unless explicitly included in the frozen feature dictionary.

3.4 Machine-Learning and Comparative Feature Foundations

Supervised machine learning in the architecture is reserved for phishing classification. SE-BRL itself is deterministic: it produces representation features from frozen evidence rules and does not require a separate multi-label behavior-identification model. The downstream experiment evaluates how different input representations affect phishing classification while controlling the classifier family and leakage boundaries.

Table 3.2 defines the four final feature configurations used across the methodology.

Table 3.2.

Final experimental feature configurations

Configuration	Eligible evidence environment	Input representation	Purpose
E1	Content-bearing email or webpage	Conventional modality-specific features only	Conventional phishing baseline
E2	Same eligible records as E1/E3	Deterministic SE-BRL support and availability states only	Independent contribution of SE-BRL representation
E3	Same eligible records as E1/E2	Conventional features + deterministic SE-BRL representation	Incremental value of the fused representation
U1	Stand-alone URL / engineered technical records	Dataset-defined approved technical features only	Separate technical benchmark; no SE-BRL contribution claim

Note. E1, E2, and E3 within a modality use identical eligible records and controlled split/preprocessing conditions. U1 remains separate because its feature schemas and evidence semantics differ.

Table 3.2 makes the representation contrast explicit. E2 no longer contains predicted behavioral confidences or out-of-fold outputs from a dimension classifier; it contains deterministic SE-BRL state values. Consequently, E3 differs from E1 by the addition of the frozen SE-BRL representation rather than by an additional learned behavioral model.

Table 3.3 summarizes the four candidate phishing-classification families and their scientific roles.

Table 3.3.
Candidate phishing-classification families and analytical roles

Model family	Role in the study	Primary analytical value
Logistic Regression	Primary confirmatory candidate; also participates in implementation selection	Interpretable linear baseline; principal E3-versus-E1 confirmatory contrast
Linear SVM	Candidate comparator	Sparse high-dimensional linear comparison
Random Forest	Candidate comparator	Nonlinear feature-interaction comparison
XGBoost	Candidate comparator	Boosted-tree comparison for nonlinear/structured relationships

Note. All four families are genuinely trained and evaluated during research-time experimentation. Only the classifier selected by the frozen development-stage selection procedure is integrated for runtime use in the corresponding modality branch.

As shown in Table 3.3, Logistic Regression anchors the main representation claim but is not automatically declared the deployment winner. Linear SVM, Random Forest, and XGBoost provide comparative evidence and also remain eligible under the predeclared final implementation selection process.

3.5 Experimental Validity in Security Machine Learning

Security-machine-learning evaluation is vulnerable to leakage from duplicates, near-duplicate templates, campaigns, domains, source-specific formatting, and repeated access to final test data (Arp et al., 2022; Bhuiyan & Bhuiyan, 2026; Pajola et al., 2025). The study therefore audits records and assigns conservative groups before partitioning. No group may cross the development, calibration, and sealed-test partitions.

Learned preprocessing, feature selection, class weighting, and hyperparameter tuning are fitted only inside the appropriate development folds. The design does not require out-of-fold behavioral-dimension predictions because deterministic SE-BRL does not train a behavioral model. Out-of-fold predictions remain applicable to learned phishing classifiers or probability-calibration procedures where required by the frozen protocol. The sealed test is opened once only after the implementation classifier, preprocessing pipeline, calibrator, operational thresholds, SE-BRL rules, and decision logic are frozen.

3.6 Explainability and Probability Calibration

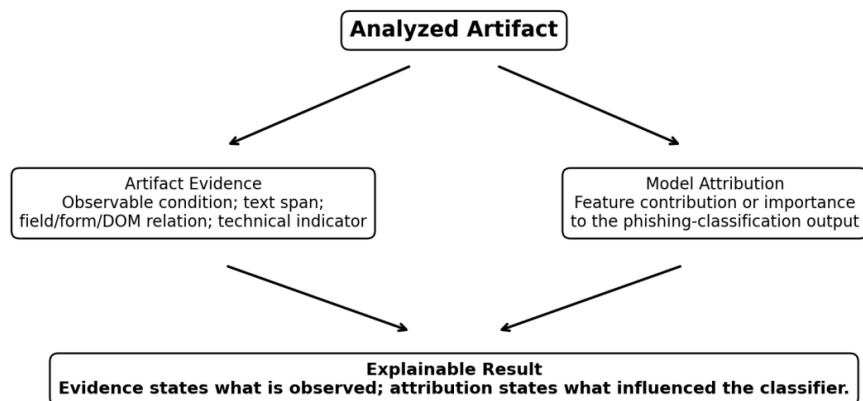
3.6.1 Artifact Evidence and Model Attribution

The architecture separates artifact evidence from model attribution because the two answer different questions. Artifact evidence states what observable text, field relation, webpage element, URL property, or technical condition supports a finding. Model attribution states which predictive features influenced the phishing classifier. A large feature contribution does not establish behavioral evidence, and the presence of behavioral evidence does not imply that it strongly influenced the classifier.

Figure 3.2 presents the separation of artifact evidence and model attribution.

Figure 3.2.

Separation of Artifact Evidence and Model Attribution



Note. Artifact evidence is factual and rule/evidence-centered; model attribution is classifier-centered. Neither channel substitutes for the other.

As shown in Figure 3.2, both channels originate from the analyzed artifact but remain independently addressable until the explainable result is assembled. This organization prevents feature importance from being presented as proof that a behavioral dimension is supported.

3.6.2 Probability Calibration

A discrimination score is not automatically a reliable probability. A model may rank phishing artifacts correctly while being overconfident or underconfident. Probability calibration maps model outputs to estimates that better correspond to observed frequencies. Sigmoid and isotonic calibration are common post-hoc methods, while reliability curves, Brier score, and expected calibration error assess calibration quality (Guo et al., 2017; Silva Filho et al., 2023). Calibration is necessary because the downstream risk mechanism interprets phishing likelihood together with evidence coverage, requested-action severity, corroboration, and uncertainty. Calibration does not create evidence and does not replace classification metrics; it supports more defensible threshold and decision use.

3.7 Evaluation Under Class Imbalance

Phishing datasets may be imbalanced, so no single metric sufficiently characterizes performance. PR-AUC emphasizes positive-class precision and recall and is therefore the primary discrimination measure, whereas ROC-AUC summarizes ranking across thresholds but can appear optimistic when negatives dominate (Saito & Rehmsmeier, 2015). Precision, recall, and F1 describe positive-class errors; balanced accuracy and Matthews correlation coefficient provide class-balanced summaries; Brier score and expected calibration error evaluate probability reliability. The four deterministic SE-BRL dimensions are not evaluated as supervised behavioral labels; they are verified through rule-conformance, evidence-state, modality-boundary, evidence-reference, schema, repeatability, and output-consistency tests.

Table 3.4 consolidates the predictive, calibration, robustness, and deployment-efficiency measures used in later model evaluation.

Table 3.4.

Evaluation measures for classification, calibration, robustness, and deployment feasibility

Evaluation dimension	Measures	Role
Primary discrimination	PR-AUC	Primary model-comparison criterion under class imbalance
Supporting classification	Precision, Recall, F1, ROC-AUC, Balanced Accuracy, MCC	Complementary discrimination evidence
Calibration	Brier score, Expected Calibration Error, reliability analysis	Probability quality for downstream risk processing
Deployment efficiency	p50 and p95 model-inference latency; throughput; memory/model size where measurable	Required model-selection and feasibility evidence
Robustness / validity	Source-stratified errors; source-only and normalized-format baselines; ablations; repeated seeds	Shortcut and sensitivity analysis

Note. Inference latency is model-only unless explicitly labeled end-to-end. End-to-end latency is measured separately during system validation.

Table 3.4 separates model quality from operational cost. PR-AUC remains the primary discrimination metric, while latency and throughput are explicit criteria in the predeclared model-selection procedure rather than being reported only as downstream system metrics.

3.8 Context-Induced Artifact Risk and Adaptive Defense

Context-induced risk is a study-specific artifact-level decision construct, not a psychological profile or estimate of organizational loss. It interprets calibrated phishing probability together with the severity of the requested outcome, availability and coverage of required evidence, independent technical and behavioral corroboration, and reproducible uncertainty conditions. Requested outcomes progress from no consequential request to a general call-to-action, credential or sensitive-data request, and financial-action request. The decision engine produces four bounded states: informational, elevated, high, or review required. Review required has priority when input is invalid, evidence is insufficient or conflicting, parsing fails, the modality is unsupported, or model and schema versions are incompatible. Otherwise, frozen probability and evidence rules determine the risk state. A deterministic decision table is

preferable to an unconstrained weighted score because it is versioned, auditable, and directly testable against expected scenarios. Risk-adaptive defense maps each state to an approved response: a noninterruptive notice, a precautionary warning with independent-verification guidance, a stronger interruption for credential or financial danger, or an explicit review-required message. The component does not generate new security instructions through an unconstrained language model. Its technical evaluation concerns rule correctness, evidence-response correspondence, failure handling, and integration—not whether users understand or follow the warning.

3.9 Secure Proof-of-Concept Architecture

The proof-of-concept browser extension enables artifact submission and presents classification, evidence, risk, and defensive responses. Using Manifest V3, it applies least-privilege permissions, explicit invocation, sanitized rendering, and separation of webpage content from extension logic to reduce attack surface (Chrome Developers, 2023a, 2023b). It does not access passwords, form values, cookies, attachments, or browsing history. The backend performs schema validation, modality routing, feature extraction, SE-BRL assembly, phishing inference, calibration, evidence and attribution retrieval, risk processing, and response selection. URLs are analyzed as strings or archived fields without live crawling. All components are versioned and must produce compatible outputs before presentation. Secure development follows lifecycle controls, including dependency locking, input validation, content security policies, model/rule hashes, restricted logging, transient processing, and explicit failure states (Souppaya et al., 2022). Malformed inputs, incompatible versions, or service failures must fail closed or return review-required states rather than generate incomplete risk decisions.

3.10 Technical Synthesis

The technical foundations establish SE-BRL as a modality-aware, evidence-linked, deterministic behavioral representation within a broader phishing-detection architecture. Four primary dimensions organize twelve subordinate indicators, while Supported, Absent, and

Unavailable states encode both observed support and evidence availability. Conventional, SE-BRL-only, fused, and U1 technical representations are evaluated under audited datasets and leakage-controlled conditions. Four phishing-classification families are compared, Logistic Regression anchors the principal representation contrast, and final implementation selection is governed by predeclared predictive, calibration, latency, throughput, resource-efficiency, and reproducibility criteria. Explainability separates factual artifact evidence from model attribution, calibration supports bounded artifact-level decisions, and deterministic rules translate analytical outputs into auditable defensive responses. Chapter 4 specifies the dataset audit, deterministic SE-BRL verification, candidate-model evaluation and selection, calibration, risk rules, secure integration, project management, and reproducibility procedures used to implement these foundations.

CHAPTER 4

METHODOLOGY

This chapter presents the research and software-development methodology for designing, implementing, and evaluating SE-BRL within a modality-aware phishing-detection architecture. The final methodology is artifact-centered and follows a strict sequence: authoritative dataset acquisition and provenance capture; dataset audit, screening, grouping, and freeze; deterministic SE-BRL rule implementation and verification; controlled E1/E2/E3/U1 construction; candidate phishing-classifier evaluation; development-only final model selection using predeclared predictive, calibration, latency, throughput, and resource criteria; probability and operational threshold freeze; secure integration; and one-time sealed-system evaluation. No expert validation or human annotation of the behavioral dimensions is part of the current methodology.

4.1 Research Environment and Research Roles

The study will be conducted in a controlled computational environment consisting of a designated experiment workstation, a version-controlled source repository, isolated Python and Node.js dependency environments, and a Chromium-based browser supporting Manifest V3 extensions. The primary candidate classifiers are CPU-compatible to support academic reproducibility. Before the first frozen run, processor, memory, operating system, browser, interpreter, package, dependency, random-seed, and repository-commit information will be captured in a versioned environment report.

The units of analysis are individual phishing and legitimate digital artifacts from three evidence environments: content-bearing emails, content-bearing webpages, and stand-alone URL or engineered URL/webpage records. Email and webpage artifacts may enter deterministic SE-BRL processing only when required evidence fields are assessable. URL-only and

engineered records remain in technical U1 benchmarks and do not receive unsupported language-dependent SE-BRL values.

The study does not recruit end users or measure human behavior. It also does not use cybersecurity experts, independent annotators, or researcher annotations to establish behavioral-dimension ground truth. Behavioral dimensions are operationalized through the frozen literature-derived rule specification and verified through software-oriented tests. If an institutional ethics or administrative determination is required by university policy, it concerns the handling of research materials and system procedures rather than expert validation of the behavioral dimensions.

4.2 Research Instruments and Sources of Data

4.2.1 Sources of Data

The selected sources separate content-bearing artifacts from engineered technical benchmarks to preserve modality-specific evidence and prevent unsupported behavioral inference. Email content will be obtained from the Zenodo Phishing Email Curated Datasets release, which preserves traceable snapshots of Nazario, Enron, SpamAssassin, CEAS-08, and related corpora (Champa, 2023; Champa et al., 2024a, 2024b). Nazario will supply phishing-positive email content after record-level audit. Enron and explicitly labeled SpamAssassin ham records will supply legitimate content. Spam and CEAS-08 categories remain semantically distinct and are retained for auxiliary hard-negative or source analyses unless a frozen audit decision supports a different role. Source-only baselines, normalized-format ablations, source-stratified errors, and a source-class-confounding limitation will be reported where phishing and legitimate sources are correlated with class.

Content-bearing webpage artifacts will be obtained from the Phishing Website Dataset, which reports 10,395 sites, including 5,151 phishing and 5,244 legitimate websites, and provides HTML, visible text, CSS, screenshots, and metadata (Putra, 2023). Screenshots may be retained as audit/verification material but are not machine-learning inputs unless a visual

extractor is separately approved before protocol freeze. Technical benchmarks will use PhiUSIIL, containing 235,795 URL/webpage records with 54 features and labels 0 for phishing and 1 for legitimate, and UCI Phishing Websites, containing 11,055 instances with 30 engineered features (Mohammad & McCluskey, 2012; Prasad & Chandra, 2024). PhiUSIIL and UCI are evaluated independently because their schemas and feature semantics differ.

Table 4.1 records source-reported/pre-audit information and methodological roles. It does not report final retained counts, which remain pending until Stage 3 is executed.

Table 4.1.

Dataset inventory, provenance, and methodological role

Source Snapshot	Evidence environment	Source-reported content / count	Class use after audit	Primary role	Key limitation / control
Nazario.csv; Zenodo 8339691	Email content	Subject/body and available fields; source count verified from frozen snapshot	Phishing-positive records that pass audit	Email E1/E2/E3 + deterministic SE-BRL	Age/header variation; source-class confounding audit
Enron.csv; Zenodo 8339691	Email content	Organizational messages; source count verified from frozen snapshot	Legitimate records that pass privacy/duplicate audit	Email E1/E2/E3	Era/style/source artifacts
SpamAssassin; Zenodo 8339691	Email content	Ham/spam labels; source count verified from frozen snapshot	Explicit ham may enter legitimate class; spam remains separate	Email legitimate support + robustness/source analysis	Spam is not silently mapped to phishing or legitimate
CEAS-08; Zenodo 8339691	Email content	Ham/spam corpus; source count verified from frozen snapshot	Original semantics retained; auxiliary unless frozen audit supports inclusion	Hard-negative source analysis	Mixed semantics
Phishing Website Dataset; Zenodo 8041387	Content-bearing webpage	10,395 sites: 5,151 phishing; 5,244 legitimate; HTML/text/CS	Repository labels after audit	Web E1/E2/E3 + deterministic SE-BRL	Large archives; inactive/partial resources; template/domain duplicates

Source Snapshot	Evidence environment	Source-reported content / count	Class use after audit	Primary role	Key limitation / control
PhiUSIIL; UC 967	Engineered URL/web	S/screenshots /metadata 235,795 records; 54 features; 100,945 phishing; 134,850 legitimate	0 phishing; 1 legitimate after schema/label audit	U1-Ph technical benchmark	No guaranteed content-bearing SE-BRL evidence; identifier/leakage controls
UCI Phishing Websites; UC 327	Engineered webpage/URL	11,055 records; 30 engineered features	Repository class definition retained after audit	U1-UCI technical benchmark	Older engineered representation ; separate schema

Note. Counts are source-reported/pre-audit unless explicitly identified otherwise. Access dates, hashes, licenses, local filenames, original schemas, exclusions, and final retained counts are recorded in the source manifest and Stage 3 audit artifacts.

Table 4.1 preserves the distinction between source inventories and modeling data. No dataset is described as audited or frozen until the prescribed Stage 3 controls have been executed and the attrition counts reconcile.

Table 4.2 is the audit reporting template. It fixes, in advance, the exact eligibility rules, predictor boundaries, exclusion classes, and post-audit quantities that must be reported for every source while deliberately leaving all unexecuted audit values unpopulated.

Table 4.2.

Dataset audit plan, record eligibility, feature use, and post-audit reporting placeholders

Dataset	Audit state + eligibility	Permitted fields/features	Excluded / audit-only	Final role
Nazario	Pending Stage 3 audit. Retain only inspectable phishing-positive email records passing provenance, label, parser, content-availability, privacy, duplicate and grouping controls.	Frozen approved email fields for E1 plus deterministic SE-BRL evidence fields; exact names finalized from audited schema.	Source/file IDs; group/split IDs; target label; raw dataset/source name; leakage-prone identifiers.	Email E1/E2/E3
Enron	Pending Stage 3 audit. Retain only inspectable legitimate email records passing provenance, privacy, content-availability, duplicate and grouping controls.	Frozen approved content and email-structure fields.	Source/file IDs; group/split IDs; target label; raw source identifiers.	Email E1/E2/E3
SpamAs sassin	Pending Stage 3 audit. Explicit ham may enter the legitimate pool; spam remains separate unless a frozen auxiliary protocol applies.	Frozen approved content-bearing email fields for admitted records.	Unapproved spam rows from primary binary pool; source/audit/group IDs; target as predictor.	Email support + robustness/source analysis
CEAS-08	Pending Stage 3 audit. Preserve original ham/spam semantics; auxiliary inclusion must follow the frozen protocol.	Frozen approved email fields for records admitted to auxiliary analysis.	Unapproved primary-pool mapping; source/audit/group IDs; target as predictor.	Auxiliary hard-negative / source analysis
Phishing Website Dataset	Pending Stage 3 audit. Retain archived sites with usable content passing provenance, resource, parser, content, template/domain duplicate and grouping controls.	Frozen approved visible-text, HTML/DOM, form/input, link/resource, embedded-URL and brand-domain derived fields.	Screenshot pixels unless separately approved; source/archive IDs; group/split IDs; target as predictor.	Web E1/E2/E3
PhiUSIIL	Pending Stage 3 audit. Retain rows passing provenance, schema, class-definition, missingness, duplicate/domain-grouping and leakage controls.	Frozen approved dataset-defined lexical, structural and host/web technical predictors.	FILENAME/raw identifiers; source/audit/group/split IDs; target as predictor; content-dependent SE-BRL.	U1-Ph
UCI Phishing Websites	Pending Stage 3 audit. Retain rows passing provenance, schema, class-definition, missingness, duplicate/group and leakage controls.	Frozen approved engineered predictors from audited feature dictionary.	Target/class as predictor; audit/group/split/source IDs.	U1-UCI

Note. This table is intentionally prospective. Bracketed fields are controlled placeholders, not missing methodology: each is populated only from Stage 3 artifacts such as `source_manifest.csv`, `screening_log.csv`, `dataset_attrition.csv`, `label_mapping.yaml`, `duplicate_groups.csv`, `group_manifest.csv`, `availability_matrix.csv`, and `split_manifest.csv`. No post-audit number is inserted before those artifacts exist and reconcile.

As shown in Table 4.2, the proposal already fixes exactly what the final audit must report without fabricating any outcome. The placeholders distinguish unknown empirical values from already-specified methodological content: eligibility rules, feature boundaries, exclusion classes, audit artifacts, and final experimental roles are fully specified now, while counts and allocations are populated only after Stage 3 execution.

4.2.2 Research Instruments

The study uses four instrument classes: dataset-audit instruments; deterministic SE-BRL rule and verification instruments; executable experiment/model-selection configurations; and functional, security, privacy, performance, failure, and reproducibility matrices. Each instrument produces a versioned audit artifact so that results can be regenerated from the same source snapshots without undocumented manual decisions. Table 4.3 summarizes these instrument classes.

Table 4.3.

Research instrument and audit-artifact matrix

Instrument	Principal content	Use	Frozen audit output
Dataset manifest and screening instruments	Source metadata, hashes, schemas, labels, availability, exclusions, grouping	Provenance, eligibility, attrition, leakage and safety audit	source_manifest.csv; screening_log.csv; dataset_attrition.csv; label_mapping.yaml; group_manifest.csv
SE-BRL rulespecification and verification suite	Dimension/indicator definitions, modality boundaries, rule logic, evidence locators, test cases	Deterministic execution verification	sebrl_rules_vX.Y.yaml; rule_test_cases.csv; rule_conformance_report.csv; repeatability_report.csv
Experiment and model-selection configurations	Split/group IDs, fold-local preprocessing, candidate search spaces, seeds, metrics, latency/throughput benchmarks, selection rule	E1/E2/E3/U1 development and final selection	split_manifest.csv; configs/*.yaml; model_selection_report.csv; run_report.json
System and reproducibility matrices	Requirements, threats, failures, expected results, performance/security/privacy/release criteria	Secure integration, validation and clean reproduction	test_reports/; release_checklist.csv; protocol_deviation_log.md

Note. Behavioral annotation assignments, adjudication logs, expert-rating sheets, CVI reports, and behavioral-classifier artifacts are not instruments in this methodology.

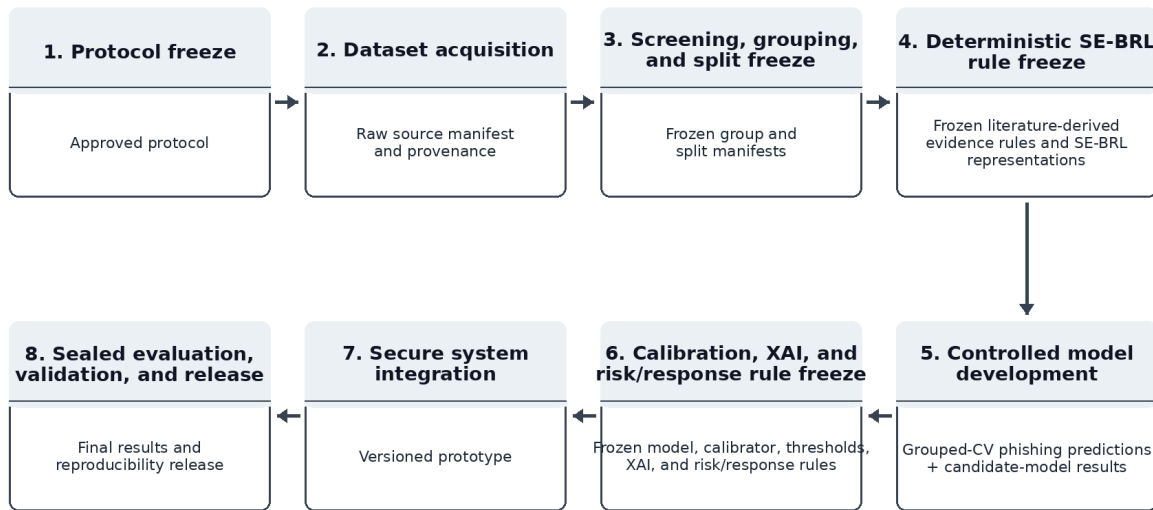
Table 4.3 defines executable, versioned artifacts that can be independently reproduced. The rule package verifies the implementation against a frozen operational specification without claiming expert behavioral ground truth.

4.3 Research Procedure and Pipeline

The research procedure follows eight stages with a versioned output from each stage. Figure 4.1 summarizes the eight-stage procedure and its versioned outputs.

Figure 4.1.

Overall research procedure and stage outputs



Each stage produces a frozen/versioned output; the sealed test remains inaccessible until the analytical pipeline is frozen.

Note. Dataset auditing precedes deterministic SE-BRL execution and candidate-model development; the sealed test remains inaccessible until the analytical pipeline is frozen.

Figure 4.1 establishes the dependency order of the study. In particular, dataset audit and rule verification are completed before model selection, while calibration, risk rules, system integration, and one-time sealed evaluation occur only after the selected classifier and required research artifacts have been frozen.

4.3.1 Stage 1 - Research Planning and Protocol Freeze

The researchers translate the five research questions and corresponding objectives into the requirements traceability matrix, define artifact and system boundaries, and approve the

experimental protocol. The protocol distinguishes deterministic SE-BRL from the phishing classifiers, model attribution, artifact evidence, probability calibration, context-induced risk, decision rules, and prototype interface. There is no separately trained behavioral-dimension classifier.

Before sealed-test access, the protocol freezes the official dataset sources; provenance and audit rules; SE-BRL dimensions, subordinate indicators, modality boundaries, and deterministic rule specification; grouping and split logic; E1/E2/E3/U1 definitions; four candidate classifier families and search budgets; primary and supporting metrics; model-selection criteria including p50/p95 inference latency and throughput; calibration procedure; risk/response rules; security controls; and system-validation plan. A later change requires a dated deviation record, technical justification, joint approval, and confirmation that the sealed test has not been accessed.

4.3.2 Stage 2 - Dataset Collection and Provenance Capture

Each source is acquired from the repository specified in the protocol and retained in read-only form. The researchers calculate SHA-256 hashes locally, assign a `source_snapshot_id`, and store repository identifiers, DOI/URL, license, access date, original filename, source-reported count, schema documentation, and class definition. No filtering, feature extraction, relabeling, or model-based decision occurs during acquisition. Potentially active URLs and attachment contents are not opened or executed; the study uses archived artifacts and precomputed technical datasets only. Derived files are written to separate versioned directories using deterministic filenames.

4.3.3 Stage 3 - Dataset Verification, Audit, Screening, Grouping, and Freeze

Stage 3 determines whether each source record is eligible before any feature configuration or candidate model is evaluated. Source-level audit verifies provenance, label semantics, content availability, licensing, privacy constraints, schema compatibility, and evidence-environment suitability. Record-level audit identifies malformed or empty artifacts,

unsupported labels, missing required fields, exact duplicates, near-duplicate templates, repeated campaigns or domains, and other predeclared exclusion conditions while retaining original IDs and reason codes.

Exact duplicates are detected through normalized-content hashes. Near-duplicate email and HTML templates are clustered with a documented MinHash or SimHash configuration when appropriate, and URL/technical records are grouped by registrable domain and normalized path where the source permits. Campaign, template, domain, source, exact-duplicate, and near-duplicate relationships are reconciled into one conservative group_id before partitioning. A group cannot cross development, calibration, or sealed-test partitions. Spam and other malicious classes remain distinct until an approved label mapping is frozen.

Table 4.4 defines the acceptance evidence and recorded outputs for this audit.

Table 4.4.

Dataset audit, screening, attrition, and leakage-control protocol

Control point	Decision rule	Method	Acceptance evidence	Recorded output
Provenance	Retain only auditable repository snapshots with documented labels/access conditions	Manifest + repository audit	DOI/URL, date, hash, schema, label definition, license present	source_manifest.csv
Label semantics	Keep phishing, legitimate, spam, suspicious and other-malicious labels distinct until frozen mapping	Documentation/schema audit	Mapping frozen before partitioning	label_mapping.yaml
Modality/content eligibility	SE-BRL branch requires inspectable email or webpage evidence; U1 follows approved technical schema	Availability + parser audit	Required fields recorded by modality	availability_matrix.csv
Malformed/incomplete	Exclude only through predeclared rules, never because of model outcome	Parser validation + reason codes	Original ID and exclusion reason preserved	screening_log.csv
Exact duplicates	One normalized artifact cannot cross partitions	SHA-256 / normalized-content hash	Canonical record + duplicate family	duplicate_groups.csv

Control point	Decision rule	Method	Acceptance evidence	Recorded output
Near duplicates / campaigns / domains	Related records remain in one group	Documented MinHash/Sim Hash + domain/template rules	group_id assigned before split	group_manifest.csv
Source-class confounding	Mandatory source-only, normalized-format and source-stratified audits where source correlates with class	Source prediction + ablation	Performance/errors reported by source	source_confounding_report.csv
Attrition	Counts reconcile from acquisition through final eligibility and split	Automated stage summary	No unexplained count difference	dataset_attrition.csv

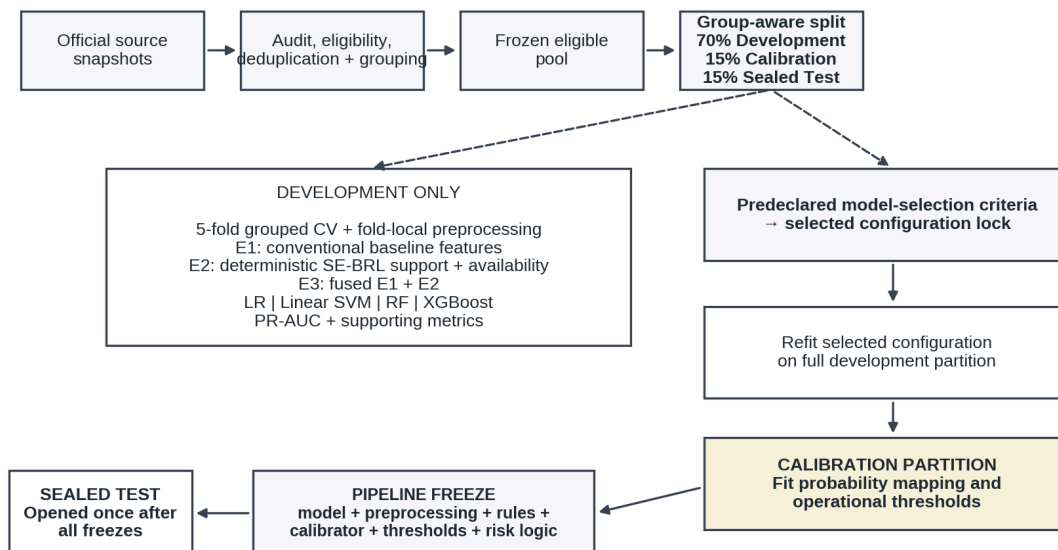
Note. Dataset attrition contains no annotation stage. All exclusions are rule-based, reason-coded, and executed before model development.

Table 4.4 provides the evidence necessary to defend the final modeling pool. A dataset is marked Audited and Frozen only after the source count, exclusions, duplicate/group decisions, final eligible count, and partition allocation reconcile. Until that point, reported counts remain source-reported or pending audit.

Figure 4.2 shows the barrier between audited datasets, development-stage model selection, calibration, and the untouched sealed test.

Figure 4.2.

Dataset partitioning, candidate-model selection, calibration, and sealed-test barriers



Note. SE-BRL is deterministic; no out-of-fold behavioral-dimension prediction step is used. Grouped out-of-fold predictions apply only to learned phishing classifiers where required by the frozen protocol.

As shown in Figure 4.2, final model selection is made from development-stage evidence only. The calibration partition is reserved for the frozen calibration/operational-threshold procedure, and the sealed test is opened once after the implementation classifier, preprocessing, rule specification, calibrator, thresholds, and decision rules are frozen.

4.3.4 Stage 4 - Deterministic SE-BRL Rule Specification and Evidence Representation

Stage 4 operationalizes the four artifact-level behavioral dimensions through twelve subordinate evidence indicators using predefined lexical, structural, relational, and requested-action rules derived from the reviewed literature and authoritative cybersecurity references. The rule package contains explicit operational definitions, qualifying evidence conditions, exclusions, modality requirements, evidence-locator requirements, and version identifiers. It does not create human annotations and does not train a behavioral-dimension model.

Table 4.5 summarizes the dimensions, subordinate indicators, principal evidence types, and modality support.

Table 4.5.

SE-BRL dimensions, subordinate indicators, evidence types, and modality support

Dimension	Subordinate indicators	Representative observable evidence	Email	Webpage	URL-only / engineered
Pressure and Threat Cues	Urgency; Fear/Threat; Scarcity	Deadline/termination/loss statements; limited-time or scarcity cues	Assessable when content exists	Assessable when content exists	Unavailable without content
Lure and Attention Cues	Curiosity; Reward/Lure	Unexpected-benefit, prize, reward or curiosity-inducing content	Assessable when content exists	Assessable when content exists	Unavailable without content
Trust and Identity	Authority; Impersonation; Brand	Authority claims; identity/brand	Assessable when fields/content exist	Assessable when	Only limited technical/brand relations;

Dimension	Subordinate indicators	Representative observable evidence	Email	Webpage	URL-only / engineered
Manipulation	Exploitation; Confidentiality/ Isolation	mismatch; secrecy/isolation language; brand-domain relations		fields/content exist	language states unavailable
Requested Action and Consequence	Call-to-Action; Credential/Sensitive-data Request; Financial-action Request	Links/buttons/forms or explicit requests for credentials, sensitive data, payment, transfer, or other consequential action	Assessable when content/links exist	Assessable when content/forms exist	Language-dependent states unavailable

Note. “Assessable” means the required evidence fields are present. It does not imply that the dimension is Supported.

Table 4.5 operationalizes the four-dimension/twelve-indicator structure through observable rule inputs. Each dimension may be Supported only when qualifying evidence is found; otherwise it is Absent if assessable or Unavailable if the modality/fields cannot support the determination.

Verification of the rule package is limited to software/specification measures. Table 4.6 defines the final verification protocol.

Table 4.6.

SE-BRL rule verification and evidence-state validation protocol

Verification target	Method	Acceptance evidence	Frozen output
Rule-conformance correctness	Predefined positive, negative, and boundary test cases	Observed rule result matches expected result	rule_conformance_report.csv
Evidence-state correctness	Test Supported / Absent / Unavailable cases	State/availability mapping matches specification	evidence_state_tests.csv
Modality-boundary compliance	Run unsupported/incomplete modality cases	Unsupported evidence never becomes Absent or Supported	modality_boundary_tests.csv
Evidence-reference completeness	Validate text span / field / form / DOM / technical locator	Every Supported result has valid traceable locator	evidence_reference_report.csv
Schema conformity	Validate long-form and fixed-width records	Required fields/types/version identifiers present	schema_validation_report.csv

Verification target	Method	Acceptance evidence	Frozen output
Deterministic repeatability	Repeat identical inputs under same frozen version	Identical output and evidence references	repeatability_report.csv
Output consistency	Cross-check dimension, indicator, availability and evidence records	No contradictory state/evidence combinations	output_consistency_report.csv

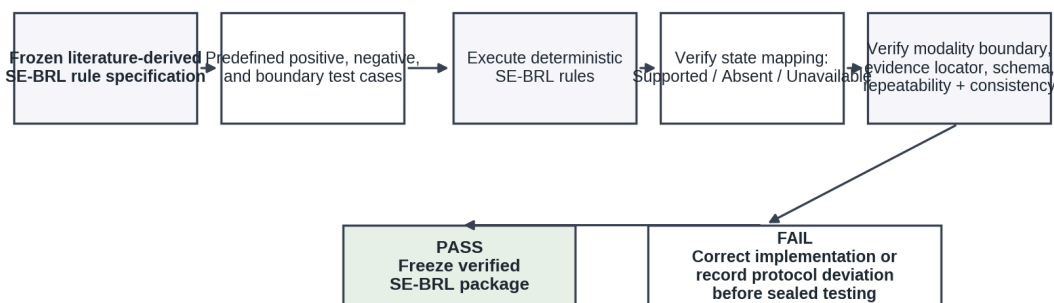
Note. These tests verify the implementation against the frozen specification; they do not establish expert-annotated behavioral ground truth, inter-annotator reliability, or supervised behavioral-model performance.

As shown in Table 4.6, the verification evidence is executable and reproducible. The study can therefore claim conformance to the frozen operational specification without claiming that human experts validated the behavioral dimensions as psychological constructs or ground-truth labels.

Figure 4.3 summarizes the Stage 4 verification flow.

Figure 4.3.

Deterministic SE-BRL rule verification workflow



Note. Human annotation, expert review scores, CVI, agreement statistics, and adjudication are intentionally absent from the methodology.

Figure 4.3 demonstrates that the specification is frozen before execution tests are interpreted. Failures lead to correction of the implementation or a documented protocol deviation before sealed testing; they do not create a post-hoc behavioral labeling process.

4.3.5 Stage 5 - Modality-Specific Model Development, Evaluation, and Final Selection

Stage 5 evaluates phishing-classification models using the frozen eligible datasets and representation configurations. Email and webpage experiments are modality-specific because their conventional feature spaces differ. U1-Ph and U1-UCI remain independent technical benchmarks and do not evaluate the contribution of SE-BRL. The order within Stage 5 is fixed: candidate models are declared first, model-selection criteria are declared second, and model evaluation/final selection is performed third. The E1/E2/E3 configurations answer the representation research question, whereas final implementation model-family selection is a separate engineering decision. For content-bearing email and webpage operation, the proof-of-concept uses the fused E3 representation so that SE-BRL is actually present in the implemented architecture; E1 and E2 remain controlled experimental baselines. Consequently, the four classifier families compete for final Email-E3 and Web-E3 implementation, while each U1 branch conducts its own model-family selection on its frozen technical feature space.

4.3.5.1 Candidate Phishing-Classification Models

Four candidate model families are evaluated: Logistic Regression, Linear SVM, Random Forest, and XGBoost. Logistic Regression is the primary confirmatory classifier for the principal E3-versus-E1 representation contrast because holding the classifier family constant strengthens attribution of observed differences to the representation. Linear SVM, Random Forest, and XGBoost are genuine comparative candidates and are trained/evaluated under the same eligibility, grouping, preprocessing, and search controls. No Dimension Identifier is used because SE-BRL does not train a behavioral-dimension classifier.

Table 4.7 defines the final E1/E2/E3/U1 experimental configurations before candidate-model evaluation.

Table 4.7.

Experimental configurations and model scope

Experiment	Eligible artifacts	Input representation	Model scope	Purpose / fairness control
Email-E1	Same eligible email records used by E2/E3	Email conventional text, relation, header/structure and embedded-URL derived features	Email-specific candidate classifier	Conventional baseline; same groups/folds/search conditions
Email-E2	Same Email-E1/E3 records	Deterministic SE-BRL support + availability values	Email-specific candidate classifier	Independent SE-BRL contribution; no behavioral ground-truth features
Email-E3	Same Email-E1/E2 records	Email-E1 + deterministic Email-E2 representation	Email-specific candidate classifier	Incremental representation value under identical controls
Web-E1	Same eligible webpage records used by E2/E3	Visible text; DOM/forms/links/structure; embedded-URL and brand-domain derived features	Web-specific candidate classifier	Conventional baseline; screenshots excluded from ML unless separately approved
Web-E2	Same Web-E1/E3 records	Deterministic SE-BRL support + availability values	Web-specific candidate classifier	Independent SE-BRL contribution
Web-E3	Same Web-E1/E2 records	Web-E1 + deterministic Web-E2 representation	Web-specific candidate classifier	Incremental representation value
U1-Ph	PhiUSIIL only	Approved dataset-defined engineered URL/web features	Independent technical benchmark	Technical performance; no SE-BRL contribution claim
U1-UCI	UCI Phishing Websites only	Approved dataset-defined engineered features	Independent technical benchmark	Secondary technical comparison; no raw pooling with PhiUSIIL

Note. The same modality-specific eligible records are used across E1-E3. Deterministic SE-BRL does not require out-of-fold behavioral-dimension predictions.

Table 4.7 ensures that the primary representation comparison does not confound a feature-set change with a change in artifact eligibility. Candidate families may be compared

within each configuration, while the confirmatory E3-versus-E1 statement remains anchored to Logistic Regression.

Table 4.8 specifies candidate-family search controls, calibration handling, explainability, and the required deployment-efficiency measurements.

Table 4.8.

Candidate phishing-classification models, evaluation controls, and deployment-efficiency measures

Model	Role	Search space / budget	Imbalance control	Probability / explanation handling	Efficiency measures
Logistic Regression	Primary confirmatory candidate; eligible for final implementation selection	$C = .01, .1, 1$, 10; L2; max_iter = 2000	None or balanced class weight selected inside grouped CV	Phishing calibrator according to frozen protocol; standardized coefficients and local $x_j\beta_j$ contributions	p50/p95 model latency; throughput; memory/model size
Linear SVM	Candidate comparator	$C = .01, .1, 1$, 10	None or balanced class weight	Signed decision contributions; sigmoid mapping when a calibrated probability is required	p50/p95 model latency; throughput; memory/model size
Random Forest	Candidate nonlinear comparator	300/600 trees; depth None/20/40; leaf 1/3/5; max_features sqrt(.5)	balanced or balanced_subsample as frozen in CV	TreeSHAP / feature importance; calibration when selected and required	p50/p95 model latency; throughput; memory/model size
XGBoost	Candidate boosted-tree comparator	depth 3/6; eta.03/.1; subsample.8/1; colsample.8/1; child weight 1/5; lambda 1/5	fold-local scale_pos_weight where applicable	TreeSHAP; calibration when selected and required	p50/p95 model latency; throughput; memory/model size

Note. All candidate hyperparameters and budgets are frozen before final evaluation. Efficiency is benchmarked on the same documented workstation and software environment.

As shown in Table 4.8, latency and throughput are measured for every candidate rather than only for the eventual prototype. This makes inference speed and latency explicit inputs to the predeclared model-selection procedure.

4.3.5.2 Model-Selection Criteria

Model selection is predeclared as a deterministic decision procedure rather than a discretionary review of whichever candidate appears strongest after experimentation. The selection unit is the classifier family within a fixed implementation representation: Email-E3 for email, Web-E3 for webpages, and the corresponding frozen U1 feature space for each technical benchmark. E1 and E2 remain experimental baselines and are not allowed to become alternate deployment representations after results are observed. This separation prevents model-family selection from becoming confounded with representation selection.

Candidate evidence is derived only from grouped development-stage out-of-fold predictions and the frozen efficiency benchmark. PR-AUC is the primary discrimination statistic. Precision, recall, F1-score, ROC-AUC, balanced accuracy, and Matthews correlation coefficient are reported as supporting measures rather than substituted opportunistically for PR-AUC. For models requiring probabilistic downstream risk processing, cross-fitted development predictions are used to assess calibration behavior through Brier score, Expected Calibration Error, and a reliability analysis generated under a fixed binning procedure declared in the protocol.

Uncertainty is incorporated into selection rather than comparing point estimates alone. The development protocol uses the already specified paired group-preserving bootstrap procedure with 2,000 resamples and 95% confidence intervals for candidate comparisons. A predeclared practical-equivalence tolerance for PR-AUC, represented in the protocol as $[\delta_{PR}]$, is frozen before candidate results are used for final selection. Its numerical value is intentionally not fabricated in this proposal; it must be justified by the revised SRS or development protocol and approved before sealed-test access. Candidates whose predictive performance is practically indistinguishable under the frozen rule proceed to calibration and

deployment-efficiency tie-breaking rather than being ranked from negligible point-estimate differences.

Deployment efficiency is benchmarked under one frozen environment. Model-only p50 and p95 latency, throughput in artifacts per second, peak or measured inference memory where available, serialized model size, batch size, thread configuration, warm-up procedure, and timed-repetition count are recorded. The exact benchmark repetition parameters appear as [N_warmup] and [N_bench] until the benchmark protocol is frozen; they are populated before model evaluation and cannot be changed after viewing candidate results. End-to-end application latency remains a separate system metric and is not substituted for model-only latency during classifier-family selection.

Table 4.9 formalizes the selection hierarchy, its evidence source, the direction of preference, and the exact point at which any still-unknown numeric tolerance must be frozen.

Table 4.9.

Predeclared classifier-family selection criteria and deterministic decision hierarchy

Order	Criterion	Evidence / statistic	Predeclared decision rule	Freeze / selection role
0	Protocol-validity gate	Complete grouped-CV run; no partition/group leakage; reproducible preprocessing; required predictions, calibration evidence and benchmark outputs present.	Exclude any candidate with unresolved leakage, incompatible schema/pipeline, unrecoverable execution failure, or missing required selection evidence.	Mandatory eligibility gate; no numeric placeholder.
1	Hard approved requirements	Any minimum recall, precision, p95 latency, memory or other release requirement formally approved in the revised SRS/protocol.	Every applicable hard requirement must be met before ranking. If no numeric hard limit has been approved, none is invented.	Placeholders: [REQ_MIN_RECALL], [REQ_MIN_PRECISION], [REQ_MAX_P95], [REQ_MAX_MEMORY].
2	Primary discrimination	PR-AUC from pooled grouped-development out-of-fold predictions.	Higher PR-AUC is preferred. A secondary metric cannot replace PR-AUC after results are observed.	Primary ranking criterion; fixed before experiments.

Order	Criterion	Evidence / statistic	Predeclared decision rule	Freeze / selection role
3	Uncertainty + practical equivalence	2,000 paired group-preserving bootstrap resamples; 95% CI for PR-AUC differences.	Apply the frozen [delta_PR] equivalence rule. Practically equivalent candidates proceed to secondary criteria rather than being separated by trivial point-estimate differences.	[delta_PR] justified and frozen before selection; protects against overinterpretation.
4	Probability quality	Cross-fitted Brier score, ECE and fixed-protocol reliability analysis for candidates requiring calibrated probability.	Within the practically equivalent predictive set, prefer better calibration under the frozen comparison rule; do not trade material PR-AUC loss for a small calibration gain.	ECE method/bins fixed; [delta_cal] only if an explicit tolerance is approved.
5	Inference efficiency	p95 model-only latency; p50 descriptive; throughput; memory/model size under one frozen environment.	Among candidates still comparable, prefer lower p95 latency; then higher throughput; then lower memory/model size.	Freeze [N_warmup], [N_bench], batch size, threads, hardware and software before benchmarking.
6	Interpretability + reproducibility tie-break	Attribution availability; dependency count; artifact size; deterministic rerun under frozen seeds/configuration.	If still indistinguishable, prefer the simpler reproducible operational bundle. LR is preferred only if it remains in the tied set after every prior gate.	Final tie-break only; no post-hoc preference.
7	Freeze and lock	Selected model family, hyperparameters, preprocessing, feature dictionary, training manifest, dependencies, environment record and hashes.	Freeze before calibration/threshold finalization and before sealed-test access. Sealed-test results cannot trigger reselection.	Irreversible selection boundary recorded in model_selection_report.csv.

Note. Bracketed items are controlled protocol placeholders, not unspecified methodology. A bracket is replaced only when the revised SRS or frozen development protocol supplies the value before candidate results are used for final selection. The sealed test is never used to populate or revise these values.

Table 4.9 makes the decision auditable because every candidate follows the same ordered gates and tie-breaks. The framework gives PR-AUC scientific primacy, incorporates uncertainty through paired group-preserving resampling, and allows calibration and deployment efficiency to decide only among candidates that are not materially separated by the primary

predictive criterion. This prevents a faster but materially weaker detector, or a marginally higher point estimate with poor calibration or latency, from being selected through an undocumented trade-off.

4.3.5.3 Controlled Evaluation and Final Model Selection

Within the 70% development partition, candidate models are tuned and evaluated by grouped cross-validation using identical eligible records, group assignments, fold-local preprocessing boundaries, candidate-specific but predeclared search budgets, seeds, and metrics. For Email and Web implementation selection, the decision procedure in Table 4.9 is applied only to the corresponding E3 candidate results; E1/E2 remain scientific controls. For U1-Ph and U1-UCI, the same hierarchy is applied independently to candidates trained on each frozen technical feature space. Candidate statistics are computed from development-stage predictions only, and paired comparisons preserve the frozen groups used to prevent template, campaign, or domain leakage.

After the selection criteria are applied, exactly one implementation classifier is selected and refitted/frozen for each applicable branch according to the protocol. The decision log records every candidate entering each gate, every exclusion, the primary PR-AUC comparison, the paired uncertainty result, any practical-equivalence determination, calibration evidence, efficiency benchmark, tie-break reason if required, and the final selected family. Model family, hyperparameters, preprocessing pipeline, feature dictionary, training manifest, seed information, dependency versions, environment record, and model hash are versioned. The sealed test is not used to decide which model wins. Logistic Regression remains the primary confirmatory model for the E3-versus-E1 research contrast even if a different candidate is selected for implementation.

4.3.5.4 Operational Computations and Measurement Rules

Figures 4.4 through 4.9 define the study-specific deterministic representation and deployment-efficiency computations used by the methodology.

Figure 4.4 defines the deterministic evidence-state encoding for one behavioral dimension.

Figure 4.4.

Deterministic Supported, Absent, and Unavailable state encoding

$$\text{Supported} = (1, 1) \quad \text{Absent} = (0, 1) \quad \text{Unavailable} = (0, 0)$$

First coordinate = support state; second coordinate = availability state.

As shown in Figure 4.4, Unavailable is encoded separately from Absent. This prevents missing or unsupported modality evidence from being interpreted as a negative observation.

Figure 4.5 defines the fixed-width vector supplied to E2 and appended to conventional features in E3.

Figure 4.5.

Fixed-width deterministic SE-BRL representation vector

$$\mathbf{z}_{\text{SE-BRL}} = [s_1, s_2, s_3, s_4, a_1, a_2, a_3, a_4]^T$$

Four support values and four availability values form the fixed-width SE-BRL vector.

Figure 4.5 shows that each of the four dimensions contributes a support value and an availability value. Evidence text and locators remain in the long-form audit record rather than being confused with the fixed-width predictive representation.

Figure 4.6 defines evidence coverage as a modality-specific availability ratio.

Figure 4.6.

Evidence-coverage computation

$$C = \frac{\text{available required evidence fields}}{\text{total required evidence fields for the modality}}$$

Evidence coverage is an availability measure, not a phishing probability.

As shown in Figure 4.6, coverage measures how much required evidence is assessable. It does not estimate phishing probability or behavioral susceptibility and is interpreted only within the frozen modality requirements.

Figure 4.7 defines the incremental change in an evaluation metric when the fused E3 representation is compared with the conventional E1 baseline.

Figure 4.7.
Incremental E3-versus-E1 metric difference

$$\Delta M = M_{E3} - M_{E1}$$

The incremental change for metric M compares the fused representation with the conventional baseline.

Figure 4.7 provides a common representation for reporting the direction and magnitude of E3-versus-E1 changes. Statistical uncertainty is reported through the frozen paired group-preserving procedure rather than inferred from the sign of the difference alone.

Figure 4.8 defines model-only inference latency, which is benchmarked independently from end-to-end application latency.

Figure 4.8.
Model-only inference-latency computation

$$L_{\text{model}} = t_{\text{prediction end}} - t_{\text{prediction start}}$$

Model-only latency excludes parsing and feature construction.

As shown in Figure 4.8, model latency begins after a model-compatible feature vector is ready and ends when the classifier prediction completes. Parsing, feature construction, calibration, attribution, risk processing, API handling, and rendering are evaluated separately as end-to-end system latency.

Figure 4.9 defines inference throughput under the same frozen benchmark environment.

Figure 4.9.
Model-inference throughput computation

$$\text{Throughput} = \frac{N_{\text{artifacts}}}{t_{\text{inference}}}$$

where $N_{\text{artifacts}}$ = number of artifacts processed,
 $t_{\text{inference}}$ = total inference time.

Throughput is reported as artifacts processed per second under the *frozen* benchmark environment.

Figure 4.9 expresses model inference speed as artifacts processed per second. p50/p95 latency and throughput are reported together because a model may have acceptable median latency but poor tail behavior or vice versa.

Table 4.10 consolidates the final evaluation and interpretation plan.

Table 4.10.

Evaluation, statistical, model-selection, and interpretation plan

Outcome	Primary measure	Secondary evidence	Comparison / uncertainty	Interpretation control
SE-BRL software verification	Rule-conformance correctness	Evidence-state, modality-boundary, evidence-reference, schema, repeatability, output-consistency tests	Expected-versus-actual deterministic test cases	Conformance to frozen operational specification only; no expert behavioral-validity claim
Phishing classification	PR-AUC	Precision, Recall, F1, ROC-AUC, Balanced Accuracy, MCC	Paired group-preserving bootstrap / frozen CI procedure where applicable	Primary confirmatory representation contrast: E3 vs E1 using Logistic Regression
Model selection / deployment feasibility	Table 4.9 deterministic hierarchy applied to E3 (Email/Web) or frozen U1 representation	Paired 2,000-resample group bootstrap; [delta_PR] equivalence rule; Brier/ECE; p50/p95 latency; throughput; memory/model size	Development-only evidence under one frozen benchmark environment	Sealed test cannot select, populate selection thresholds, or retune model
Calibration	Brier score	ECE + reliability analysis	Frozen calibration procedure	Calibration applies to phishing classifier, not SE-BRL dimensions

Outcome	Primary measure	Secondary evidence	Comparison / uncertainty	Interpretation control
Explainability	Evidence-attribution separation and correctness	Locator correctness; attribution stability; sign/rank consistency where appropriate	Repeated seeds / nonsemantic perturbations as frozen	Evidence remains factual record; attribution remains model-centered
Generalization / shortcuts	Source-stratified errors + source-only baseline	Availability-only, modality-only, normalized-format and feature ablations	Performance deltas + frozen uncertainty procedure	No unsupported external generalization claim
System validation	Critical requirement pass	End-to-end latency, memory, hostile-input/failure/security/privacy/reproducibility results	Automated expected-versus-actual checks	All critical and fail-closed tests must pass

Note. Any numeric thresholds not already justified by the SRS or frozen development protocol remain to be specified before sealed-test access; none are fabricated in the proposal.

Table 4.10 aligns RQ1 with software verification and RQ2 with both representation comparison and final model selection. It also separates model-only efficiency from end-to-end system performance and explicitly prevents the sealed test from becoming a model-selection instrument.

4.3.6 Stage 6 - Explainability, Calibration, Risk, and Response Rule Freeze

Explainability maintains two independent channels. The artifact evidence record identifies the text span, field relation, webpage element, URL feature, or technical indicator supporting a behavioral or technical finding. Model attribution identifies features influencing the phishing classifier. Linear models use standardized coefficients and local contributions where applicable; tree models use the frozen tree-attribution method such as TreeSHAP (Lundberg & Lee, 2017). Attribution never substitutes for deterministic evidence.

Probability calibration applies to the selected phishing classifier, not to SE-BRL dimensions. The frozen protocol compares or applies the approved calibration mapping using development-stage predictions and the calibration partition as specified. Operational low/high probability boundaries and downstream risk rules are frozen before sealed-test access. If a

classifier cannot produce a valid calibrated output or if component/schema versions conflict, the system returns Review Required rather than fabricating a risk state.

The risk mechanism uses precedence-based deterministic rules rather than an unconstrained weighted score. Table 4.11 defines the bounded decision states without fabricating numerical thresholds that have not yet been frozen.

Table 4.11.

State	Illustrative governing condition under frozen rules	Required system behavior	Interpretation boundary
Review Required	Parsing failure, unsupported/conflicting evidence, incompatible component versions, or unresolved uncertainty condition	Return explicit review-required message; do not issue a complete risk decision	Failure/uncertainty state, not a human-risk score
High	Frozen high-risk conditions satisfied, particularly when calibrated phishing evidence is corroborated by severe credential/financial requested-action evidence	Use strongest approved interruption and independent-verification guidance	Threshold values are frozen before sealed test
Elevated	Suspicious calibrated outcome or corroborated artifact evidence warrants caution but High conditions are not satisfied	Use precautionary warning and verification guidance	Artifact-level condition only
Informational	No higher-precedence state is satisfied under the frozen logic	Provide noninterruptive result, limitations, and evidence summary	Does not certify safety or future benign behavior

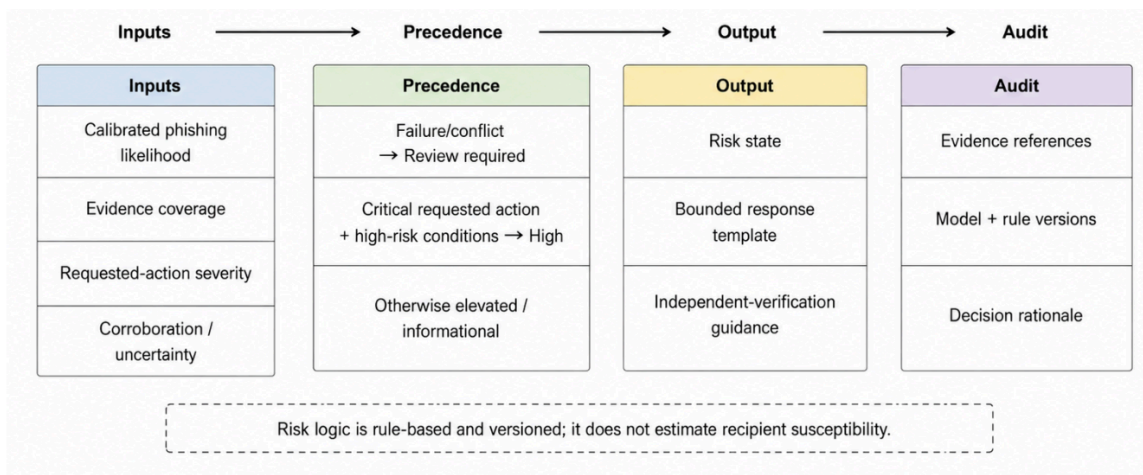
Deterministic context-induced risk and response matrix**Note.** *The matrix describes precedence and response semantics. Exact numeric probability thresholds, if used, are selected/frozen according to the calibration protocol and are not invented in this proposal.*

Table 4.11 keeps the risk layer auditable by tying each state to explicit conditions and bounded response behavior. The state describes the analyzed artifact and system confidence conditions, not recipient susceptibility or attacker intent.

Figure 4.12 visualizes the same precedence-based risk and intervention flow.

Figure 4.12.

Deterministic context-induced risk and intervention flow



As shown in Figure 4.12, failure and unresolved conflict are handled before ordinary risk escalation, preventing incomplete analytical output from being presented as a valid security decision.

4.3.7 Stage 7 - Secure System Integration

Stage 7 integrates the frozen deterministic SE-BRL rule package, selected phishing classifier, preprocessing pipeline, applicable calibration object, evidence record, attribution configuration, risk rules, response rules, and presentation contracts. Only the selected implementation classifier for the routed modality is loaded for operational inference; the remaining candidate models are retained as research artifacts rather than executed as a runtime ensemble.

Security controls follow least privilege, schema validation, bounded payloads, no live crawling, sanitized rendering, version/hash verification, restricted logging, and fail-closed or Review Required behavior for incompatible or partial results (Chrome Developers, 2023a, 2023b). Table 4.12 summarizes the principal security/privacy/failure controls.

Table 4.12.

Security, privacy, trust-boundary, and failure-control matrix

Asset / boundary	Threat or failure	Control	Validation evidence	Residual / fallback
Extension content / service worker	Markup/script injection; excessive permissions	Manifest V3, strict CSP, escaped rendering, explicit invocation, minimum permissions	Permission and malicious-markup tests	Sanitized output or fail closed
API gateway	Malformed, oversized, replayed or abusive request	Typed schema, payload/rate/time limits, CORS/private endpoint configuration	Boundary/fuzz/oversized payload tests	HTTP failure; no fabricated risk output
URL input	SSRF or dangerous scheme	No live fetch; normalize supported URL strings only	file://localhost/private-IP/malformed cases	Reject with non-sensitive reason
Models / schemas / calibrators / rules	Version mismatch or tampering	Bundle hash + compatibility manifest	Altered/incompatible bundle tests	Fail closed / Review Required
Credentials / personal data	Unintended collection or retention	No cookies/form values; minimization; transient processing	Payload inspection + retention audit	Delete transient content and document issue
Audit database / logs	Raw sensitive content leakage	Non-sensitive audit schema; no raw-artifact logging by default	Log review and injection tests	Purge and document incident
Service outage	Partial or stale output	Health checks, timeouts, transactional result assembly	Terminate component during test run	Explicit unavailable/failure state

Note. Controls are prototype validation requirements and are not claims of production-grade security.

This is important because a correct phishing model combined with a stale rule package, incompatible schema, or partial response can still produce an unsafe or misleading result.

4.3.8 Stage 8 - Sealed System Validation and Reproducibility Release

Stage 8 opens the sealed test partition once and executes the already frozen evaluation pipeline. The system generates final artifact parsing results, modality routes, deterministic SE-BRL outputs, E1/E2/E3/U1 phishing predictions where applicable, calibration metrics, evidence and attribution records, risk states, bounded interventions, confidence intervals under

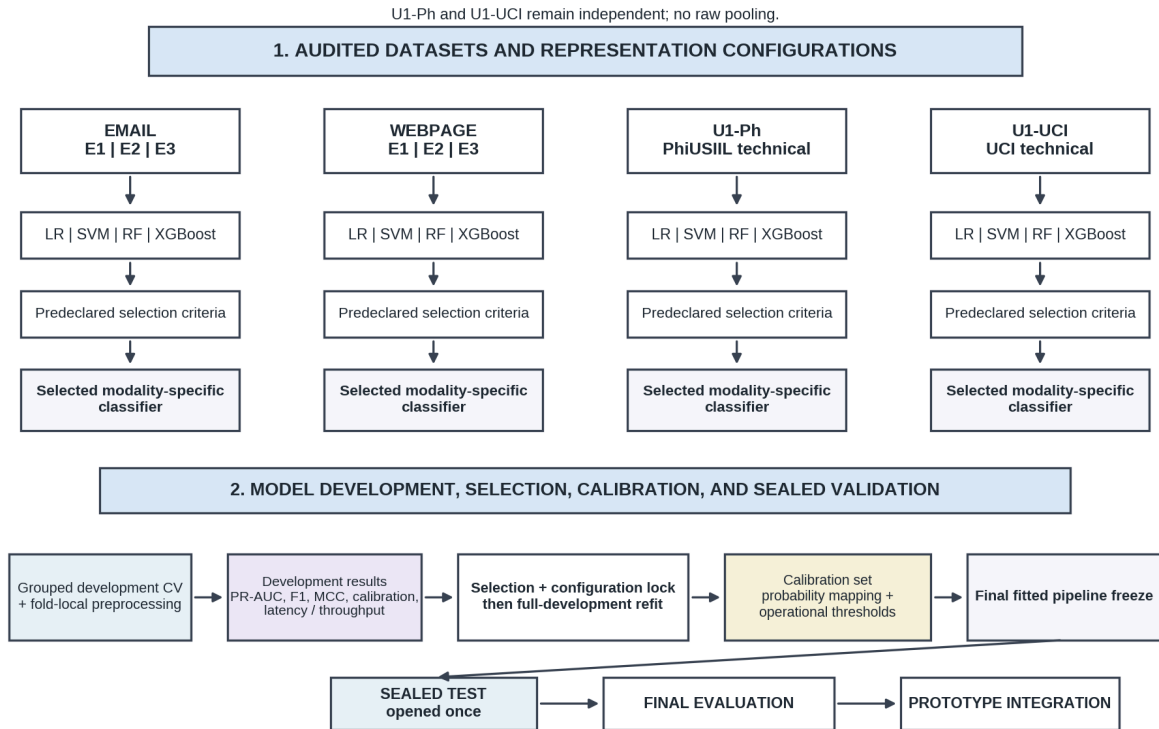
the frozen statistical plan, source-stratified errors, model-efficiency measures, and system logs. Post-test defects are documented but do not permit returning to model-family selection, hyperparameter tuning, feature revision, threshold revision, or SE-BRL rule revision based on sealed-test performance.

Functional and integration validation compares expected and actual outputs. Release requires passing all critical security and fail-closed tests, version compatibility across components, reproducibility of split assignments, and a clean rerun from the frozen package. This one-time sealed-test rule preserves the final partition as an unbiased estimate of the already selected system rather than a decision aid for model selection.

4.4 Research Conceptual Framework

The conceptual framework links the five research questions and objectives to the final pipeline. Audited modality-specific datasets and the literature-derived rule specification are the research inputs. Deterministic SE-BRL and conventional feature configurations are evaluated through controlled candidate classifiers; the selected classifier produces calibrated phishing output while artifact evidence and model attribution remain separate; deterministic decision logic produces bounded risk/response states; and the proof-of-concept validates integration, security, performance, and reproducibility. Figure 4.10 depicts these relationships.

Figure 4.10.
Research conceptual framework of the SE-BRL study



As shown in Figure 4.10, it summarizes the study's controlled modeling workflow across email, webpage, and U1 technical modalities. It shows how audited datasets are partitioned into development, calibration, and sealed-test sets; how LR, SVM, RF, and XGBoost are evaluated using predefined criteria; and how the selected model is frozen, calibrated, tested once on the sealed set, and then integrated into the prototype.

4.5 Analysis and Design

4.5.1 Overall System Architecture and Data Flow

The architecture processes an artifact only after schema validation, sanitization, and modality routing. Content-bearing email and webpage artifacts proceed to their conventional feature branch and deterministic SE-BRL branch. Stand-alone URL and engineered records proceed to the approved U1 technical pathway only. Candidate classifiers are research-time alternatives; after development-only selection and freeze, the operational pipeline invokes only

the selected modality-specific phishing classifier. Evidence and attribution remain separately addressable throughout downstream processing.

Figure 4.11 presents the final data flow and makes the deterministic SE-BRL feature-producing layer explicit.

Figure 4.11.

Overall SE-BRL architecture and data flow

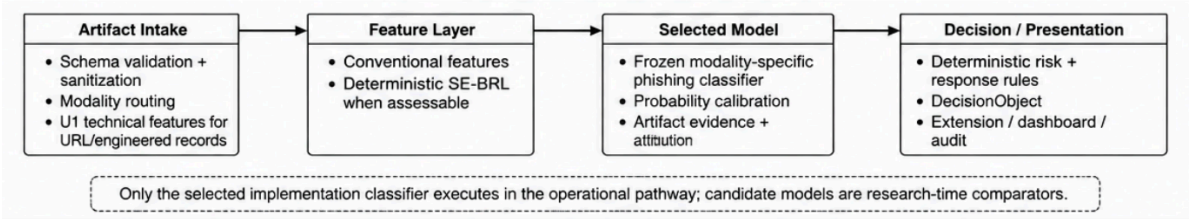


Figure 4.11 clarifies that deterministic SE-BRL is a feature-producing evidence layer, not a second classifier. The selected phishing classifier consumes the appropriate E1/E2/E3 or U1 representation, after which calibration, evidence/attribution assembly, decision logic, and presentation execute under frozen interfaces.

4.5.2 Modality-Aware Processing and SE-BRL Schema

Email processing preserves subject/body content, normalized sender/reply relations, available headers or routing/authentication fields, formatting indicators, embedded URLs, and attachment metadata without opening attachment contents. Webpage processing preserves archived visible text, HTML/DOM structure, forms and input types, link/resource relations, and brand-domain relations. URL/engineered processing preserves approved lexical, structural, host-related, and dataset-provided technical values already present in the source; no live lookups are required. Source identifiers and raw dataset names are not predictive features.

Table 4.13 summarizes the final feature and evidence schema.

Table 4.13.

Feature and deterministic SE-BRL schema summary

Category	Representative fields	Transformation boundary	E1	E2	E3	Audit-only / excluded
Email conventional	Subject/body text; sender/reply relations; available headers/structure; embedded-URL derived features	Vocabulary, scaling, imputation and learned selection fitted inside development fold	Yes	No	Yes	Raw source name/file path, group IDs, target label
Webpage conventional	Visible text; DOM/forms/links/structure ; embedded URL; brand-domain relations	Parser fixed; learned text/numeric transforms fitted inside fold	Yes	No	Yes	Screenshot pixels unless separately approved; source/audit IDs
URL technical	Approved lexical, structural, host/web and dataset-provided engineered values	Dataset-specific frozen pipeline; no live lookup	U1 only	No	No	Content-depend ent SE-BRL; source/audit IDs
SE-BRL predictive	Four deterministic support values + four availability values	Frozen rulespecification and fixed-width mapping	No	Yes	Yes	Raw evidence text/locators not supplied as undocumented predictive values
SE-BRL evidence audit	Dimension, indicator, state, evidence type/locator/hash, rule version	Stored in long-form evidence record	No	No	No	Explanation/audi t record only
Common control	Optional modality indicator only for approved pooled secondary analysis	Identical treatment across compared configurations; ablation required	Condi tional	Condi tional	Condi tional	Dataset source and target class label

Note. The schema contains no behavioral calibrator, learned dimension confidence, annotator field, adjudication field, or behavioral ground-truth label.

As shown in Table 4.13, the E2/E3 SE-BRL predictive values are deterministic support and availability states. Evidence locators remain in the long-form record for explanation and audit, preventing the fixed-width model input from being confused with the complete evidence package.

4.5.3 Explainability, Risk, Decision, and Intervention Design

The explanation layer joins the artifact evidence record with model attribution without collapsing the two. A user-facing result may state that Requested Action and Consequence is Supported because a credential-request form or text span was found and separately state whether an associated feature materially influenced the phishing classifier. Each explanation record identifies the artifact, model/preprocessing/rule version, feature name/value where applicable, attribution method/value, and linked evidence reference. The calibration mapping is reported separately from attribution.

The risk component applies the precedence in Figure 4.12 and Table 4.11 rather than an unconstrained weighted score. The decision engine returns one of four frozen states and selects only approved response templates. It does not generate security instructions through an unconstrained language model, and its validation concerns expected-versus-actual decision behavior rather than human warning effectiveness.

4.5.4 Browser Extension and Backend Integration Design

The browser extension uses Manifest V3 and explicit user invocation. It obtains only the artifact fields required by the supported route, sanitizes display output, and communicates with the FastAPI backend through schema-valid bounded requests. The backend validates modality and schema, performs feature construction and deterministic SE-BRL execution, invokes the selected frozen classifier and calibrator, joins evidence with attribution, applies deterministic risk/response logic, and returns a versioned DecisionObject. URLs are processed as strings or archived fields without live crawling (Chrome Developers, 2023a, 2023b).

Figure 4.13.

Browser extension-backend sequence and trust boundaries

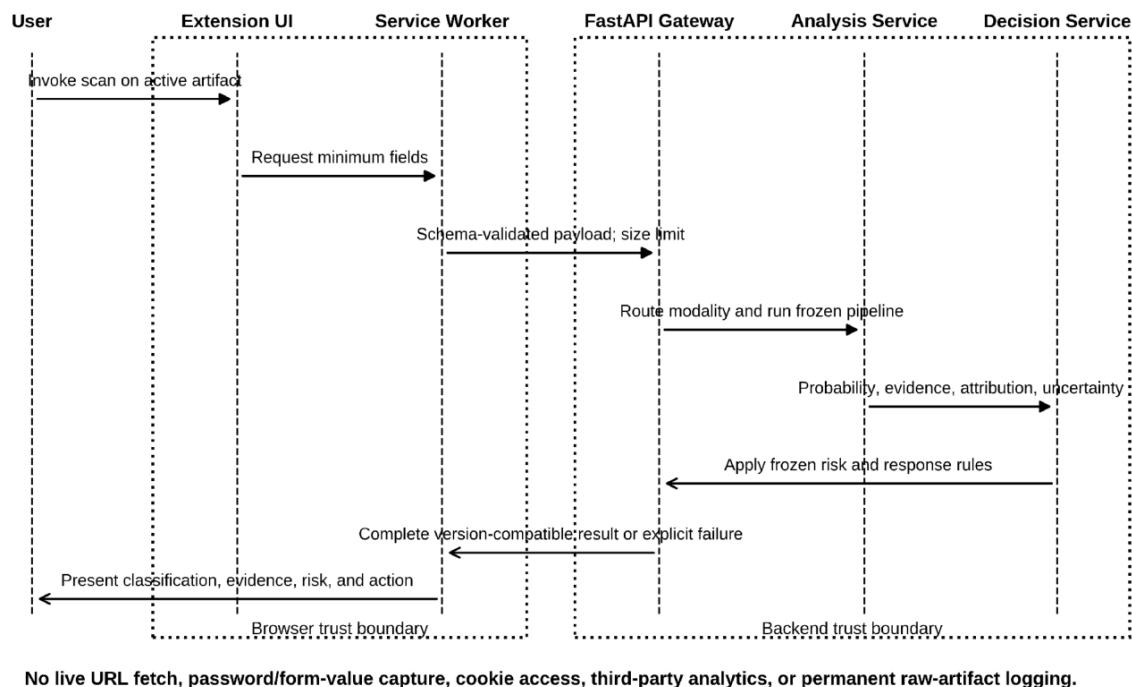


Figure 4.13 emphasizes validation at every trust boundary. Active artifact content is treated as untrusted, analytical components must be version-compatible, and partial or stale results cannot be rendered as complete risk decisions.

Table 4.14 provides the corresponding trust-boundary matrix.

Table 4.14.
Compact threat model and trust-boundary matrix

Trust boundary / asset	Security objective	Principal threat	Control evidence
Active artifact → extension content script	Treat page/email content as untrusted and minimize collection	Markup injection, hidden values, excessive scraping	Permission matrix, CSP, escaped rendering, payload inspection
Extension service worker → API gateway	Accept only intentional, schema-valid, bounded requests	Malformed/replayed/oversized payload, endpoint abuse	OpenAPI/Pydantic schema, size/rate/time limits, endpoint configuration
API gateway → analytical bundle	Use compatible and untampered model/rule/schema artifacts	Model/rule/schema substitution or mismatch	Bundle hashes, compatibility manifest, fail-closed tests

Trust boundary / asset	Security objective	Principal threat	Control evidence
Analysis → decision service	Preserve complete evidence, attribution, calibration and uncertainty	Partial/stale result interpreted as valid	Transactional result assembly, versioned payload, outage tests
Backend → audit database/logs	Retain reproducibility metadata without raw sensitive content	PII/artifact leakage	Audit schema, log review, retention/deletion record

Note. The matrix is a prototype threat model and does not imply production-grade endpoint or gateway security.

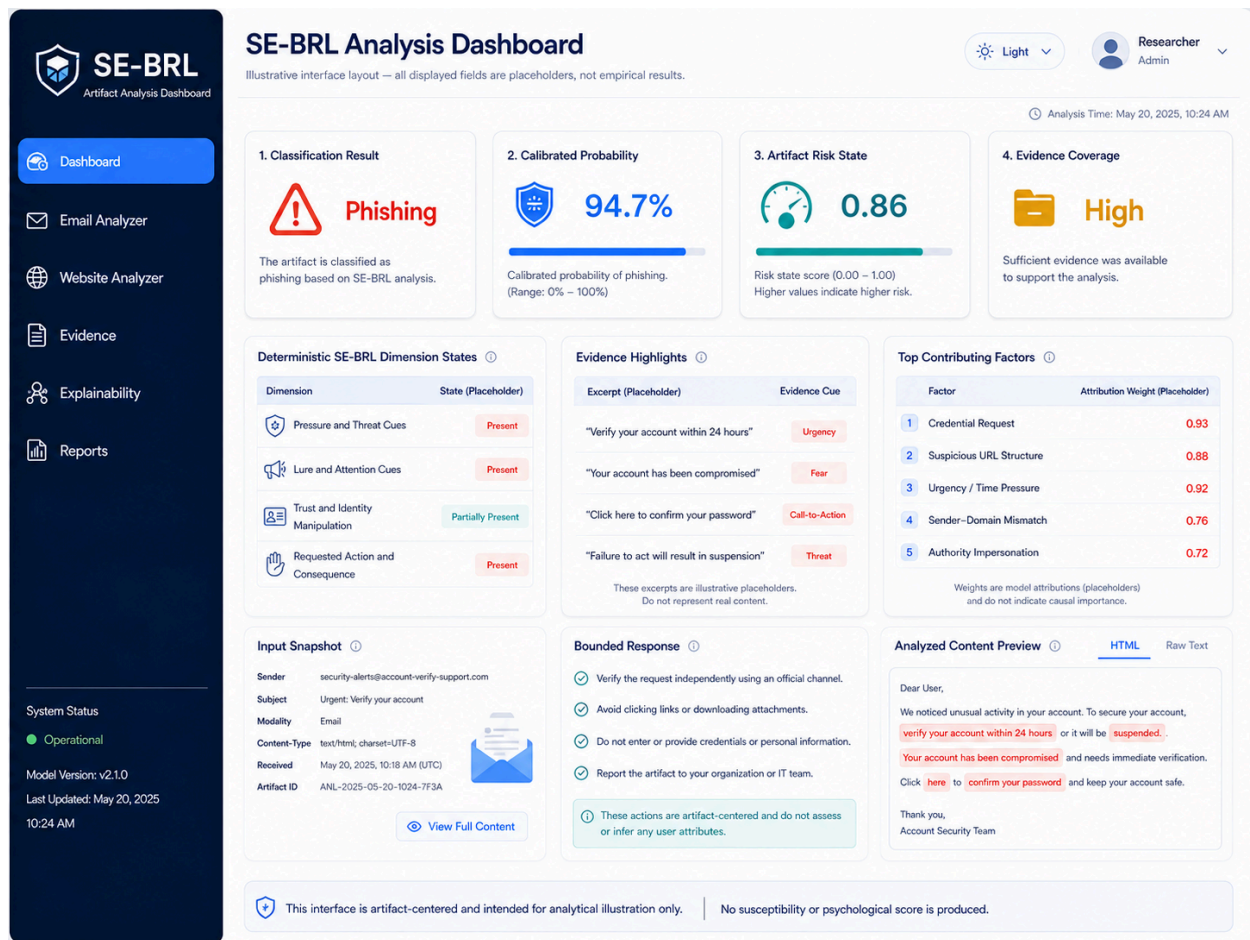
Table 4.14 translates the architecture into testable security objectives. Each boundary has observable control evidence, allowing security validation to be treated as a release requirement rather than as an informal design aspiration.

4.5.5 Web Dashboard and Result Presentation Design

The supporting dashboard presents the routed artifact type, classifier result, calibrated probability, deterministic SE-BRL dimension states and evidence references where assessable, model attribution, risk state, bounded response, limitations, and component versions. It does not display a “susceptibility level,” psychological score, or other user-centered inference that lies outside the artifact-level scope. Figure 4.14 provides an illustrative layout only; placeholder fields are not empirical results.

Figure 4.14.

Illustrative MVP analysis dashboard



Note. Interface fields are illustrative placeholders and do not represent empirical results.

Figure 4.14 presents the artifact-centered dashboard layout, with evidence and attribution shown as distinct channels and no susceptibility-oriented output. Evidence and attribution are displayed as distinct panels, and limitations/version metadata remain visible so that users can distinguish observed support from model influence and system uncertainty.

4.5.6 Use-Case Model

The operational user personas are the Demonstration End User, Researcher/Administrator, and Evaluator/Tester. The Demonstration End User submits supported artifacts and views bounded results; the Researcher/Administrator manages approved research artifacts and controlled experiment configurations; and the Evaluator/Tester

executes expected-versus-actual functional, security, performance, and reproducibility tests. Cybersecurity experts and behavioral annotators are not operational user personas under this scope.

Figure 4.15 presents the use-case model.

Figure 4.15.

SE-BRL system use-case model

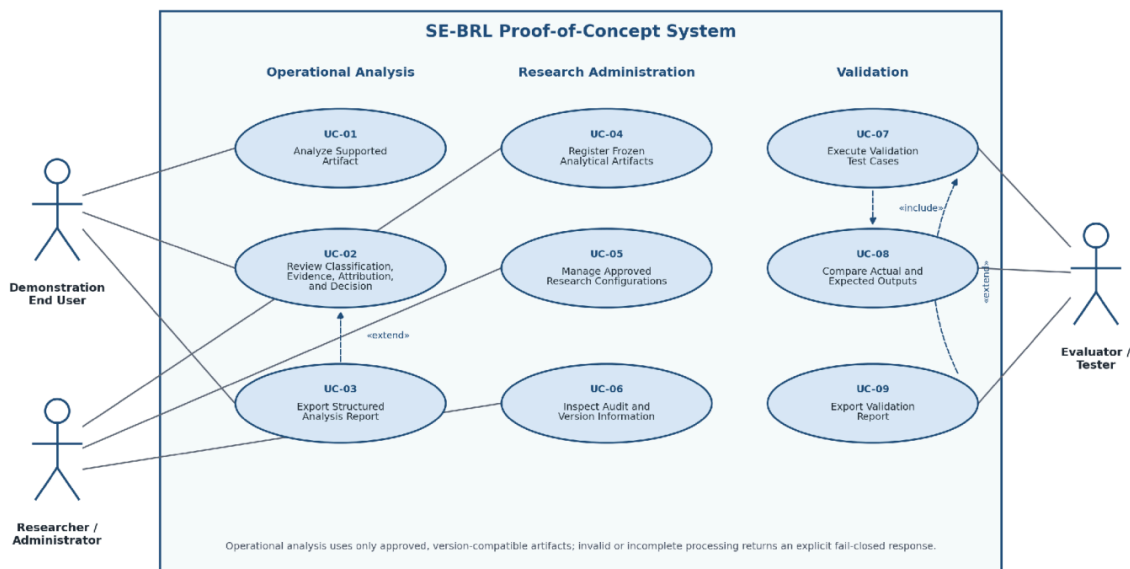


Figure 4.15 aligns the manuscript and SRS around the same three operational personas; expert-validator and behavioral-annotator roles are outside the methodology.

4.6 Development Model

The proof-of-concept follows controlled Agile iterations while preserving frozen research boundaries. Iteration 1 freezes sources, dataset-audit rules, schemas, privacy controls, grouping, and split protocol. Iteration 2 implements and verifies the deterministic SE-BRL rule package and finalizes eligible audited datasets. Iteration 3 develops E1/E2/E3/U1 features, evaluates the four candidate phishing classifiers, benchmarks latency/throughput, applies the predeclared model-selection rule, and freezes the selected classifier, calibration, explainability, and risk-response artifacts. Iteration 4 integrates the secure API and extension/dashboard, executes hostile-input, functional, security, privacy and performance tests, and performs the

one-time sealed final evaluation. Core research questions, metrics, split logic, SE-BRL rule definitions, selection criteria, calibration sequence, and final-test rules require formal change control.

4.7 Development Approaches

The study combines four complementary approaches. Deterministic, modality-aware evidence processing operationalizes SE-BRL without human behavioral ground truth. Supervised machine learning supports controlled comparison of conventional, deterministic SE-BRL, fused, and technical representations. Explainable AI provides model-centered attribution while an independent evidence record preserves artifact-centered meaning. Modular secure software engineering provides schema validation, trust-boundary controls, reproducible artifacts, bounded responses, and fail-closed behavior. These approaches are evaluated independently enough to avoid treating a successful classifier as proof of behavioral validity or a correct evidence rule as proof of model performance.

4.8 Software Development Tools and Reproducibility Environment

The implementation uses version-controlled Python and Node.js environments, scikit-learn-compatible tooling for classical classifiers, XGBoost for the boosted-tree candidate, the approved attribution package for tree-model explanations, FastAPI for backend interfaces, Chromium/Manifest V3 for the browser extension, and versioned configuration files for splits, preprocessing, rule specifications, calibration, risk logic, and tests. Exact package versions, lock files, operating-system and hardware information, seeds, repository commit/tag, model hashes, rule hashes, schema versions, and benchmark settings are frozen in the reproducibility package. Official project documentation is used for implementation-specific interfaces, while all empirical claims are produced only by the frozen study pipeline.

Table 4.15 summarizes the tool categories and their bounded purpose.

Table 4.15.

Software development tools and reproducibility environment

Tool category	Examples / environment	Purpose	Reproducibility control
Version control	Git repository + protected final tag	Source, configuration and artifact history	Commit/tag and deviation log
Data / modeling	Python; pandas; scikit-learn; XGBoost	Audit, preprocessing, candidate training, metrics and calibration	Pinned environment, seeds, config files
Explainability	Linear contributions / approved tree-attribution implementation	Model-centered attribution only	Method/version stored with output
Backend	FastAPI + typed schemas	Routing, analysis orchestration, DecisionObject API	OpenAPI/schema version + integration tests
Browser / frontend	Chromium Manifest V3 extension + supporting dashboard	Explicit artifact submission and bounded presentation	Permission/CSP review + build lock
Testing / audit	Automated unit, integration, security, performance and clean-rerun scripts	Expected-versus-actual validation	Versioned test reports + release checklist

Note. Specific pinned versions are recorded in the frozen environment report and are not fabricated before implementation freeze.

Table 4.15 ensures that each tool is tied to a controlled methodological role. Environment metadata is part of the experiment evidence because latency, throughput, and reproducibility cannot be meaningfully interpreted without the execution context.

4.9 Project Management

4.9.1 Schedule and Timeline

The project schedule follows the final pipeline: dataset acquisition and audit precede SE-BRL verification; candidate-model evaluation and efficiency benchmarking precede final model selection; and model/calibration/risk freeze precedes secure integration and the one-time sealed test. Figure 4.16 presents the eight-week sequence.

Figure 4.16.

Eight-week project schedule, research freezes, and final evaluation sequence

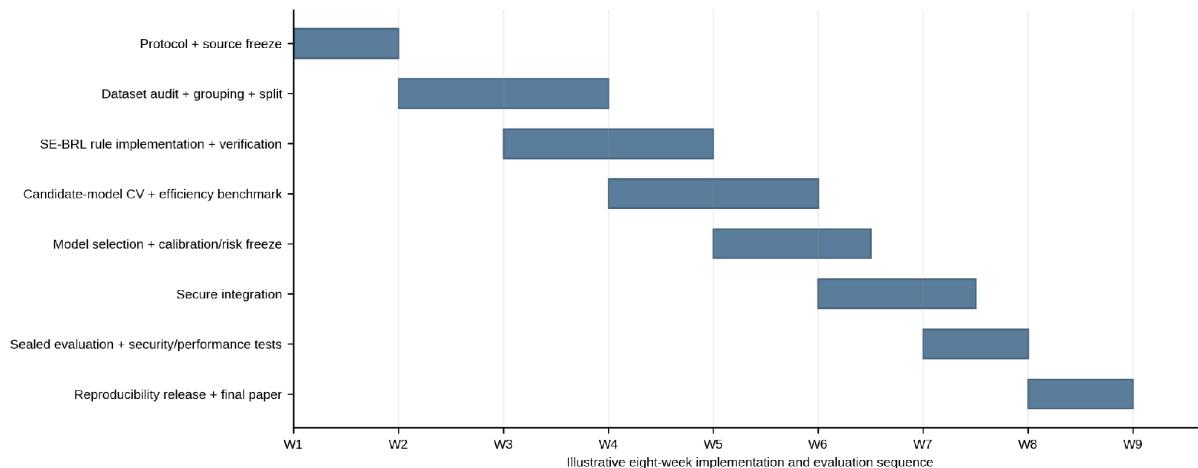


Figure 4.16 sequences dataset audit, deterministic rule verification, candidate-model evaluation, efficiency benchmarking, model selection, secure integration, sealed evaluation, and reproducibility release. The sequence protects the sealed test by placing it after all model/rule/calibration freezes.

4.9.2 Responsibilities

Both researchers maintain joint control over protocol changes, sealed-test access, and the final release package. Responsibilities are divided for execution efficiency but independently checked for high-impact artifacts. No responsibility includes behavioral annotation or adjudication. Table 4.16 defines the assignment.

Table 4.16.

Work area	Primary responsibility	Independent check	Required evidence
Dataset acquisition, provenance, audit, grouping and split	Modeling lead	Other researcher verifies manifests/count reconciliation	Hashes, source manifest, screening/attrition/group/split logs
SE-BRL rule implementation and verification	Joint / designated rule lead	Other researcher reruns test suite	Rule spec, test vectors, conformance/repeatability reports
Candidate models, metrics, efficiency and selection	Modeling lead	Other researcher reproduces selection	Configs, CV results, latency/throughput

Work area	Primary responsibility	Independent check	Required evidence
Calibration, XAI, risk and response rules	Joint	decision from frozen reports Cross-check expected-versus-actual scenarios	results, model-selection log Calibrator/rule versions, explanation/risk tests
Extension, backend, dashboard and security tests	Prototype lead	Modeling lead verifies analytical contracts	Build artifacts, integration/security/performance reports
Sealed evaluation and release	Joint	Mutual sign-off before test access and final tag	Access record, final test report, release checklist

Researcher responsibilities and independent checks **Note.** *No blinded annotation, behavioral adjudication, or expert behavioral-validation responsibility is included.*

Table 4.16 makes the audit trail explicit: high-impact artifacts have both a primary executor and an independent check. This supports reproducibility without introducing human behavioral labeling into the methodology.

4.9.3 Budget and Cost Management

The study prioritizes open-source software and existing academic hardware. The existing project budget is retained because the methodology does not require paid expert validation or annotation. Table 4.17 summarizes the planned direct costs.

Table 4.17.
Planned project budget

Item	Estimated Cost (PHP)	Purpose / Control
Printing and documentation	₱2,145	Printing and binding of the proposal/final manuscript and other required research documentation
Backup / storage	₱500	Storage of versioned datasets, code, experiment outputs, documentation, and redundant research artifacts
Internet / data access	₱800	Dataset acquisition, repository access, documentation retrieval, and development-related connectivity
Local transport / project coordination	₱1,200	Transportation for university requirements, submission, meetings, and project coordination
Contingency	₱700	Allowance for minor unforeseen research and documentation expenses
Laptops / existing computing hardware	₱0	Researchers' existing laptops are used for development, experimentation, analysis, and

Item	Estimated Cost (PHP)	Purpose / Control
Open-source software	₱0	documentation; no new hardware purchase is required Development and analysis primarily use open-source or freely available software; no software-license purchase is assumed

Note. Budget values are retained from the submitted proposal and do not assert actual expenditures.

Table 4.17 reflects the bounded academic prototype rather than a production deployment. Any material budget deviation is documented under project change control.

4.9.4 Reproducibility and Release Checklist

The minimum reproducibility package excludes annotation, adjudication, CVI, behavioral-model calibration, and Dimension Identifier artifacts. Table 4.18 defines the final package required for clean rerun and audit.

Table 4.18.

Minimum reproducibility package

Artifact	Required content	Freeze point	Verification
Dataset cards, source manifest and attrition files	DOIs/URLs, dates, hashes, schemas, labels, licenses, exclusions, retained IDs, privacy controls	End of Stage 3	Counts reconcile; hashes/licenses/source records match
Group and split manifests	artifact_id, group_id, modality, partition, grouping reason	End of Stage 3 before model work	No group crosses partitions
SE-BRL rule package and verification	Versioned dimensions/indicators/rules, modality boundaries, test cases, evidence locators, conformance/repeatability reports	End of Stage 4	Deterministic test suite passes; no expert/annotation artifacts required
Data and feature dictionaries	Field definitions, transformations, missingness, fitting boundary, E1/E2/E3/U1 membership	Before Stage 5 modeling	Pipeline/dictionary consistency test
Candidate-model and selection package	Search configs, seeds, CV results, calibration evidence, p50/p95 latency, throughput, resource measures, selection decision log	End of Stage 5	Independent rerun reproduces selected model decision

Artifact	Required content	Freeze point	Verification
Selected models, calibrators and thresholds	Model/preprocessing files, parameters, hashes, calibrator, operational thresholds	End of Stage 6	Compatibility and hash tests pass
XAI, evidence, risk and response artifacts	Attribution configuration, evidence-record schema, risk rules, approved templates, scenario suite	Before sealed test	Expected rule/explanation audits pass
System/security/privacy/performance tests	Functional, integration, hostile-input, permission, privacy, failure, model-efficiency/end-to-end performance reports	End of Stage 8	All critical/fail-closed tests pass
Statistical-analysis package	Primary contrast, frozen resampling/CI procedure, correction/error analysis	End of Stage 8	Results regenerate from frozen predictions
Protocol deviation and release record	Dated deviations, approvals, repository tag, lock files, README, model/dataset cards, sealed-test access record	Final submission	Independent clean-install reproduction documented

Note. The reproducibility package intentionally excludes expert-rating forms, human behavioral annotations, adjudication logs, CVI/agreement reports, learned behavioral confidence files, and Dimension Identifier artifacts.

Table 4.18 closes the methodological audit trail. A clean rerun must be able to reconstruct dataset eligibility, split assignments, deterministic SE-BRL states, candidate-model results, final model selection, calibration, explanations, decisions, and final system tests from frozen artifacts without undocumented manual judgment.

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Appendices

Appendix A

Agreement Letter for Thesis Proposal Defense



UNIVERSITY
of SAN CARLOS



DEPARTMENT OF COMPUTER, INFORMATION SCIENCES AND MATHEMATICS

AGREEMENT LETTER FOR THESIS PROPOSAL DEFENSE

This is an agreement between the Department of Computer, Information Sciences and Mathematics and the undersigned proponents of the thesis proposal entitled **SE-BRL: A Modality-Aware Social Engineering Behavior Representation Architecture for Explainable Phishing Detection and Risk-Adaptive Defense**

The undersigned agrees to submit the following:

1. 3 complete copies of thesis/capstone proposal document; and
2. 1 copy of sworn and recommendation letter.

These are to be submitted on **July 27, 2026, before 5:00 PM at the department office.**

Failure to do so will automatically earn the proponents a grade of **5.0** for the **CS3201N - Thesis Proposal** this semester.

Signed this **24th** day of **July, 2026.**

Handwritten signature of Vera Marie Therese A. Orbello.

VERA MARIE THERESE A. ORBELLO

Handwritten signature of Florlita A. Orbello.

FLORLITA A. ORBELLO

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CHRISTIAN V. MADERAZO, MEng
Thesis Proposal Adviser

Appendix B

Sworn Statement and Recommendation for Thesis Proposal Defense



SWORN STATEMENT AND RECOMMENDATION FOR THESIS PROPOSAL DEFENSE

I certify and swear under the procedures imposed for CS 3201N (CS Thesis 1) that I have reviewed thoroughly the Thesis Proposal Document entitled: **SE-BRL: A Modality-Aware Social Engineering Behavior Representation Architecture for Explainable Phishing Detection and Risk-Adaptive Defense** guided by the standard format and contents done by **Dhanwil Nick M. Alcover** and **Vera Marie Therese A. Orbello**

As Thesis Adviser, I highly recommend the above-mentioned students to proceed to their Thesis Proposal Defense on **July 31 to August 5, 2026**

Certified and Recommended by:


CHRISTIAN V. MADERAZO, MEng
Thesis Proposal Adviser

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 2. Programming (C, Java, React.js, TypeScript, SQL, Python, JS, HTML5, CSS3, Tailwind)

EDUCATIONAL BACKGROUND

1. Education Level: College

Year: 2024 - Present

School: University of San Carlos

Tertiary

STRENGTHS/TRAITS & SKILLS

1. Proactive in developing innovative software solutions
2. Hands-on experience in full-stack application development
3. Strong analytical, research, and problem-solving abilities
4. Effective technical presentation and communication skills
5. Capable of managing multiple projects within strict deadlines

CAREER OBJECTIVES

To pursue opportunities in software development and artificial intelligence where I can apply my skills in full-stack development, system design, and technical research while continuously improving my expertise in emerging technologies.

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1. Office Applications (Word, Excel, PowerPoint)

2. Programming (C, Java, React.js, TypeScript, SQL, JS, HTML5, CSS3, Tailwind)

EDUCATIONAL BACKGROUND

1. Education Level: College

Year: 2024 - Present

School: University of San Carlos

STRENGTHS/TRAITS & SKILLS

1. Strong initiative in developing and leading software projects
2. Hands-on experience in full-stack application development
3. Strong technical research and problem-solving skills
4. Effective presentation and project communication skills
5. Able to manage multiple tasks under strict deadlines

CAREER OBJECTIVES

To pursue opportunities in software development and artificial intelligence where I can apply my experience in full-stack development, research, and system design while continuously expanding my technical expertise.

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Software Requirements Specification

IEEE 830 / ISO/IEC/IEEE 29148 Format

Social Engineering Behavior Representation Layer (SE-BRL): A Modality-Aware Social Engineering Behavior Representation Architecture for Explainable Phishing Detection and Risk-Adaptive Defense

Version 3.0 - Final Revised Scope

13 August 2026

By

Dhanwil Nick Maringuran Alcover

Vera Marie Therese A. Orbello

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Faculty Adviser

Department of Computer, Information Sciences and Mathematics

University of San Carlos

Revision History

Table R.1

Document revision record

Date	Version	Description	Author
2026-07-26	0.1	Initial template population and requirements mapping.	D. N. M. Alcover; V. M. T. A. Orbello
2026-07-26	1.0	Baseline SRS covering the research pipeline and proof-of-concept prototype.	D. N. M. Alcover; V. M. T. A. Orbello
2026-07-27	2.0	Condensed baseline aligned with the prior proposal scope and deterministic risk-state precedence.	D. N. M. Alcover; V. M. T. A. Orbello
2026-08-11	3.0	Final revision aligned with deterministic literature-derived SE-BRL, no expert/behavioral annotation, candidate-model selection, audited datasets, and latency/throughput requirements.	D. N. M. Alcover; V. M. T. A. Orbello

Note. Version 3.0 supersedes conflicting current-method statements in earlier SRS versions.

Status note: This SRS specifies planned and required behavior. It does not assert completed empirical results, a selected winning classifier, achieved performance, production-grade security, or demonstrated changes in human behavior. Where proposal-stage values remain pending execution, the implementation shall record and freeze the actual audited result rather than infer it.

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Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and nonfunctional requirements for the Social Engineering Behavior Representation Layer (SE-BRL) research prototype, version 3.0. The system operationalizes observable artifact evidence through a frozen, literature-derived, deterministic representation layer; compares conventional, deterministic SE-BRL, and fused phishing representations; produces explainable outputs; and applies deterministic artifact-level risk-adaptive responses.

The SRS is the implementation and verification baseline for the browser extension, web dashboard, FastAPI backend, deterministic SE-BRL module, phishing-classification pipeline, calibration and decision components, dataset-audit artifacts, and reproducibility package. Supervised machine learning is reserved for phishing classification. The SRS does not define expert behavioral validation, behavioral ground-truth annotation, a separately trained behavioral classifier, or learned behavioral confidence.

1.2 Document Conventions

Table 1.1 defines the requirement and evidence-state terms used throughout this SRS.

Table 1.1

Document conventions

Convention	Meaning
shall	Mandatory and verifiable requirement.
should / may	Recommended or permitted behavior; not release-blocking unless promoted by controlled revision.
REQ-4.x-n	Functional requirement identifier.
NFR-5.x-n	Nonfunctional requirement identifier.
BR-n	Business or research-governance rule.
High / Medium / Low	High is essential to correctness or safety; Medium supports reporting/quality; Low is optional.
Supported / Absent / Unavailable	Qualifying evidence is present; evidence is assessable but absent; or the modality/required fields do not permit assessment.

The deterministic evidence states are semantic system outputs, not human behavioral labels or learned dimension probabilities.

1.3 Intended Audience and Reading Suggestions

Table 1.2 maps the principal audiences to the SRS sections most relevant to implementation and verification.

Table 1.2

Intended audience and reading guidance

Audience	Recommended Sections	Primary Use
Research proponents / developers	2-6 and appendices	Implement the bounded architecture and preserve the approved scope.
ML implementers	2.5, 4.2-4.6, 4.9, 5	Build leakage-controlled features, candidate classifiers, calibration, efficiency benchmarks, attribution, and frozen artifacts.
Frontend / integration developers	3, 4.1, 4.5-4.8, 5.2-5.3	Implement extension/dashboard contracts, safe rendering, decision presentation, and fail-closed behavior.
Evaluator / Tester	4-5 and appendices	Derive functional, boundary, deterministic-rule, security, performance, and reproducibility tests.
Adviser / panel	1-2, 4.3-4.9, 5.5, appendices	Verify scope alignment, model-selection ordering, claim boundaries, and traceability.

The named operational roles in this table do not create expert validator or behavioral annotator roles.

1.4 Product Scope

SE-BRL is an artifact-centered, modality-aware phishing-analysis architecture. Behavioral evidence is produced only from supported content-bearing email and webpage fields using frozen deterministic rules. Bare URLs and engineered technical records remain in independent U1 technical branches and shall not receive unsupported language-dependent SE-BRL states.

1.4.1 In-Scope

- Email artifact processing and approved email feature extraction.
- Webpage artifact processing and approved content, DOM, form, link, and brand-domain feature extraction.
- URL and engineered technical processing for independent U1 benchmarks.
- Deterministic SE-BRL execution for supported content-bearing modalities with four behavioral dimensions and twelve subordinate evidence indicators.

- E1 conventional-only, E2 deterministic SE-BRL-only, E3 fused, and U1 technical feature construction; development evaluation of four candidate phishing-classification families; selected-classifier runtime inference.
- Phishing-probability calibration and frozen operational risk-threshold processing.
- Separate presentation of artifact evidence and model attribution.
- Deterministic Review Required / High / Elevated / Informational state assignment and bounded protective/verification guidance.
- Manifest V3 browser extension, local FastAPI backend, React dashboard, structured reports, and versioned interfaces.
- Functional, integration, security, privacy, failure, model-efficiency, end-to-end performance, and reproducibility testing.
- Model-only p50/p95 inference latency, throughput, end-to-end latency, benchmark-environment, and resource reporting.

1.4.2 Out-of-Scope

- Expert validation of the four behavioral dimensions or twelve subordinate indicators.
- Expert/researcher behavioral annotation as ground truth, adjudication, inter-rater reliability, I-CVI/S-CVI, or Krippendorff agreement for the present study.
- Supervised behavioral-dimension classification, learned behavioral confidence, behavioral probability calibration, OOF behavioral prediction, or per-dimension learned thresholds.
- Human susceptibility, psychological profiling, demographic inference, attacker-intent inference, or user-compliance probability.
- Human warning comprehension/effectiveness, click reduction, learning, or behavior-change measurement.
- Behavioral inference from bare URLs or engineered technical records without required content evidence.
- Live web crawling, malware/attachment-content execution, remote resource loading, unrestricted credential/form-value collection, or autonomous navigation.

- Production-grade endpoint/network protection, automatic quarantine/reporting, or universal future-attack detection.

1.5 References

Table 1.3 lists the governing and implementation references used by this SRS.

Table 1.3

References

Ref	Document	Version / Date	Source
[1]	ISO/IEC/IEEE 29148 - Requirements Engineering	2018	ISO/IEEE standards catalog
[2]	SE-BRL Final Compiled Proposal, Chapters 1-4	Current approved revision	Internal thesis manuscript
[3]	NIST AI Risk Management Framework 1.0	2023	NIST AI 100-1
[4]	NIST Secure Software Development Framework 1.1	2022	NIST SP 800-218
[5]	Republic Act No. 10173 - Data Privacy Act	2012	Official Gazette of the Philippines
[6]	OWASP API Security Top 10	2023	OWASP Foundation
[7]	Chrome Extensions / Manifest V3 Documentation	Project reference	Chrome for Developers
[8]	FastAPI, scikit-learn, XGBoost, and SHAP documentation	Pinned releases	Official project documentation

Implementation versions shall be pinned and recorded in the release package rather than assumed from this SRS.

2. Overall Description

2.1 Product Perspective

SE-BRL is a controlled research prototype, not a replacement for an email-security gateway, safe-browsing service, malware scanner, or endpoint-control platform. It complements conventional phishing analysis by preserving deterministic behavioral evidence, modality support, exact evidence references, calibrated phishing output, model attribution provenance, and deterministic response rationale.

The development lifecycle and operational lifecycle are separated. Development verifies and freezes dataset sources and groups, executes deterministic SE-BRL rule verification, evaluates Logistic Regression, Linear SVM, Random Forest, and XGBoost under controlled development folds, selects the

implementation classifier using predeclared criteria, freezes the selected model and benchmark environment, calibrates probability and operational thresholds, and only then opens the sealed test once. Operational inference loads only compatible frozen artifacts and never retrains online.

2.2 Product Functions

- Accept and validate user-selected email, webpage-content, URL, or approved research-record inputs; route each input by modality and field availability.
- Parse MIME/text/HTML/headers/links/forms/URLs without executing active content or fetching destinations.
- Generate modality-specific conventional features and deterministic SE-BRL states with evidence references for supported email/webpage artifacts.
- Execute the selected frozen phishing classifier and calibrator at runtime; preserve E1/E2/E3/U1 as controlled research configurations.
- Keep artifact evidence and model attribution as independent explanation channels.
- Apply deterministic Review Required, High, Elevated, or Informational rules and bounded defensive/verification actions.
- Provide extension, dashboard, report, audit, performance, test, and reproducibility outputs without operational raw-content persistence by default.

Figure 2.1 summarizes the client, local-service, and frozen-artifact boundaries used by the proof-of-concept.

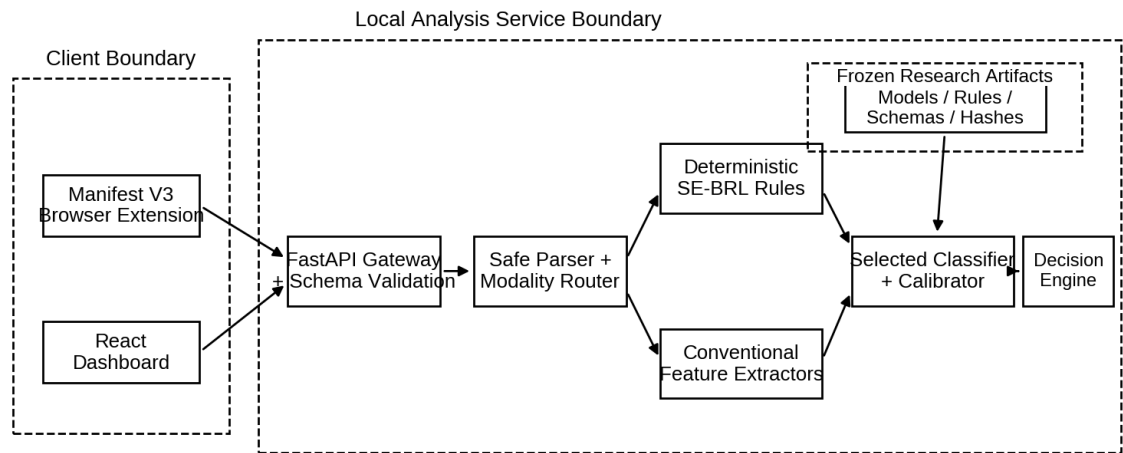


Figure 2.1. Condensed product context and trust boundaries.

As shown in Figure 2.1, deterministic SE-BRL rules and conventional feature extractors feed the selected phishing-classification path, while the frozen artifact registry remains a separate integrity-controlled source. The client cannot directly change models, rules, calibrators, or thresholds.

2.3 User Persona

Table 2.1 defines the three operational personas used consistently across the SRS and use-case model.

Table 2.1

User personas

User Persona	Characteristics	Privileges / Access
Demonstration End User	May be nontechnical; requires clear evidence, limitations, and safe next steps.	May invoke an explicit scan, review results, and export an approved report; no dataset, model, threshold, or rule administration.
Researcher / Administrator	Technically proficient; manages approved research-critical components and configurations.	May register frozen artifacts, run approved research scripts, inspect non-sensitive audit/version records, and export research evidence; no silent post-freeze changes.
Evaluator / Tester	Understands expected outputs, security boundaries, deterministic-rule tests, and validation protocols.	May submit test artifacts, inspect full DecisionObjects, run approved suites, and export validation evidence; shall not alter sealed labels or frozen bundles.

No Expert Validator or Behavioral Annotator persona exists in the final operational or methodological scope.

2.4 Operating Environment

Table 2.2 defines the baseline execution environment without requiring specialized hardware for operational inference.

Table 2.2

Operating environment

Component	Required Environment	Notes
Browser extension	Current project-supported Chromium desktop browser.	Manifest V3; least-privilege active-tab or explicitly selected artifact access.
Dashboard	Modern desktop browser on Windows, macOS, or Linux.	React-based local interface for laptop/desktop demonstration.
Analysis API	Python runtime with pinned dependencies and FastAPI-compatible local service.	Loopback binding by default; versioned JSON contracts.
Data / ML pipeline	Python, Pandas, NumPy, scikit-learn, XGBoost, approved explanation libraries.	Final seeds, dependencies, benchmark environment, hashes, and model-selection records archived.
Storage	Local filesystem and lightweight metadata store.	Raw, derived, audit, split, model, result, and report artifacts separated.
Optional computation	GPU or academic cloud runtime for supplementary training.	Not required for baseline CPU inference or defense demonstration.

No camera, biometric device, sensor, barcode scanner, production mail gateway, or specialized inference accelerator is required.

2.5 Design and Implementation Constraints

- The unit of analysis shall remain the digital artifact; the system shall not profile a recipient or infer attacker intent.
- Deterministic SE-BRL shall execute only where the modality and required fields support the frozen rule specification; unsupported evidence shall be Unavailable, not Absent.
- URL-only and engineered records shall remain technical-only; heterogeneous raw schemas shall not be appended into a universal behavioral table or score.
- Raw research data shall remain immutable; provenance checks, transformations, exclusions, group assignment, and split creation shall be scripted and versioned.

- Duplicates, near duplicates, threads, campaigns, domains, templates, or equivalent grouping units shall remain within one partition.
- Learned preprocessing, model fitting, model selection, calibration, and threshold selection shall not access sealed-test labels.
- Operational inference shall load only frozen checksum-valid rules, feature schemas, selected classifier(s), calibrator(s), thresholds, and API contracts; no online retraining is permitted.
- Untrusted content shall remain inert; no remote resource loading, link opening, script execution, attachment-content execution, or live crawling is permitted.
- Only actually executed and documented components may be reported as implemented or validated.

2.6 Assumptions and Dependencies

Table 2.3 records dependencies that may affect availability, evaluation, or operational behavior.

Table 2.3

Assumptions and dependencies

Assumption / Dependency	Effect on the Product
Dataset access and licensing	Each retained source must pass provenance, label, content/schema, privacy, and license review before model development.
Content availability	Missing text, HTML, identity, or structural fields constrain deterministic SE-BRL assessment and must not be treated as ordinary absence.
Frozen artifact bundle	Inference depends on compatible feature schemas, selected model(s), calibrator(s), SE-BRL rules, thresholds, and decision rules.
Local analysis service	An unreachable or failed service produces an explicit failure and no guessed or partial risk output.
Browser permission	The user grants only the permission needed for the current scan; mailbox-wide access is not assumed.
Calibration outcome	Operational thresholds are frozen from the predeclared calibration protocol; unattainable targets are reported rather than fabricated.
Webpage evidence pool	Webpage E1/E2/E3 requires audited content-bearing artifacts; otherwise the sample remains technical-only or unsupported for deterministic SE-BRL.

Assumptions shall be reported as limitations where they materially constrain an evaluation or operational result.

3. External Interface Requirements

3.1 User Interfaces

Interfaces shall visually and semantically separate phishing prediction, artifact evidence, model attribution, decision state, defensive action, and limitations. Suspicious text and URLs shall be rendered inert, and Review Required shall remain distinct from definitive tiers.

Table 3.1

User-interface requirements

Interface	Required Elements	Required Behavior
Extension panel	Scan control, modality, service status, concise class/probability, decision state, supported SE-BRL dimensions, primary safe action, dashboard link.	Analysis begins only after an explicit gesture; unrelated tabs/messages are not read.
Dashboard input	Separate Email, Webpage Content, and URL/Research Record modes; validation hints; Analyze and Clear controls.	Selected mode controls required fields and routing; unsupported combinations are rejected before inference.
Dashboard result	Class, calibrated probability, technical findings, SE-BRL states/evidence, attribution, decision state, actions, limitations, versions.	Evidence and attribution remain separate; active HTML and suspicious destinations remain inert.
System status / report	Loaded component versions, compatibility state, run ID, model-only and end-to-end latency fields, structured export.	No UI-based model training, model selection, or threshold editing is exposed in the bounded MVP.
Error / limitation view	Stable error code, affected component/stage, safe retry or manual-verification guidance.	No default class, probability, tier, or intervention is shown after critical failure.

Keyboard operation, visible focus, readable contrast, meaningful text labels, and non-color-only status cues are required for the demonstration interfaces.

3.2 Hardware Interfaces

Table 3.2 lists the hardware interfaces required by the prototype.

Table 3.2

Hardware interfaces

Interface	Purpose	Constraint
Workstation storage	Stores datasets, manifests, frozen bundles, reports, and backups.	Paths are configurable; no hardware-specific device dependency.
Keyboard, pointing device, display	Provides artifact entry and visual review.	Desktop/laptop baseline; touch-only use is not required.

Interface	Purpose	Constraint
Optional GPU / cloud accelerator	May reduce approved training time.	Optional; final inference/demo remains CPU-capable.

Hardware performance reporting shall identify the actual benchmark environment used for model selection and system validation.

3.3 Software Interfaces

Table 3.3 defines the principal versioned software boundaries.

Table 3.3

Software interfaces

Interface	Purpose	Data Exchanged
Extension / Dashboard ↔ FastAPI	Submit a user-selected artifact and receive status/result.	UTF-8 JSON over versioned REST; no executable content.
API ↔ Parser / Feature Modules	Invoke safe modality-specific normalization and extraction.	Validated internal objects with run ID, modality, evidence map, and schema version.
API ↔ Frozen Model Registry	Load selected preprocessing objects, models, calibrators, thresholds, and hashes.	Local serialized artifacts and compatibility manifest.
API ↔ Deterministic SE-BRL Module	Produce four dimension states, subordinate indicators, availability, and evidence references.	Versioned deterministic evidence records; no learned behavioral confidence.
Decision Engine ↔ Frozen Risk Rules	Resolve coverage, severity, corroboration, conflict, state, and defense.	DecisionObject fields plus versioned rule ID.
Research Pipeline ↔ Storage	Record provenance, audit, grouping, splits, configurations, candidate results, efficiency benchmarks, selection decision, and verification evidence.	Local research files and metadata; operational raw content not persisted by default.
Git / repository	Version source, configuration, documentation, and release tags.	Nonsecret source and permitted artifacts only.

Candidate models not selected for runtime remain research artifacts and are not all executed during operational inference.

3.4 Communications Interfaces

- Local client-server communication shall use versioned REST endpoints and UTF-8 JSON.
- Loopback HTTP may be used for local development. Any non-loopback deployment shall require HTTPS with TLS 1.2 or higher, authentication, and a renewed threat review.
- CORS shall be restricted to approved origins. Requests shall be schema-validated, size-limited, rate-limited, and time-bounded.

- Responses shall use escaped text, inert/defanged URLs, stable error codes, and no active HTML.
- The baseline shall not require SMTP, SMS, persistent WebSockets, message queues, VPN tunnels, live reputation APIs, or third-party generative-AI services.

4. System Features

Features are ordered from intake through deterministic representation, phishing classification, decision, presentation, failure handling, and reproducibility. Each requirement is independently testable unless an explicit dependency is stated.

4.1 Artifact Intake, Validation, and Modality Routing

4.1.1 Description and Priority

This feature accepts an explicitly submitted artifact, validates the selected mode and required fields, assigns traceability metadata, and routes only to supported processing. Priority: High.

4.1.2 Stimulus/Response Sequences

1. The user selects Email, Webpage Content, URL, or approved Research Record and submits the available fields.
2. The system validates payload size, schema, allowed scheme, encoding, and declared modality; it assigns a unique run ID and artifact hash.
3. The router sends content-bearing email/webpage artifacts to eligible conventional and deterministic SE-BRL branches, and URL/engineered records to the technical-only U1 branch.
4. Invalid, mismatched, or insufficient input is rejected or marked Review Required before a definitive analytical output is assembled.

4.1.3 Functional Requirements

Table 4.1 specifies artifact-intake and routing requirements.

Table 4.1

Artifact-intake functional requirements

ID	Requirement Description	Priority
REQ-4.1-1	The system shall begin analysis only after an explicit user action.	High

ID	Requirement Description	Priority
REQ-4.1-2	The system shall support distinct Email, Webpage Content, URL, and approved Research Record input contracts.	High
REQ-4.1-3	The system shall assign a unique run identifier, declared modality, schema version, and nonreversible artifact hash to each accepted request.	High
REQ-4.1-4	The system shall reject dangerous schemes, malformed encodings, oversized payloads, and modality-field mismatches with stable error codes.	High
REQ-4.1-5	The system shall route URL-only and engineered records to U1 technical processing without assigning unsupported language-dependent SE-BRL states.	High
REQ-4.1-6	The system shall represent unassessable behavioral evidence as Unavailable rather than Absent.	High
REQ-4.1-7	The system shall not automatically fetch, open, or navigate to a submitted destination.	High

These requirements enforce modality boundaries before any deterministic representation or learned phishing classification is executed.

4.2 Safe Parsing and Modality-Specific Feature Extraction

4.2.1 Description and Priority

This feature converts untrusted artifacts into inert, normalized, evidence-preserving representations and generates conventional features appropriate to each modality. Priority: High.

4.2.2 Stimulus/Response Sequences

1. The router invokes the corresponding email, webpage, URL, or engineered-record parser.
2. The parser extracts permitted fields, neutralizes active content, and produces a normalized object with source-to-normalized evidence mappings.
3. Conventional feature extractors operate on the normalized object using the frozen feature schema.
4. Warnings, excluded fields, parser version, and feature-schema version are returned for later DecisionObject assembly.

4.2.3 Functional Requirements

Table 4.2 defines safe parsing and feature-extraction requirements.

Table 4.2

Parsing and feature-extraction requirements

ID	Requirement Description	Priority
REQ-4.2-1	The system shall parse email subject/body, available sender/reply relations, headers, links, formatting, and attachment metadata without opening attachment contents.	High
REQ-4.2-2	The system shall parse supplied webpage text/HTML, DOM/form structure, input types, links, and brand-destination relations without executing scripts or loading remote assets.	High
REQ-4.2-3	The system shall normalize URLs as strings and extract lexical, domain, path, query, and source-provided technical values without network retrieval.	High
REQ-4.2-4	The system shall preserve a traceable mapping between normalized evidence and original field, token span, element, form, or relation.	High
REQ-4.2-5	The system shall apply only modality-approved conventional feature definitions and shall not substitute missing fields with class-dependent defaults.	High
REQ-4.2-6	The system shall report parser warnings, excluded content, preprocessing version, and feature-schema version.	Medium
REQ-4.2-7	The system shall escape or remove active markup before any content is rendered in a client interface.	High

Safe parsing preserves the evidence needed by deterministic rules without turning the analyzer into a crawler or content-execution environment.

4.3 Deterministic SE-BRL Rule Execution and Evidence Representation

4.3.1 Description and Priority

This feature applies the frozen literature-derived SE-BRL rule specification to eligible content-bearing email and webpage artifacts. The four behavioral dimensions are deterministic representation states; the twelve original behaviors remain subordinate evidence indicators. No supervised behavioral classifier is trained. Priority: High.

4.3.2 Stimulus/Response Sequences

1. The system loads the checksum-valid deterministic SE-BRL rule specification and modality support matrix.
2. For each eligible artifact, rules inspect only approved evidence fields and record any qualifying subordinate indicators with exact evidence references.

3. Each dimension receives Supported, Absent, or Unavailable according to rule satisfaction and evidence availability.
4. The system emits a versioned long-form evidence record and a fixed-width deterministic state vector for E2/E3 feature construction.

Figure 4.1 formalizes the deterministic state encoding used by the implementation.

$$\text{Supported} = (1, 1) \quad \text{Absent} = (0, 1) \quad \text{Unavailable} = (0, 0)$$

First coordinate = support state; second coordinate = availability state.

Figure 4.1. Deterministic evidence-state encoding.

As shown in Figure 4.1, support and availability are encoded separately. Unavailable evidence is therefore not collapsed into ordinary absence and cannot be misread as evidence against a dimension.

Figure 4.2 defines the fixed-width deterministic representation consumed by E2 and E3..

$$\mathbf{z}_{\text{SE-BRL}} = [s_1, s_2, s_3, s_4, a_1, a_2, a_3, a_4]^T$$

Four support values and four availability values form the fixed-width SE-BRL vector.

Figure 4.2. Fixed-width deterministic SE-BRL representation vector

As shown in Figure 4.2, the four support values and four availability values form the model-compatible SE-BRL vector. Artifact identifiers, source names, raw evidence text, and phishing labels are excluded from this predictive vector.

4.3.3 Functional Requirements

Table 4.3 specifies deterministic SE-BRL execution and evidence requirements.

Table 4.3

Deterministic SE-BRL requirements

ID	Requirement Description	Priority
REQ-4.3-1	The system shall represent Pressure and Threat Cues using urgency, fear/threat, and scarcity subordinate indicators.	High

ID	Requirement Description	Priority
REQ-4.3-2	The system shall represent Lure and Attention Cues using curiosity and reward/lure subordinate indicators.	High
REQ-4.3-3	The system shall represent Trust and Identity Manipulation using authority, impersonation, brand exploitation, and confidentiality/isolation subordinate indicators.	High
REQ-4.3-4	The system shall represent Requested Action and Consequence using call-to-action, credential/sensitive-data request, and financial-action request subordinate indicators.	High
REQ-4.3-5	A Supported dimension shall require at least one rule-compliant subordinate indicator and traceable evidence reference; otherwise it shall not be asserted.	High
REQ-4.3-6	The system shall distinguish assessable absence from unavailable evidence through explicit support and availability values.	High
REQ-4.3-7	The E2/E3 predictive vector shall exclude artifact IDs, dataset/source names, raw evidence text, evidence locators, phishing labels, and any human behavioral labels.	High
REQ-4.3-8	Every deterministic SE-BRL record shall include rule-specification version, modality-support version, and schema version.	High
REQ-4.3-9	Repeated execution of the same frozen rules on the same normalized artifact shall produce the same states and evidence references, subject only to explicitly versioned normalization changes.	High

Rule verification shall use predefined software/data test cases for conformance, evidence-state correctness, modality boundaries, evidence-reference completeness, schema validity, and repeatability; it shall not be described as expert behavioral validation.

4.4 Phishing Classification, Candidate Selection, and Controlled Feature Comparison

4.4.1 Description and Priority

This feature separates research-time candidate evaluation from runtime inference. Four candidate phishing-classification families are evaluated under controlled development conditions; the selected implementation classifier is frozen before calibration/threshold freeze and before the sealed test is opened. Priority: High.

4.4.2 Experimental Feature Configurations

Table 4.4 defines the feature configurations used for controlled comparison.

Table 4.4

Experimental feature configurations

Configuration	Feature Input	Scope / Control
E1	Conventional modality-specific features only.	Email and webpage; same eligible artifacts and grouped partitions used for E2/E3.
E2	Deterministic SE-BRL state representation only.	Email and webpage content-bearing artifacts; no behavioral classifier or predicted dimension confidence.
E3	Conventional features + deterministic SE-BRL state representation.	Principal fused representation; compared against E1 with classifier held constant for the confirmatory contrast.
U1	Engineered technical features only.	Independent technical benchmarks; schemas remain source-specific and are not pooled into a universal raw feature space.

Note. E2 and E3 consume deterministic states from Section 4.3; they do not require OOF behavioral predictions. The configuration definitions preserve a fair representation comparison without introducing a separate behavioral-learning task.

4.4.3 Candidate Classifiers and Predeclared Selection Criteria

Table 4.5 lists the four candidate phishing-classification families that must be trained and evaluated before implementation selection.

Table 4.5

Candidate phishing-classification models

Candidate	Role	Calibration / Explanation	Efficiency Evidence
Logistic Regression	Candidate and primary confirmatory classifier for the E3-vs-E1 representation contrast.	Probabilistic output; standardized coefficients and/or SHAP-compatible attribution.	p50/p95 model-only latency, throughput, memory/model size where feasible.
Linear SVM	Sparse/high-dimensional candidate comparator.	Decision scores calibrated when operational probability is required; linear attribution.	Same benchmark conditions as all candidates.
Random Forest	Nonlinear interaction candidate comparator.	Probability calibration as required; tree-based attribution.	Same benchmark conditions as all candidates.

Candidate	Role	Calibration / Explanation	Efficiency Evidence
XGBoost	Boosted-tree/high-capacity candidate comparator.	Probability calibration as required; TreeSHAP-compatible attribution.	Same benchmark conditions as all candidates.

Note. No candidate is declared the empirical deployment winner before the controlled development-stage selection procedure is executed.

Table 4.6 specifies the predeclared hierarchy used for development-stage model selection.

Table 4.6

Predeclared model-selection criteria

Criterion Class	Measure	Decision Role
Predictive adequacy	PR-AUC (primary); precision, recall, F1, ROC-AUC, balanced accuracy, MCC.	Establish discrimination and class-specific adequacy under the same grouped folds.
Calibration	Brier score, ECE, reliability analysis where probabilistic output is required.	Required when downstream risk rules consume calibrated probability.
Latency	p50/median and p95 model-only inference latency (ms/artifact).	Deployment-feasibility criterion measured on the same frozen environment.
Speed	Throughput (artifacts/second).	Deployment-feasibility criterion measured under the same benchmark conditions.
Resources	Memory consumption and model artifact size where feasible.	Secondary feasibility evidence.
Interpretability / reproducibility	Explanation compatibility and deterministic pipeline reproduction.	Tie-breaking/implementation consideration; not a substitute for predictive adequacy.

Note. Selection uses a predeclared hierarchy, not a post-hoc weighted score invented after results are observed.

Figure 4.3 shows the required order from audited data to runtime implementation.

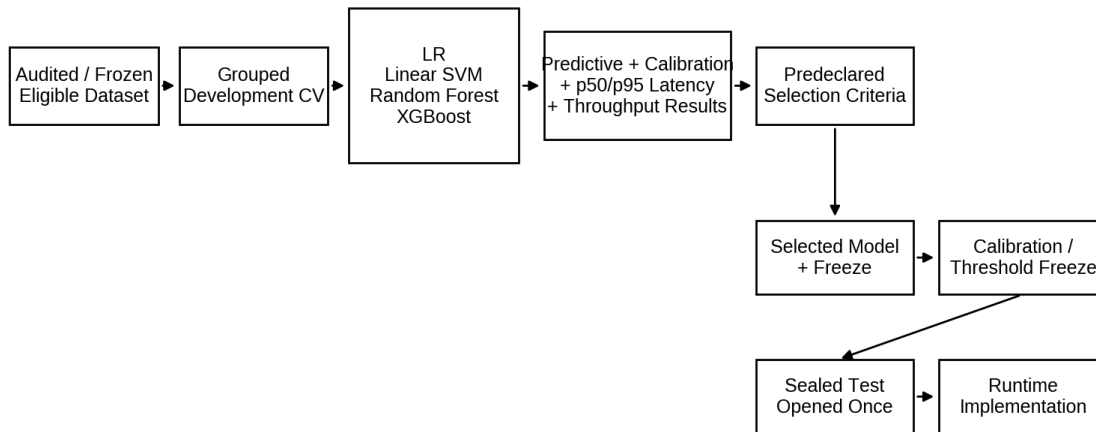


Figure 4.3. Candidate model evaluation, selection, freeze, calibration, and sealed-test workflow.

As shown in Figure 4.3, model-family and hyperparameter selection occur entirely within development evaluation. The selected model is frozen before calibration/threshold freeze, and the sealed test is opened once only after those decisions are fixed.

4.4.4 Operational Computations and Controlled Comparison

Evidence coverage is an artifact-level availability measure used to prevent low-information outputs from being treated as well-supported decisions. Figure 4.4 defines the computation.

$$C = \frac{\text{available required evidence fields}}{\text{total required evidence fields for the modality}}$$

Evidence coverage is an availability measure, not a phishing probability.

Figure 4.4. Evidence coverage.

Figure 4.4 expresses coverage as the fraction of modality-required evidence fields that are available and valid. The frozen decision rules may use a modality-specific minimum, but the SRS does not invent an unsupported numeric threshold.

For a metric M evaluated under identical controls, Figure 4.5 defines the incremental contribution of the fused E3 representation relative to E1.

$$\Delta M = M_{E3} - M_{E1}$$

The incremental change for metric M compares the fused representation with the conventional baseline.

Figure 4.5. Incremental contribution of the fused representation.

As shown in Figure 4.5, the contribution is reported as a direct E3-minus-E1 difference under matched classifier, split, and preprocessing conditions; it does not imply a guaranteed improvement.

4.4.5 Functional Requirements

Table 4.7 specifies classification, development-selection, and controlled-comparison requirements.

Table 4.7

Classification and controlled-comparison requirements

ID	Requirement Description	Priority
REQ-4.4-1	Runtime inference shall load only checksum-valid, version-compatible feature schemas, selected classifier(s), and calibrator(s).	High
REQ-4.4-2	The system shall output a predicted class, calibrated phishing probability, selected-model identifier, calibration identifier, and configuration identifier.	High
REQ-4.4-3	E1, E2, and E3 research comparisons shall use identical eligible artifacts, group-aware partitions, classifier family, and tuning budget within each confirmatory contrast.	High
REQ-4.4-4	Exactly four candidate classifier families shall be evaluated for selection: Logistic Regression, Linear SVM, Random Forest, and XGBoost.	High
REQ-4.4-5	Model family/hyperparameter selection shall use development-stage evidence only, with 5-fold grouped cross-validation, fold-local learned preprocessing, fixed seeds, and controlled search budgets.	High
REQ-4.4-6	Dataset groups shall be assigned before the 70% development / 15% calibration / 15% sealed-test partitioning; no group shall cross partitions.	High
REQ-4.4-7	The selected implementation classifier and benchmark environment shall be frozen before calibration/threshold freeze and sealed-test access.	High
REQ-4.4-8	PhiUSIIL and UCI technical benchmarks shall remain independent and shall not be represented as one universal raw feature space.	High
REQ-4.4-9	The system shall not present an uncalibrated margin or score as an operational phishing probability.	High

ID	Requirement Description	Priority
REQ-4.4-10	The sealed test shall not be used to choose model family, hyperparameters, feature schema, calibration method, or thresholds.	High

These controls distinguish confirmatory representation comparison from empirical implementation selection while preserving a one-time final sealed evaluation.

4.5 Explainability and DecisionObject Assembly

4.5.1 Description and Priority

This feature assembles a schema-valid DecisionObject and preserves artifact evidence and model attribution as noninterchangeable explanation channels. Priority: High.

4.5.2 Stimulus/Response Sequences

1. The API receives parser, conventional-feature, deterministic SE-BRL, selected-classifier, calibrator, and attribution outputs.
2. The assembler validates completeness, finiteness, and component compatibility.
3. Artifact evidence is recorded as observed support; model attribution is recorded as model-centered influence.
4. The DecisionObject is forwarded to the deterministic decision engine only when all critical fields are valid.

4.5.3 Functional Requirements

Table 4.8 defines explainability and DecisionObject requirements.

Table 4.8

Explainability and DecisionObject requirements

ID	Requirement Description	Priority
REQ-4.5-1	The DecisionObject shall contain run ID, artifact hash, modality, class, calibrated probability, technical findings, deterministic SE-BRL records, attribution, decision fields, actions, limitations, and component versions.	High
REQ-4.5-2	Artifact evidence shall identify the supporting text span, field relation, webpage element, form, URL feature, or technical indicator when available.	High
REQ-4.5-3	Model attribution shall identify the feature, value, contribution, explanation method, and selected-model version and shall be labeled as attribution rather than factual evidence.	High

ID	Requirement Description	Priority
REQ-4.5-4	The system shall not display a Supported SE-BRL finding without a valid evidence reference.	High
REQ-4.5-5	An observed deterministic SE-BRL finding may be displayed even when its model attribution is low, provided the evidence record is valid.	High
REQ-4.5-6	The assembler shall reject missing critical fields, incompatible versions, nonfinite probability values, or invalid state combinations.	High
REQ-4.5-7	Structured exports shall preserve schema version and explanation provenance.	Medium

Evidence answers what is present in the artifact; attribution answers what influenced the selected phishing classifier. The interface shall not collapse these meanings.

4.6 Context-Induced Risk Assessment and Risk-Adaptive Defense

4.6.1 Description and Priority

This feature maps calibrated phishing probability, evidence coverage, requested-outcome severity, independent corroboration, and uncertainty conditions to a deterministic artifact-level state and bounded defensive response. It does not compute a weighted psychological or personalized risk score. Priority: High.

4.6.2 Operational Thresholds

If the final protocol retains precision-constrained and recall-constrained operational thresholds, they shall be selected only from the calibration partition and frozen before sealed testing. Figure 4.6 defines the high threshold.

$$\tau_H = \min\{\tau : \text{Precision}(\tau) \geq P_{\min}\}$$

Figure 4.6. High operational probability threshold.

The high threshold is the smallest calibration-set threshold meeting the predeclared minimum precision target. If the target is unattainable, the system shall report that condition rather than invent a threshold.

Figure 4.7 defines the low threshold used to preserve the predeclared minimum recall target.

$$\tau_L = \max\{\tau : \text{Recall}(\tau) \geq R_{\min}\}$$

Figure 4.7. Low operational probability threshold.

The low threshold is the largest calibration-set threshold that still meets the predeclared recall target. The thresholds govern artifact-level decision rules only; they are not behavioral-dimension thresholds.

4.6.3 Ordered Decision Logic

Figure 4.8 summarizes the required precedence of Review Required, High, Elevated, and Informational states.

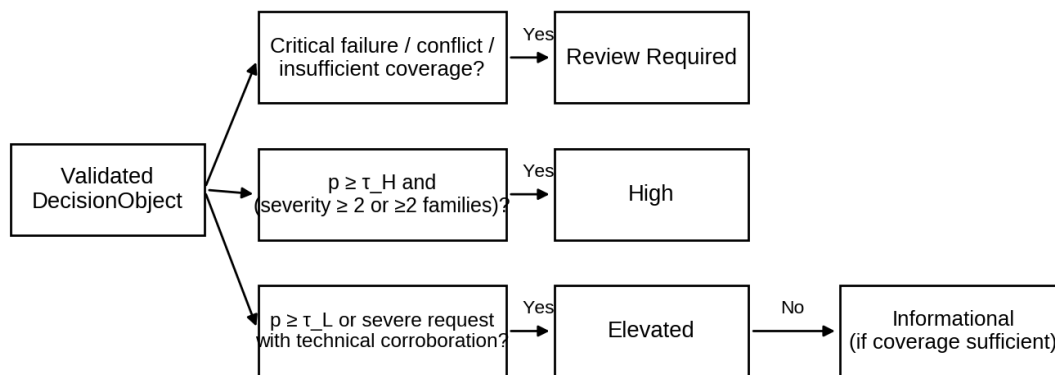


Figure 4.8. Ordered risk-state precedence.

As shown in Figure 4.8, invalid, incompatible, conflicting, or insufficient-evidence conditions are evaluated before definitive risk tiers. The final state follows ordered deterministic rules rather than a weighted aggregate score.

4.6.4 Functional Requirements

Table 4.9 specifies risk and response requirements.

Table 4.9

Risk and response requirements

ID	Requirement Description	Priority
REQ-4.6-1	The system shall encode requested-outcome severity as 0 none, 1 general call-to-action, 2 credential/sensitive-data request, or 3 financial-action request; the maximum observed severity shall apply.	High
REQ-4.6-2	Corroboration shall count independent evidence families: behavioral language, identity/authentication, URL/destination, form/structure, and brand-domain relation.	High
REQ-4.6-3	Invalid input, incompatible versions, parser/model failure, evidence conflict, distribution warning, or coverage below the frozen modality minimum shall produce Review Required before any definitive tier.	High
REQ-4.6-4	High shall require calibrated probability at or above the frozen high threshold and either severity at least 2 or corroboration from at least two independent evidence families.	High
REQ-4.6-5	Elevated shall apply when calibrated probability meets the frozen low threshold or when severity at least 2 has at least one corroborating technical family, unless a higher-priority rule applies.	High
REQ-4.6-6	Informational shall require probability below the low threshold, sufficient evidence coverage, no conflict, and no severe request.	High
REQ-4.6-7	The rule order, not a weighted aggregate score, shall determine the final state.	High
REQ-4.6-8	Actions shall be advisory and shall not automatically navigate, submit data, transfer funds, quarantine content, or report a user.	High

The risk engine consumes calibrated phishing output plus validated artifact-level evidence and uncertainty fields; it does not infer user susceptibility.

4.7 Browser Extension and Dashboard Presentation

4.7.1 Description and Priority

This feature provides the bounded user-facing validation platform: a concise Manifest V3 extension panel and a React dashboard for complete evidence, attribution, decision, and report presentation. Priority: High.

4.7.2 Functional Requirements

Table 4.10 defines extension and dashboard requirements.

Table 4.10

Extension and dashboard requirements

ID	Requirement Description	Priority
REQ-4.7-1	The extension shall use least-privilege Manifest V3 permissions and access only the active tab or explicitly selected message/artifact fields required for the scan.	High

ID	Requirement Description	Priority
REQ-4.7-2	The dashboard shall provide distinct input modes whose field validation matches the API contracts.	High
REQ-4.7-3	The result view shall present prediction, calibrated probability, technical evidence, deterministic SE-BRL evidence/state, model attribution, decision state, actions, limitations, and versions in separate labeled regions.	High
REQ-4.7-4	Suspicious URLs, HTML, scripts, and evidence snippets shall be rendered as escaped, inert content.	High
REQ-4.7-5	Review Required shall be visually and textually distinct from Informational, Elevated, and High.	High
REQ-4.7-6	The interface shall not expose dataset administration, model training/selection, or threshold editing to a Demonstration End User.	Medium
REQ-4.7-7	Controls shall be keyboard operable and status shall not rely on color alone.	Medium
REQ-4.7-8	The dashboard may export a schema-versioned non-executable JSON or report representation of the current result.	Medium

Presentation requirements remain descriptive and functional; the study does not measure human warning comprehension or behavior change.

4.8 Error, Conflict, and Review-Required Handling

4.8.1 Description and Priority

This feature prevents partial, stale, incompatible, unsupported, or low-coverage analytical outputs from being presented as valid phishing-risk decisions. Priority: High.

4.8.2 Functional Requirements

Table 4.11 specifies fail-closed and Review-Required requirements.

Table 4.11

Error and Review-Required requirements

ID	Requirement Description	Priority
REQ-4.8-1	A critical parser, feature, deterministic SE-BRL, selected-model, calibrator, attribution, decision-rule, or schema failure shall prevent definitive risk-state generation.	High
REQ-4.8-2	Version or checksum mismatch shall fail closed and identify the incompatible component without exposing secret paths or credentials.	High
REQ-4.8-3	Conflicting evidence channels or probability/state boundary ambiguity shall produce Review Required according to the frozen rules.	High
REQ-4.8-4	Unsupported modality evidence shall be marked Unavailable and shall not be imputed as behavioral absence.	High
REQ-4.8-5	Partial component results shall not be merged with stale values from an earlier run.	High
REQ-4.8-6	Every failure response shall contain a stable error code, affected stage, human-readable limitation, and safe next step.	Medium

ID	Requirement Description	Priority
REQ-4.8-7	The system shall not silently fall back to a default class, probability, state, or intervention.	High
REQ-4.8-8	End-user error details shall exclude stack traces, raw content, secrets, and sensitive filesystem information.	High

Fail-closed behavior applies equally to deterministic rules and learned classifier components; missing analytical evidence shall not be hidden behind a default risk state.

4.9 Audit, Versioning, and Reproducibility

4.9.1 Description and Priority

This feature preserves sufficient non-sensitive evidence to reproduce research results, verify operational compatibility, audit dataset/model decisions, and rerun principal analyses without retaining raw user-submitted content by default. Priority: High.

4.9.2 Functional Requirements

Table 4.12 specifies dataset-audit, selection-freeze, sealed-test, and reproducibility requirements.

Table 4.12

Audit, versioning, and reproducibility requirements

ID	Requirement Description	Priority
REQ-4.9-1	Each raw research-source snapshot shall be immutable and identified by source, access date, local filename, license, schema, source-reported semantics, source_snapshot_id, and SHA-256 hash.	High
REQ-4.9-2	The dataset pipeline shall record schema inventory, label verification, exclusions/reason codes, missingness, exact/near-duplicate groups, source-confounding checks, attrition, final eligible pool, and group IDs before partitioning.	High
REQ-4.9-3	Split manifests shall preserve group identifiers and 70/15/15 development, calibration, and sealed-test membership; no group shall cross partitions.	High
REQ-4.9-4	Deterministic SE-BRL release artifacts shall include rule specification, modality-support matrix, rule test cases, evidence-state tests, schema checks, evidence-reference checks, repeatability report, and version/hash records.	High
REQ-4.9-5	Candidate-model artifacts shall include controlled configurations, grouped-CV results, calibration and efficiency metrics, benchmark environment, model-selection decision log, and selected-model hash.	High
REQ-4.9-6	Frozen runtime artifacts shall include selected model(s), vectorizer/preprocessing pipeline, feature schema, calibrator, deterministic SE-BRL rules, thresholds, risk/response rules, and API schema versions/hashes.	High
REQ-4.9-7	Operational logs shall exclude raw artifact content and may retain run ID, artifact hash, modality, status, component versions, latency summaries, and error category.	High

ID	Requirement Description	Priority
REQ-4.9-8	The sealed test shall be opened once only after representation, selected model(s), calibration, thresholds, decision rules, and benchmark protocol are frozen.	High
REQ-4.9-9	Post-test defects shall be documented and shall not be silently retuned and presented as the same final experiment.	High
REQ-4.9-10	The release package shall support an independent clean rerun of dataset group/split assignment and principal metrics within a stated numerical tolerance.	Medium

The audit trail replaces older annotation/adjudication artifacts with deterministic-rule verification, dataset-audit, candidate-evaluation, efficiency-benchmark, and model-selection records.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

Model-selection efficiency and end-to-end application performance are reported separately. Figure 5.1 defines model-only inference latency as the prediction interval after a model-compatible feature vector has already been prepared.

$$L_{\text{model}} = t_{\text{prediction end}} - t_{\text{prediction start}}$$

Model-only latency excludes parsing and feature construction.

Figure 5.1. Model-only inference latency used in candidate-model evaluation.

Model-only latency excludes parsing, deterministic SE-BRL rule execution, feature construction, calibration, attribution, decision processing, API handling, and presentation. The benchmark shall report p50/median and p95 values for each applicable candidate.

Figure 5.2 defines model inference throughput under the same documented benchmark conditions.

$$\text{Throughput} = \frac{N_{\text{artifacts}}}{t_{\text{inference}}}$$

where $N_{\text{artifacts}}$ = number of artifacts processed,

$t_{\text{inference}}$ = total inference time.

Throughput is reported as artifacts processed per second under the *frozen* benchmark environment.

Figure 5.2. Inference throughput.

Throughput is reported in artifacts per second and shall not be substituted for end-to-end application latency. Cold/warm conditions, batch size, hardware, software versions, and payload class must be documented.

Table 5.1

Performance requirements

ID	Requirement	Priority
NFR-5.1-1	Interactive analysis shall terminate within a configurable safety timeout; timeout shall produce no partial definitive decision.	High
NFR-5.1-2	The default API payload limit shall be bounded and configurable for controlled research tests.	High
NFR-5.1-3	Candidate-model benchmarks shall record model-only p50/median and p95 inference latency in milliseconds per artifact on the same documented frozen benchmark environment.	High
NFR-5.1-4	Candidate-model benchmarks shall record inference throughput in artifacts per second under the same benchmark conditions.	High
NFR-5.1-5	System validation shall record end-to-end latency separately from model-only latency and shall identify the included pipeline stages.	High
NFR-5.1-6	Baseline inference shall execute on the documented CPU reference environment without a mandatory GPU.	High
NFR-5.1-7	Performance reports shall disclose hardware, OS/runtime/library versions, cold/warm conditions, payload class, memory consumption, and model artifact size where feasible.	Medium
NFR-5.1-8	Any numeric performance acceptance threshold used for release shall have a documented requirements basis and shall be frozen before sealed testing.	High

Note. The SRS requires measurement and transparent reporting; it does not claim production-scale throughput.

Performance evidence is therefore both a model-selection input and a separate system-validation output.

5.2 Safety Requirements

Table 5.2 defines the safety boundaries of the proof-of-concept.

Table 5.2

Safety requirements

ID	Requirement	Priority
NFR-5.2-1	No submitted script, active HTML, remote asset, link, or attachment content shall be executed or opened by analysis.	High
NFR-5.2-2	No automated browsing, credential submission, transaction, quarantine, deletion, or external report shall result from a risk state.	High
NFR-5.2-3	Uncertainty, unsupported modality, and insufficient evidence shall prefer Review Required over unsupported certainty.	High
NFR-5.2-4	User-facing actions shall include an independent-verification option and state relevant limitations.	High

These safeguards preserve a bounded advisory role and prevent the research prototype from acting as an autonomous enforcement system.

5.3 Security Requirements

Table 5.3 defines security requirements for the executed path.

Table 5.3

Security requirements

ID	Requirement	Priority
NFR-5.3-1	The API shall use typed schema validation, supported-scheme allowlists, payload limits, timeouts, rate limits, and restricted CORS.	High
NFR-5.3-2	The extension shall enforce a strict content security policy and least-privilege Manifest V3 permissions.	High
NFR-5.3-3	Any deployment beyond loopback shall require TLS 1.2 or higher, authentication, and a documented threat review.	High
NFR-5.3-4	Secrets, credentials, restricted datasets, and raw user artifacts shall be excluded from source control and client bundles.	High
NFR-5.3-5	Dependency locks and vulnerability audits shall block release on unresolved critical vulnerabilities affecting the executed path.	High
NFR-5.3-6	Frozen model, rule, schema, and calibration artifacts shall be checksum-verified before use.	High

Security controls apply to both learned and deterministic components because either can invalidate a decision if substituted or modified outside the frozen release.

5.4 Software Quality Attributes

Table 5.4 summarizes software quality expectations for the final proof-of-concept.

Table 5.4

Software quality attributes

Attribute	Requirement
Correctness	Every release-blocking requirement shall have expected-versus-actual verification; all critical routing, deterministic evidence-state, decision-rule, version, and fail-closed tests shall pass.
Reliability	A complete compatible DecisionObject or explicit failure shall be returned; stale or partial state shall not survive between runs.
Maintainability	Modules shall separate parsing, conventional features, deterministic SE-BRL, phishing classification, calibration, attribution, decision logic, presentation, and audit through versioned interfaces.
Reproducibility	Configurations, seeds, manifests, hashes, dependencies, benchmark environment, and selection records shall permit clean reruns of principal results.
Portability	The local prototype shall run on documented Windows, macOS, or Linux environments using project-pinned dependencies and a supported Chromium browser.
Usability	Result terminology shall be consistent, limitations visible, statuses non-color-only, and next actions concise and nontechnical.
Testability	Requirement IDs, stable schemas, deterministic rules, predefined boundary cases, and saved payloads shall support automated regression testing.
Privacy	Operational raw content shall be transient by default, and audit output shall use data minimization and hashed identifiers.

Quality claims shall be limited to the executed tests and environments documented in the final release.

5.5 Business Rules

Table 5.5 records the non-negotiable research-governance rules implemented across the requirements.

Table 5.5

Business and research-governance rules

ID	Business / Research Rule
BR-1	Observable artifact evidence and frozen deterministic rules are the only basis for SE-BRL states; phishing class labels shall not be inputs to those rules.
BR-2	The four dimensions are deterministic representation states; the twelve original behaviors are subordinate evidence indicators.
BR-3	Supported, Absent, and Unavailable are distinct states; Unavailable is never encoded as ordinary absence.
BR-4	Email and webpage E1/E2/E3 experiments are modality-specific; U1 benchmarks are independent technical branches.

ID	Business / Research Rule
BR-5	Supervised machine learning is reserved for phishing classification; no separately trained behavioral-dimension classifier exists in the final methodology.
BR-6	Candidate model selection uses development-stage evidence and predeclared criteria; the sealed test never selects the implementation classifier.
BR-7	Review Required has precedence over High, Elevated, and Informational.
BR-8	The final decision state is selected by ordered deterministic rules, not a weighted aggregate psychological risk score.
BR-9	Artifact evidence and model attribution shall never be presented as interchangeable.
BR-10	The sealed test is opened once after representation, selected model, calibration, thresholds, and decision rules are frozen.
BR-11	No claim may exceed the executed data, models, interfaces, software verification, security/privacy testing, or human-subject scope.

These rules supersede conflicting current-method statements from earlier SRS revisions.

6. Other Requirements

6.1 Data Governance and Privacy

- Public datasets shall be retained only under documented access and license conditions; restricted raw files shall not be redistributed without permission.
- Raw sources, derived data, audit artifacts, group/split manifests, deterministic-rule artifacts, candidate-model results, selected models, reports, and release records shall be stored in separated versioned directories with documented retention procedures.
- Operational analysis shall not collect passwords, form values, cookies, unrelated browsing history, attachment contents, demographics, psychological data, or persistent personalization profiles.
- Dataset audit outputs shall use reason codes and hashes where possible and shall avoid unnecessary replication of sensitive content.

6.2 Licensing and Reproducibility

- The release shall document licenses for all components and exclude artifacts that cannot be lawfully shared.
- The reproducibility package shall include source code, environment files, dataset and split records, SE-BRL rules/tests, feature definitions, model configurations/results, efficiency benchmarks, selection records, calibration/threshold versions, risk/response rules, sealed-test records, validation tests, and a rerun guide.

- Omitted artifacts shall document lawful reconstruction or access procedures.

6.3 Language and Localization

- The baseline deterministic SE-BRL rule specification shall target the language supported by the frozen literature-derived definitions and implemented test cases; English is the baseline research language unless a later controlled revision adds validated rules for another language.
- Non-English or mixed-language artifacts without supported deterministic rules shall be marked unsupported for SE-BRL or Review Required rather than force-assigned behavioral states.
- UI strings shall be externalizable; full localization is not required for the baseline release.

6.4 Required Documentation and Deliverables

- Updated SRS, schemas, SE-BRL rules, modality matrix, dataset and split records, experiment/model records, calibration, risk rules, benchmarks, test evidence, security/privacy checks, and reproducibility report.
- Proof-of-concept extension, dashboard, FastAPI service, frozen analytical and SE-BRL bundles, and sample results.
- Final results shall report audited datasets, rule verification, model evaluation and selection, sealed-test results, and system validation without post-test retuning.

6.5 Scope and Claim Boundaries

The implementation shall not reintroduce expert behavioral validation, behavioral ground-truth annotation, behavioral classifiers, learned behavioral confidence, OOF behavioral prediction, behavioral calibration, or per-dimension learned thresholds. Any future scope expansion requires an explicit SRS revision, corresponding methodology revision, and new verification plan.

Claims about phishing detection, explainability, risk response, performance, security, or usability shall remain bounded to the actual executed datasets, frozen configurations, test environments, and proof-of-concept interfaces.

Appendix A. Glossary

Table A.1

Glossary

Term	Definition
Absent	Required evidence is assessable, but no rule-compliant supporting evidence is observed.
Artifact evidence	Observable text, field relation, form/DOM element, link relation, or technical indicator supporting a finding.
Attribution	A model-centered estimate of how a feature influenced a phishing prediction; not factual artifact evidence.
Audited/Frozen dataset	A dataset state after provenance, schema, label, eligibility, duplicate/group, attrition, and split controls are executed and recorded.
Calibration	Adjustment/evaluation of phishing-classifier scores so operational probability is meaningful for downstream risk rules.
Candidate classifier	One of Logistic Regression, Linear SVM, Random Forest, or XGBoost evaluated before implementation selection.
Corroboration	Support from independent evidence families, not a count of correlated features.
DecisionObject	Versioned schema containing analysis outputs, evidence, attribution, decision fields, actions, limitations, and compatibility data.
Deterministic SE-BRL rule specification	Frozen literature-derived operational definitions, modality support, inclusion/exclusion rules, evidence requirements, and deterministic state logic.
E1	Conventional modality-specific phishing feature configuration.
E2	Deterministic SE-BRL state representation only.
E3	Fusion of E1 and deterministic E2 on the same eligible artifacts.
Evidence coverage	Proportion of modality-required analytical evidence fields that are available and valid.
Evidence state	Supported, Absent, or Unavailable deterministic assessment outcome.
Frozen artifact	Approved model, preprocessing object, schema, calibrator, deterministic rule specification, threshold, or decision-rule file that cannot change during sealed testing.
High / Elevated / Informational	Ordered definitive artifact-level decision states defined by frozen probability, severity, corroboration, and evidence conditions.
Model-only latency	Prediction interval measured after a model-compatible feature vector has been prepared.
Review Required	Precedence state for invalid, conflicting, unsupported, incomplete, incompatible, distribution-warning, or insufficient-coverage conditions.
SE-BRL	Social Engineering Behavior Representation Layer.
Sealed test	Final partition opened once only after model selection, representation, calibration, thresholds, and decision rules are frozen.
Selected implementation classifier	Candidate model family/configuration selected using predefined criteria and frozen before sealed testing.
Supported	Rule-required evidence is assessable and qualifying evidence is observed.

Term	Definition
Throughput	Number of artifacts processed by model inference per unit time under a documented benchmark environment.
U1	Independent technical benchmark for stand-alone URL or engineered records.
Unavailable	The modality or required field does not permit assessment.
Risk-adaptive defense	Deterministic selection of bounded protective and independent-verification guidance for the current artifact; not personalized learning.
Glossary terms use the final deterministic/no-expert scope and supersede obsolete learned behavioral-confidence terminology.	

Appendix B: Analysis Models

B.1 System Context and Trust Boundaries

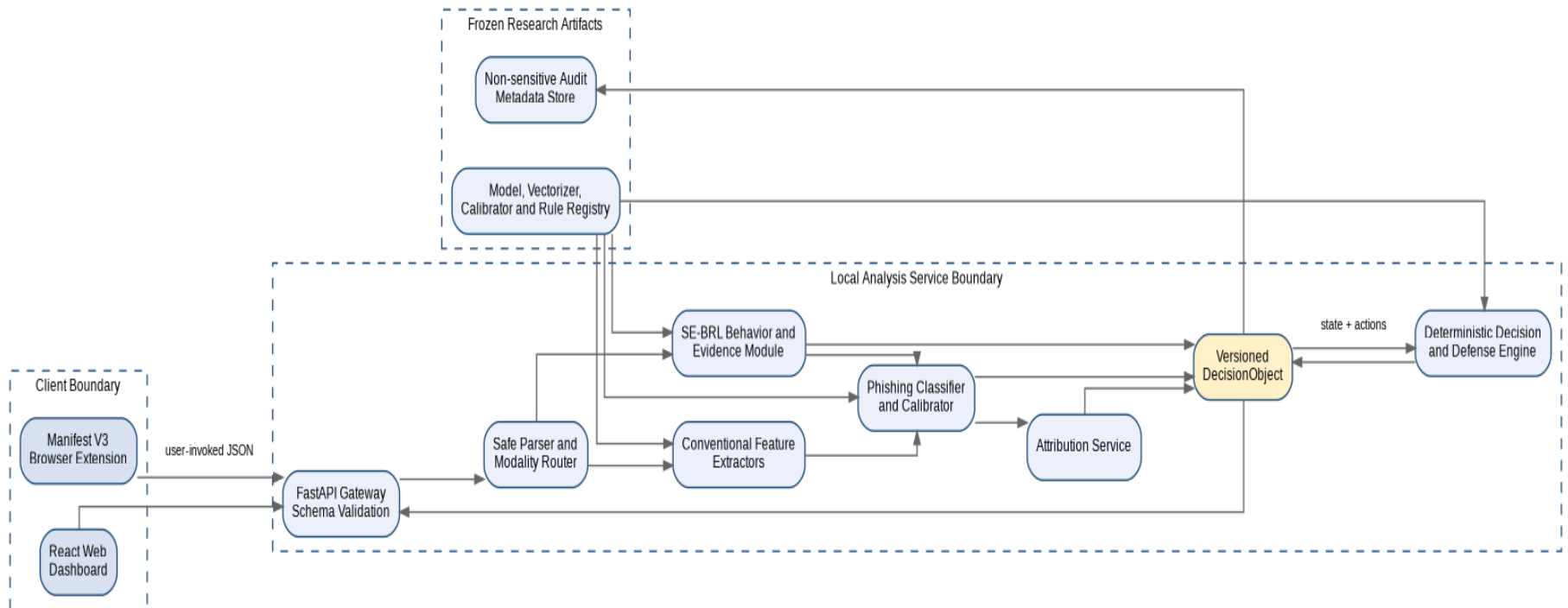


Figure B.1. SE-BRL system context, local-service boundary, and frozen-artifact boundary

B.1 Modality Routing and Technical Boundary

Figure B.1 shows how content-bearing modalities and technical-only records remain separated before the selected phishing-classification step.

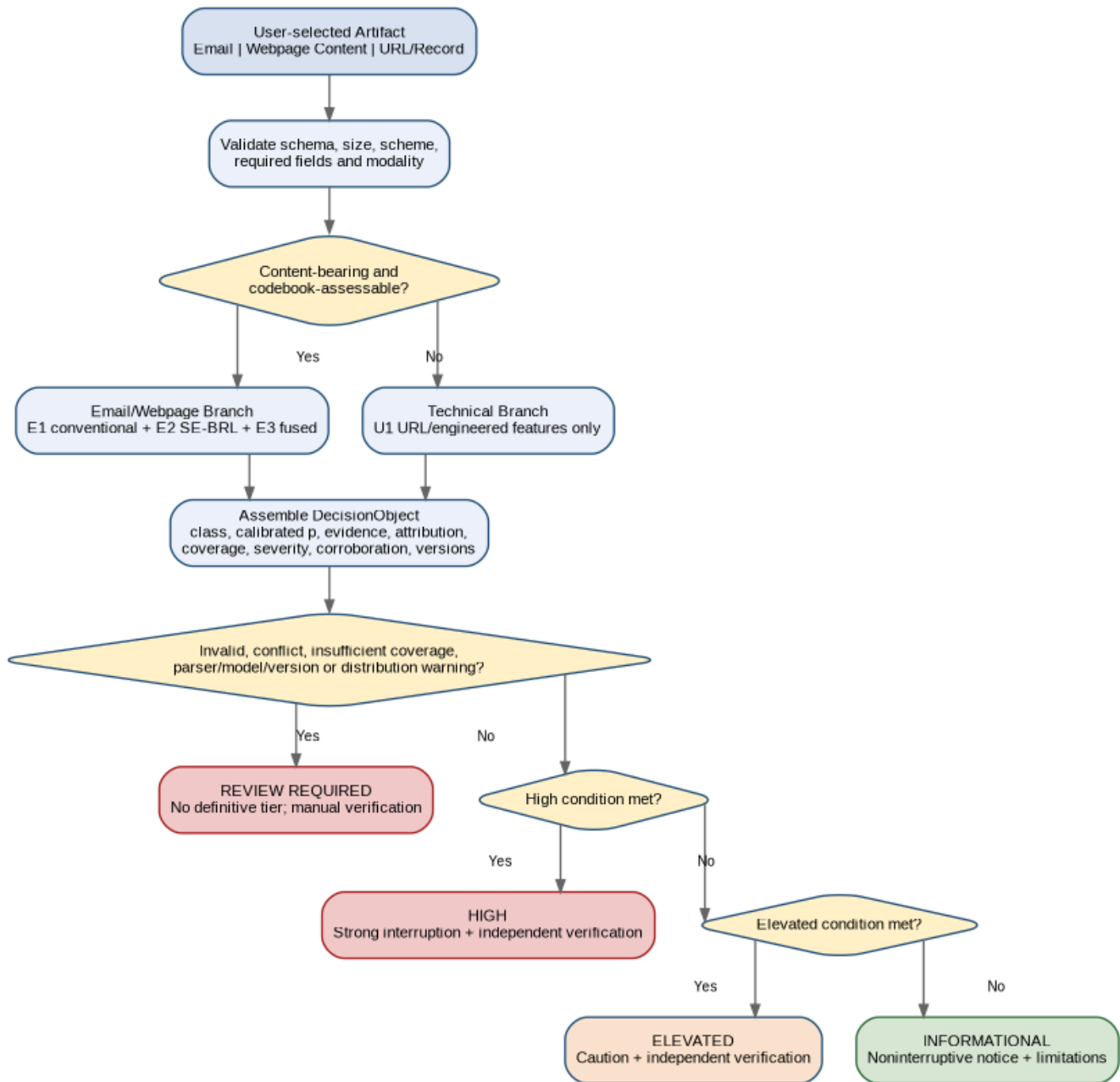


Figure B.2. Modality-aware processing boundary.

Email and webpage artifacts may contribute deterministic SE-BRL states when required content is available. Bare URLs and engineered records remain U1 technical-only and cannot receive language-dependent SE-BRL values.

B.2 DecisionObject Logical Model

Figure B.2 summarizes the logical fields that must be assembled before deterministic risk processing.

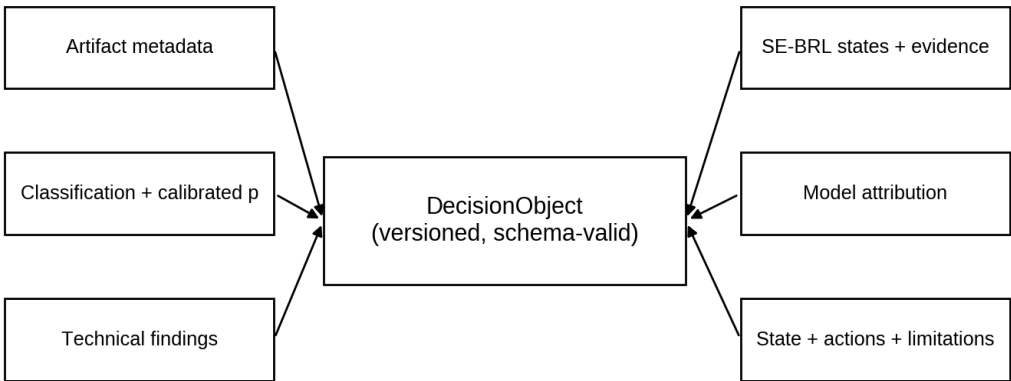


Figure B.2. Logical composition of the versioned DecisionObject.

The DecisionObject preserves artifact evidence, model attribution, prediction, decision state, and limitations as separately typed information. Schema validation rejects incompatible or incomplete combinations.

B.3 Analysis Sequence

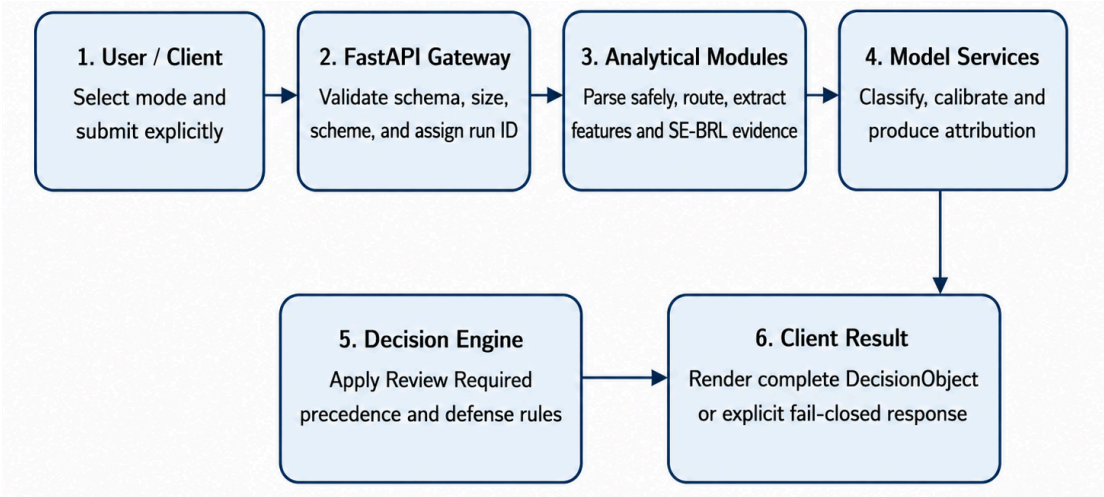


Figure B.3. User-invoked end-to-end analysis sequence

Figure B.1. SE-BRL system context, local-service boundary, and frozen-artifact boundary

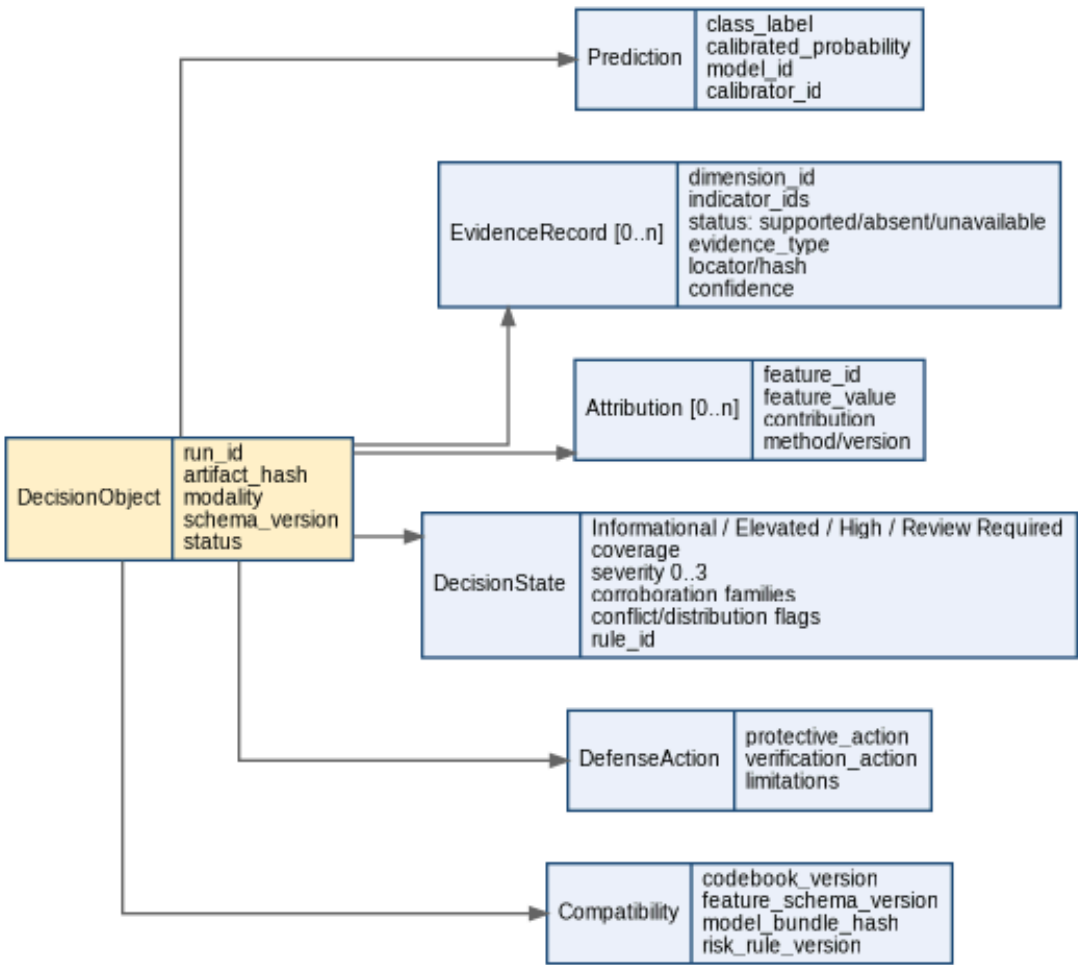


Figure B.4. Logical composition of the versioned DecisionObject

B.3 Research-Objective Traceability

Table B.1 maps the five final research objectives to the principal SRS controls.

Table B.1

Research-objective traceability

Research Objective	Primary SRS Coverage
Develop and software-verify the deterministic modality-aware SE-BRL rule specification and evidence schema.	REQ-4.1-5 to REQ-4.3-9; BR-1 to BR-5; NFR-5.2-3.
Compare E1/E2/E3 phishing representations and evaluate candidate classifier families under controlled selection criteria.	REQ-4.4-1 to REQ-4.4-10; REQ-4.9-3 to REQ-4.9-6; BR-4, BR-6, BR-10.
Separate artifact evidence from model attribution in explainable outputs.	REQ-4.5-1 to REQ-4.5-7; REQ-4.7-3; BR-9.
Map calibrated phishing output and validated artifact evidence to bounded risk-adaptive defense.	REQ-4.6-1 to REQ-4.6-8; BR-7 to BR-8; NFR-5.2.
Functionally validate the extension/backend/dashboard workflow, including security, performance, and reproducibility.	REQ-4.7 to REQ-4.9; NFR-5.1 to NFR-5.4; Section 3 interfaces.

The traceability map contains five objectives only and does not introduce an additional O6/RQ6.

Appendix C. Use-Case Model

C.1 SE-BRL Use-Case Diagram

Figure C.1 identifies the operational interactions of the three final user personas while preserving frozen research artifacts and fail-closed boundaries.

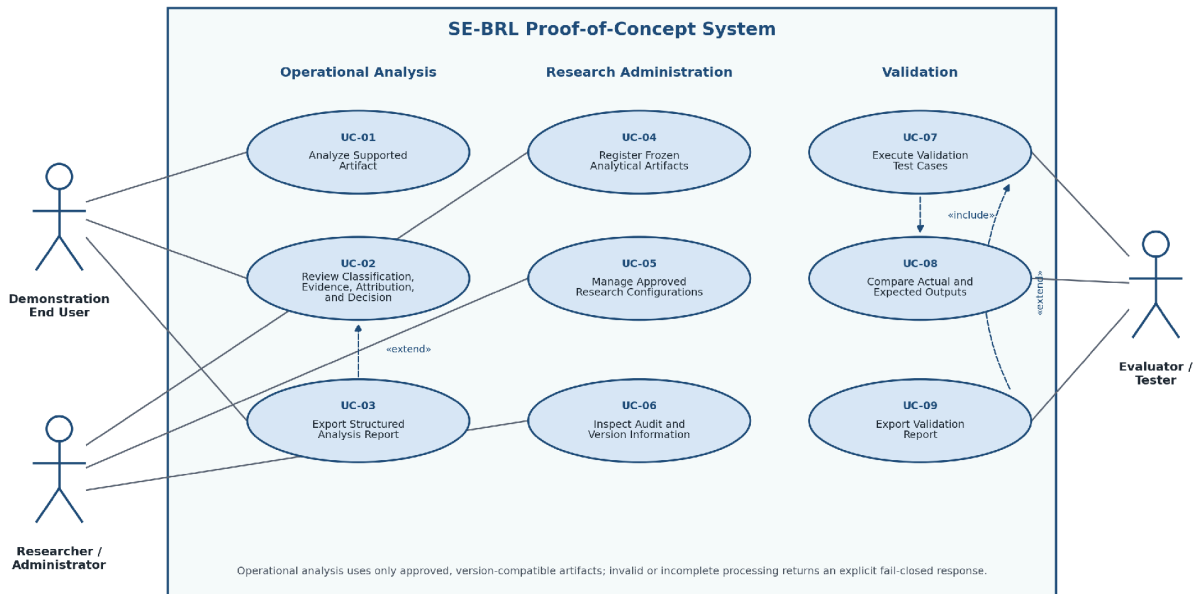


Figure C.1. SE-BRL proof-of-concept use-case diagram.

The use-case model contains no Expert Validator or Behavioral Annotator actor. Researcher / Administrator functions concern approved configuration, frozen artifacts, and audit records; Evaluator / Tester functions concern predefined verification cases.

C.2 Use-Case Descriptions

Table C.1 defines the expected outcome of each proof-of-concept use case.

Table C.1

Use-case descriptions

ID	Use Case	Primary User Persona	Description and Expected Outcome
UC-01	Analyze Supported Artifact	Demonstration End User	Validate the selected Email, Webpage Content, URL, or approved Research Record; route/process safely; return a complete versioned DecisionObject or explicit fail-closed response.
UC-02	Review Analysis Result	Demonstration End User	Present predicted class, calibrated probability, technical indicators, deterministic SE-BRL evidence/states, model attribution, decision state, bounded response, and limitations.
UC-03	Export Structured Analysis Report	Demonstration End User; Researcher / Administrator	Export analysis identifier, result fields, evidence references, component versions, and limitations; raw artifact content remains excluded by default.
UC-04	Register Frozen Analytical Artifacts	Researcher / Administrator	Register approved selected models, preprocessing objects, calibrators, deterministic SE-BRL rules, thresholds, schemas, and decision rules with checksums and compatibility metadata.
UC-05	Manage Approved Research Configurations	Researcher / Administrator	Select and record versioned dataset-audit, model-development, calibration, benchmark, or validation configurations; operational analysis shall not retrain models or modify frozen thresholds.
UC-06	Inspect Audit and Version Information	Researcher / Administrator	Inspect non-sensitive source/audit/run identifiers, warnings, failure states, and component versions required for traceability.
UC-07	Execute Validation Test Cases	Evaluator / Tester	Run approved functional, deterministic-rule, boundary, conflict, security, fail-closed, performance, integration, and end-to-end cases against frozen components.
UC-08	Compare Actual and Expected Outputs	Evaluator / Tester	Verify routing, evidence states, DecisionObject completeness, classification, decision precedence, response mapping, version compatibility, and error handling; record deviations as defects.
UC-09	Export Validation Report	Evaluator / Tester	Produce scenario IDs, pass/fail status, actual and expected results, component versions, benchmark conditions, relevant non-sensitive logs, and documented deviations.

Use cases describe software interactions and testable outcomes only; they do not add human-subject measures or expert behavioral validation.